

claude-windows / c6028e72-ebfd-4218-89a8-c4d7dd23be96

Metadata

```
- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\c6028e72-ebfd-4218-89a8-c4d7dd23be96\subagents\workflows\wf_49aab888-e10\agent-a53096ff3d7d17aec.jsonl`
- SHA-256: `d105982daa492f00043510e3fd5c6dabea43a02861c099306068711989ec6693`
- Source modified: `2026-06-08T13:21:28+00:00`
- Imported at: `2026-07-05T16:48:23+00:00`
- project: `wf_49aab888-e10`
- session_id: `c6028e72-ebfd-4218-89a8-c4d7dd23be96`
```

Transcript

1. user (2026-06-08T13:17:41.865Z)

Synthesize ONE detailed, roadmap-aligned study for how we should train our OWNED rally/highlight model reusing our labeled videos. Integrate the research + apply the verification corrections (drop/adjust any refuted/outdated claim). Be concrete and decision-useful: rank the training PLATFORMS (Vertex AI vs alternatives) through OUR privacy+cost lens; give a PHASED model path (now: a small owned TAL model on our features that beats the heuristic baseline at small n; later: Option-B end-to-end multimodal on an Apache-2.0 video base ? never Gemma 3); a model-agnostic LABEL-REUSE strategy so today's CSVs train every future generation; the licensing/privacy guardrails; and honest cost estimates. Respect every hard guardrail (Apache-2.0 bases only, no paid-LLM training data, no minors). Cite sources.

GROUNDING:

```
{
  "owned_model_goal": "Build an OWNED, on-device-capable generative video-QA model (fine-tuned from an Apache-2.0 video VLM base) for RALLY TEMPORAL SEGMENTATION and future end-to-end multimodal sport understanding. Privacy lever: local inference, user video never leaves device. The model plugs into the existing provider registry with paid Gemini as an always-ready fallback. Long-term path (Option B): collapse modular local cues (WASB trajectory, audio hit detection, pose detection) into a single fine-tuned end-to-end model. Near-term (Option A, already shipping): modular signal fusion on frozen or near-frozen bases. The moat is NOT the model weights ? it's the consented badminton-specific video?QA dataset + the eval harness. Owned model becomes primary only after scaffolding (async jobs, multi-tenant, storage/compute abstractions) is rock-solid and the golden-set eval/training corpus is healthy.",
  "constraints": [
    "Base model MUST be Apache-2.0 clean (commercial-viable, fine-tunable, derivative-wrappable). REJECT Gemma 3 (viral Google PUP), Llama Vision (700M-MAU cap + naming mandate). APPROVED bases: Qwen2.5-VL-7B/32B (Apache-2.0 at those sizes only; 3B/72B custom-licensed), Qwen3-VL-4B/8B, Gemma 4 multimodal (Apache-2.0, 2026), InternVL2.5 (MIT).",
    "HARD RULE: Training data must be Apache-2.0 or MIT open-model outputs ONLY. FORBIDDEN: any outputs from paid frontier APIs (Gemini, OpenAI, Anthropic, Claude) as training signal ? they are competing-model clauses and breach terms. Paid APIs are inference/evaluation/teacher-reference only, never training data sources. This is non-negotiable per legal.",
    "Exclude minors' video from the training pool BY DEFAULT. Training requires separate, explicit opt-in consent (distinct from processing consent). Per-user training adapters trained only on that user's own consented video; deletion = drop the adapter.",
    "Generic data schema (no coach-athlete hardcoding). Coach/athlete overlay is optional, orthogonal. Data model stays reusable for solo users, academies, power users.",
    "Training evaluation must use a held-out golden QA eval set (controlled, curated, not user-contributed). Agent-runnable training requires trustworthy pass/fail verdict (not eyeballing). Regression testing locked to golden labels to prevent metric inflation from data leakage.",
    "Video-QA fine-tuning works ONLY on labeled data. Currently available: 96 (Adarsh+Avi) + 32 (Kushagra) human-labeled rally [start,end] CSVs per match + Gemini silver labels (eval-only, not training). 5k?20k curated clip?QA pairs is the realistic training band."
  ],
  "data_assets": [
```

"Human-labeled rally [start,end] CSVs per match (Adarsh-and-Avi 96 rallies, Kushagra 32; schema: rally_number, start_time, end_time, ending_reason, sport). Source: sports-data-collector repo, curate export with commercial-license gate.",

"Per-rally ending_reason labels (winner, forced_error, unforced_error, service_fault, let) ? rally semantics feeding the QA-pair dataset.",

"Per-video WASB shuttle-trajectory CSVs (Frame, Visibility, X, Y) from frozen MIT-licensed WASB-SBDT shuttle detector. Trajectory is a primary local signal for multi-cue fusion.",

"Gemini silver labels (eval and teacher-reference ONLY, explicitly not training-data). Used to measure oracle quality and as a training-time prior (Option A modular fusion), never harvested as outputs to train the owned model.",

"Single-camera amateur GoPro footage: 1080p/4K, 60fps. 22/22 ShuttleSet matches re-acquired at best available resolution (Phase 2 complete). ~26 GB corpus licensed/owned, no user uploads.",

"Golden-set ground-truth expansion via Roora vendor (Bengaluru, initial pilot underway): 3-sport pilot (badminton, pickleball, table tennis) with validated rally CSV schema and annotation guideline.",

"Sibling rallyannotator repo (MIT, open-source, VLC-based): owner can self-produce expert-labeled eval clips. Feed via `import-local` to the collector and `--labels` loaders to the indexer."

],

"task_framing": "***Task: design a training study blueprint for the OWNED video-QA model that clarifies (1) what data is safe/available now, (2) which base model to lock, (3) which fine-tuning backend fits our constraints, and (4) how to stage the transition from heuristics ? Option A modular fusion ? Option B end-to-end owned model.**\\n\\n**Current pipeline (heuristics + frozen detectors):** WASB (frozen vision) ? trajectory ? hand-tuned windowing + net-crossing gate + distilled LogisticRegression on trajectory features. Honest baseline: threshold-tuned F1 ? 0.73 vs human labels (one match); learned model underperforms at n=2 videos. Bottleneck = golden data + WASB recall, not method.\\n\\n**Option A (now ? 6?12 months):** Modular, fully-local cues: WASB trajectory + audio hit-detection (E1 probe AMBER / leaning negative, gated on more golden labels) + pose/court serve-gating (validated but heavy). Each cue swappable, independently testable, generates training signal for the end-to-end model. Fusion happens in our rally layer, not inside frozen detectors.\\n\\n**Option B (12?24+ months, after scaffolding + healthy corpus):** Collapse modular cues into a single owned multimodal rally model fine-tuned on (frames + audio + pose/stance + ?) ? rally [start, end]. Removes recall ceiling imposed by WASB candidate windows and licensing question on academic-trained weights. Requires: (a) enough golden eval/training video (owner self-labels via rally-annotator + vendor labels), (b) end-to-end training harness (nanochat-style operability: one command, report card, deterministic, reproducible), (c) a measured improvement bar vs. the modular Option A.\\n\\n**Bridge data flow (cross-repo, ADR-002 contract):** sports-data-collector's per-match rally CSVs + manifest.json (commercial-license gate) ? indexer's golden eval set and training signal. Collector's 5-sport tables (badminton/tennis/cricket/pickleball/table-tennis) share a (video_id, ordinal, start_time, end_time) shape, so the bridge is sport-generic. Golden-set expansion funded via Roora (Bengaluru) ? initial 3-sport pilot.\\n\\n**Immediate blockers on training-study readiness:** (1) **Owned-model base selection + licensing lock** (Qwen vs Gemma 4; verify per-checkpoint commercial terms). (2) **Fine-tuning backend decision** (self-managed ms-swift on rented GPU [RunPod/Lambda/Modal] vs. managed Together/Fireworks vs. enterprise Predibase). (3) **Golden QA eval harness design** (the crux: nanochat-style pass/fail verdict). (4) **Training data schema & consent** (how to convert rally CSVs + ending-reason labels + Gemini reference ? clip?QA pairs without using paid outputs; minors exclusion). (5) **Data leakage guardrails** (`train | eval` split by match, not clip; overlap detection). (6) **Training job orchestration** (how it slots into the async job model P0, runs on ComputeBackend, fits BYO-compute).\\n\\n**Success criteria for the study:** (a) Blueprint for converting existing golden labels into 5k?20k curated clip?QA pairs. (b) Confirmed Apache-2.0 base + verified commercial DPA from the fine-tune backend. (c) Proof-of-concept golden QA eval gate that blocks a regression-quality model. (d) Staged roadmap: Option A (modular, 2?4 weeks) ? Option B e2e model (after corpus health signals, 3?6 months post-scaffold). (e) Clear cost/latency comparison (self-hosted 7B VLM vs. Gemini per-call on target workload).\\n",

"roadmap_alignment_notes": [

"**Sequencing principle (owner directive):** Make scaffolding (P0?P4 in PLATFORM_ARCHITECTURE §6) rock-solid FIRST, keep the known paid stack (Gemini) as an always-ready fallback, and only then shift focus to owned-model quality (P5). This study unblocks P5's design but does NOT start implementation until P0?P4 infrastructure is delivered.",

"**P0 (async job model, single-tenant)** is the prerequisite: owned-model fine-tuning runs as an orchestration job on ComputeBackend GPUs (§5A / DEFERRED #16). Without async + job dispatch, training is not operationalizable.",

***P1?P3 (multi-tenant, storage/compute abstraction)** enable the data flywheel: training corpus lives in Storage (§3.1), training/inference jobs dispatch to ComputeBackend (§3.2), per-user adapters live in the compute tier. Without these, the owned model is stranded on a single box.",

***P4 (rich rally/shot semantic metadata)** feeds P5: the annotation registry + candidate_segments + golden-set labels are the training-signal source. Per-rally ending_reason labels + human-correction loop ? QA-pair generation.",

***C14 (COMMERCIALIZATION.md)** tracks the owned-model licensing + training-data policy as a P1 blocker: Apache-2.0 base lock, open-model teachers only, no paid-LLM training signal, separate explicit training opt-in, minors excluded. This study must resolve C14 before any code lands.",

***ADR-004 substrate decision** (2026-06-02 update): Phase 4 offline segmenter is WASB shuttle-motion, not YOLO player-motion (YOLO @ IoU-0.5 recall too low). The owned end-to-end model (Option B) will eventually subsume WASB's role. ADR-007 audio fusion is the Option A stepping stone.",

***ADR-007 direction (accepted):** Audio hit-detection is the next rally-detection cue (Option A). It is fully local, permissive, and fuses cleanly with the distilled classifier. Gated on golden labels (E1 probe AMBER, awaits more data). This is the realistic 274 week Option A milestone while waiting for scaffold + corpus growth.",

***Golden-set expansion (ADR-006 / Rooru pilot):** Vendor-produced labels are the highest-leverage move. With Rooru on 3-sport badminton/pickleball/table-tennis, the training corpus grows month-on-month. The training study should plan for incremental re-training (CI-integrated, agent-runnable) as each batch lands.",

***Long-term moat (§5A, PLATFORM):** The defensible assets are (a) the consented, badminton-specific video?QA dataset and (b) the eval harness. Do NOT message 'we own an LLM' ? the base VLM (Qwen/Gemma) is freely downloadable. Message the dataset + eval precision instead.",

***Cost & margin (C7 / COMMERCIALIZATION):** Self-hosted 7B VLM (?16 GB VRAM) on rented GPU is cheaper per inference than Gemini only if: (a) sustained QA volume is high, (b) training corpus is defensibly large, (c) per-call API cost pain is real. Below those thresholds, paid API is cheaper than standing GPU + second stack. Benchmark before committing budget.",

***Training harness operability (§5A, PLATFORM):** The goal is 'an agent (or CI) can run a fine-tune on a fresh batch of golden-labeled video and get a trustworthy ship/no-ship verdict, unattended.' This requires: (1) one command, end-to-end (ms-swift engine, recipe-driven); (2) one complexity dial (auto-derived defaults); (3) automatic report card (held-out golden QA eval vs target); (4) reproducible + deterministic + cheap; (5) minimal/readable. Nanochat is the reference for operability (not the engine). The harness = thin glue over ms-swift + the golden eval + the provider registry."

]

}

RESEARCH:

[

{

"dimension": "Training compute platforms for an owned video-QA (rally temporal segmentation) model",

"summary": "A LoRA/QLoRA fine-tune of an Apache-2.0 7B VLM on our 5k-20k clip-QA pairs is a small job: under 24h on one A100 80GB, \$10-50 total. So the deciding factor is privacy terms and operational ergonomics, not GPU dollar-per-hour. Modal has the strongest written no-access contract; Vertex AI is the only strong-privacy option with an India region. The RTX 2070 Super 8GB is dev/inference-only, too small for real 7B-VLM training.",

"options": [

{

"name": "Self-hosted RTX 2070 Super 8GB + WSL2",

"what": "Local single-GPU fine-tuning/inference via ms-swift/Unsloth/QLoRA. Zero marginal cost, zero data egress.",

"pros": [

"Maximal privacy: data never leaves the machine - the local-first moat",

"Zero cost; ideal free dev/CI smoke-test box and 4-bit 7B inference",

"No lock-in, no DPA, no billing surprises"

],

"cons": [

"8GB too small for a real 7B-VLM QLoRA run (needs ~16-24GB with vision/video tokens; 4-bit 7B text inference alone ~7.1GB)",

"Turing lacks bf16/modern FlashAttention kernels",

"Single GPU; unsuitable for Option B end-to-end multimodal training"

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    ],
    "cost_note": "$0 marginal (owned). Dev/CI/inference only.",
    "license": "N/A (own hardware)",
    "privacy_note": "BEST possible - data never leaves device.",
    "fit_for_us": "medium"
  },
  {
    "name": "Modal (serverless GPU) - RECOMMENDED PRIMARY",
    "what": "Serverless Python-native compute, per-second billing, scale-to-zero, A100/H100.
Has a Training product.",
    "pros": [
      "Strongest privacy contract: Modal 'will never access your source code, the
inputs/outputs to your Functions, or any data you store'; inputs/outputs deleted within 7 days;
SOC2 Type II; HIPAA via BAA",
      "Per-second billing + scale-to-zero = no idle cost, perfect for retrain-on-each-
golden-batch cadence",
      "Python/recipe-driven - fits a one-command agent-runnable harness and the future
ComputeBackend abstraction",
      "Asia-Pacific regions (Singapore, Seoul); $30/mo free credits"
    ],
    "cons": [
      "No India asia-south1 region (Singapore closest) - fine for batch training but not
India residency-at-rest",
      "A100 ~$3.72/hr, H100 ~$4.29/hr - pricier per-hour than RunPod/Vast",
      "Regional multipliers 1.25x-2.5x for non-US"
    ],
    "cost_note": "A100 80GB ~$3.72/hr, H100 ~$4.29/hr per-second. Full LoRA fine-tune
~$10-$50. $30/mo free credits.",
    "license": "Commercial cloud; you keep all weights/data. Qwen2.5-VL-7B Apache-2.0 base
clean.",
    "privacy_note": "STRONGEST written no-access/short-retention terms found. Meets the hard
requirement.",
    "fit_for_us": "strong"
  },
  {
    "name": "RunPod (Secure Cloud tier) - RECOMMENDED FALLBACK",
    "what": "Rented GPU pods + serverless. Secure Cloud (vetted DCs) vs Community Cloud
(unvetted hosts).",
    "pros": [
      "Cheap A100 80GB ~$1.19-$1.29/hr, H100 ~$2.65/hr (spot ~$1.19) - best value for
longer/multi-GPU runs",
      "Secure Cloud: SOC2 Type II, ISO 27001, GDPR, HIPAA; AES-256 at rest; ToS bars hosts
inspecting pod data",
      "Per-tenant container isolation; instant data destruction on delete; serverless +
checkpointing; has a DPA"
    ],
    "cons": [
      "Community Cloud tier unvetted peer-to-peer - MUST restrict corpus to Secure Cloud
only",
      "No India/Asia datacenter - no India residency",
      "Less one-command-reproducible than Modal; USD billing"
    ],
    "cost_note": "A100 80GB ~$1.19-$1.29/hr (Secure), H100 ~$2.65/hr on-demand / ~$1.19
spot. Our fine-tune ~$3-$15.",
    "license": "Commercial cloud; you keep weights/data.",
    "privacy_note": "STRONG on Secure Cloud. MUST avoid Community Cloud for sensitive
data.",
    "fit_for_us": "strong"
  },
  {
    "name": "Google Vertex AI / GCP Compute Engine GPUs - KEEP ON BENCH",
    "what": "Managed Vertex custom-training jobs or raw Compute Engine GPU VMs. The platform
the user named.",
    "pros": [
      "Written: 'Customer data is never used by Google. We do not use customer data to train

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our models, or models used by others, without permission'",
    "India region asia-south1 (Mumbai) for Compute Engine GPUs - genuine India residency-
at-rest, unique among strong options",
    "Mature multi-GPU/checkpointing/VPC/CMEK; strong ecosystem maps to future
Storage/Compute backends"
  ],
  "cons": [
    "Most expensive + most overhead: Vertex mgmt fee (A100 ~$3.37/hr all-in, H100
~$11.27/hr); easy to overspend on always-on resources",
    "Heavyweight IAM/quota/capacity friction; slowest path to a reproducible harness",
    "gen-AI residency region list excluding asia-south1 applies to MANAGED gen-AI APIs,
not custom training on your own weights (asia-south1 GPUs ARE usable)"
  ],
  "cost_note": "A100 ~$3.37/hr all-in, H100 ~$11.27/hr; L4/T4 cheap. Highest per-hour +
overhead.",
  "license": "Commercial cloud; you keep weights/data.",
  "privacy_note": "STRONG written no-train; ONLY strong option with an India (Mumbai)
region for residency.",
  "fit_for_us": "medium"
},
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  "name": "Lambda Labs (Lambda Cloud)",
  "what": "Pure ML-focused GPU cloud, simple on-demand pricing, free egress.",
  "pros": [
    "ML-native UX, pre-baked CUDA/PyTorch; A100 80GB ~$1.99-$2.49/hr, H100
~$3.29-$4.29/hr",
    "Free data egress (vs AWS $0.09/GB, GCP $0.12/GB)",
    "Good multi-GPU clusters for Option B"
  ],
  "cons": [
    "No spot/preemptible tier",
    "Privacy terms less explicit publicly - needs DPA review before the corpus goes on
it",
    "US-only, no India/Asia region; historically tight capacity"
  ],
  "cost_note": "A100 80GB ~$1.99-$2.49/hr, H100 ~$3.29-$4.29/hr, free egress. Our fine-
tune ~$5-$20.",
  "license": "Commercial cloud; you keep weights/data.",
  "privacy_note": "UNVERIFIED - obtain/read DPA before using for consented video.",
  "fit_for_us": "medium"
},
{
  "name": "Together AI (managed fine-tuning API)",
  "what": "Managed LoRA/full fine-tuning of open models priced per training-token; hosted
serving.",
  "pros": [
    "Lowest operational burden: managed LoRA ~$0.48/M tokens (<=16B); fastest path to a
first checkpoint",
    "Supports Qwen and 'any open-source model from HF Hub'; resume-from-checkpoint",
    "SOC2 Type II, HIPAA-aligned, Asia/ME deployment; 'data and models remain fully under
your ownership'"
  ],
  "cons": [
    "Could NOT confirm a verbatim no-train clause or public DPA - BLOCKED for the
consented corpus until signed DPA obtained",
    "Token pricing opaque for VIDEO/multimodal; Qwen2.5-VL video FT support is the open
question",
    "Managed = ship data to their cloud; weakest fit with 'data never leaves device'"
  ],
  "cost_note": "LoRA ~$0.48/M tokens (<=16B); ~$8-$12/M typical. Video token cost
unverified.",
  "license": "Commercial managed; you own resulting weights (per claim) - verify
downloadability.",
  "privacy_note": "UNCONFIRMED no-train clause + no public DPA -> BLOCKED for the corpus
until signed DPA + no-train assurance.",

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    "fit_for_us": "weak"
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  {
    "name": "vast.ai (marketplace) - Secure tier only",
    "what": "Peer-to-peer GPU marketplace, cheapest rates. Community (unvetted) vs Secure Cloud (ISO-27001+ vetted DCs).",
    "pros": [
      "Cheapest anywhere: A100 80GB ~$0.67/hr, H100 ~$1.87/hr, spot to ~$0.34/hr - great for non-sensitive experiments",
      "Secure Cloud: vetted DCs (ISO 27001 min), SOC2 Type II platform-level, per-tenant containers, instant data destruction"
    ],
    "cons": [
      "Community/peer-to-peer tier unvetted - DIRECT conflict with our hard privacy requirement for consented video",
      "Price volatility; reliability varies host-to-host; no India region; rough operability"
    ],
    "cost_note": "A100 80GB ~$0.67/hr, H100 ~$1.87/hr (spot ~$0.34/hr). Cheapest by 2-5x - Secure tier only, ideally non-sensitive.",
    "license": "Marketplace; you keep weights/data.",
    "privacy_note": "Community tier FAILS the requirement. Secure Cloud acceptable for non-consented/synthetic data.",
    "fit_for_us": "weak"
  },
  {
    "name": "AWS SageMaker (EC2 P4d/P5) - Mumbai ap-south-1",
    "what": "Enterprise managed training or raw EC2 GPU. ap-south-1 (Mumbai) fully supported.",
    "pros": [
      "India residency: ap-south-1 full region with SageMaker + P4d (A100)",
      "VPC isolation by default, CMEK/KMS encryption, GDPR-aligned; can opt out of aggregated-metadata sharing",
      "P4/P5 price cuts up to 45% (2025); spot + savings plans; mature distributed training"
    ],
    "cons": [
      "High complexity + cost for a solo founder; SageMaker mgmt premium; spot interruptions",
      "Default aggregate library-usage metadata collection (opt-out is a footgun)",
      "Steep learning curve; heavy SDK lock-in; among the most expensive per-hour"
    ],
    "cost_note": "p4d (8xA100) ~$3.07-$3.43/hr/GPU-equiv; P5 (H100) ~$6.88/hr on-demand / ~$3.83 spot; plus SageMaker premium.",
    "license": "Commercial cloud; you keep weights/data.",
    "privacy_note": "STRONG (VPC+KMS+GDPR, India region) IF you opt out of metadata sharing and use VPC. Operationally heavy.",
    "fit_for_us": "weak"
  },
  {
    "name": "Azure ML - Central India",
    "what": "Enterprise managed ML on Azure NC/ND A100/H100 VMs; Central India region.",
    "pros": [
      "India residency: Central India with A100/H100",
      "Cheap spot: NC A100 v4 ~$0.82/hr spot vs ~$3.67/hr on-demand",
      "Enterprise compliance (GDPR/ISO/SOC), VPC/private endpoints, CMEK; good multi-GPU"
    ],
    "cons": [
      "Could NOT confirm an Azure ML 'we don't train on your data' statement - needs Microsoft DPA review",
      "On-demand H100 very expensive (~$6.98-$12/hr/GPU); only spot attractive",
      "Heavy enterprise tooling; high overhead; pipeline lock-in"
    ],
    "cost_note": "NC A100 v4 ~$3.67/hr on-demand / ~$0.82/hr spot; ND H100 v5 ~$6.98-$12/hr/GPU. India region available.",
    "license": "Commercial cloud; you keep weights/data.",
  }

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    "privacy_note": "Likely STRONG via Microsoft DPA + private endpoints, UNVERIFIED here -
read the DPA first.",
    "fit_for_us": "weak"
  },
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    "name": "CoreWeave / Paperspace (DigitalOcean Gradient)",
    "what": "CoreWeave = GPU-specialist for large reserved fleets; Paperspace = simpler
notebook/VM GPU cloud.",
    "pros": [
      "CoreWeave: SOC2/HIPAA/GDPR/ISO 27001; strong for very large multi-GPU H100 fleets
(Option B if big)",
      "Paperspace: no minimum commitment, notebook-friendly (A100 ~$3.09/hr, H100 ~$5.95/hr
promo)"
    ],
    "cons": [
      "CoreWeave enterprise/reserved-oriented (H100 ~$6.16/hr/GPU, A100 ~$2.70/hr) -
overkill/pricey for small jobs",
      "Paperspace mid-to-high pricing, no spot; tight capacity",
      "Neither has an India region; neither beats Modal/RunPod on privacy specificity"
    ],
    "cost_note": "CoreWeave A100 ~$2.70/hr, H100 ~$6.16/hr/GPU. Paperspace A100 ~$3.09/hr,
H100 ~$5.95/hr. No India region.",
    "license": "Commercial cloud; you keep weights/data.",
    "privacy_note": "CoreWeave has the cert stack; Paperspace less differentiated.
Acceptable, not better than Modal/RunPod.",
    "fit_for_us": "weak"
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],
"key_sources": [
  "https://modal.com/docs/guide/security",
  "https://modal.com/pricing",
  "https://modal.com/blog/region-selection-launch",
  "https://www.runpod.io/pricing",
  "https://docs.runpod.io/references/security-and-compliance",
  "https://www.runpod.io/press/runpod-meets-hipaa-and-gdpr-standards",
  "https://cloud.google.com/blog/products/ai-machine-learning/google-cloud-generative-ai-
data-residency-guarantees-for-data-stored-at-rest",
  "https://cloud.google.com/vertex-ai/pricing",
  "https://lambda.ai/pricing",
  "https://www.together.ai/fine-tuning",
  "https://docs.together.ai/docs/fine-tuning-pricing",
  "https://vast.ai/pricing",
  "https://docs.vast.ai/guides/reference/faq/security",
  "https://aws.amazon.com/sagemaker/pricing/",
  "https://docs.aws.amazon.com/sagemaker/latest/dg/data-protection.html",
  "https://azure.microsoft.com/en-us/pricing/details/machine-learning/",
  "https://learn.microsoft.com/en-us/azure/virtual-machines/sizes/gpu-
accelerated/nca100v4-series",
  "https://www.coreweave.com/pricing",
  "https://www.spheron.network/blog/gpu-cloud-pricing-comparison-2026/",
  "https://willitrunai.com/can-run/qwen-2.5-7b-on-rtx-2070-super-8gb",
  "https://kaitechup.substack.com/p/qwen25-qlora-lora-and-full-fine-tuning"
],
"recommendation": "Decouple from raw GPU price: our fine-tune (Apache-2.0 Qwen2.5-VL-7B +
LoRA/QLoRA on 5k-20k clip-QA pairs) is a sub-24h, $10-$50 job on one A100 80GB, so privacy terms
+ operational ergonomics dominate. Plan: (1) DEV/CI on the RTX 2070 Super (8GB) - free; harness
plumbing, overfit-one-batch sanity, 4-bit 7B inference, NOT real training. (2) PRIMARY = Modal -
strongest written privacy contract ('Modal will never access your source code, inputs/outputs,
or stored data'; 7-day max retention; SOC2 Type II), per-second billing with scale-to-zero
matching the retrain-on-each-golden-batch cadence, Python/recipe-driven model that slots into a
one-command agent-runnable harness and the future ComputeBackend abstraction; run a thin glue
over ms-swift on Modal A100s. (3) FALLBACK / Option-B scale-up = RunPod Secure Cloud (A100
~$1.19-$1.29/hr, SOC2+GDPR+HIPAA, contractual no-inspect) - Secure Cloud only, never Community.
(4) KEEP Vertex AI / GCP on the bench: ONLY strong-privacy choice with an India (asia-south1
Mumbai) region + written no-train-on-customer-data - adopt if/when India residency or an audited

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managed pipeline is mandated. (5) Use vast.ai/Lambda ONLY for non-sensitive synthetic/ablation runs - never the consented corpus on vast.ai community tier. (6) HARD GATE before any consented video reaches Together / Lambda / Azure: signed DPA + explicit written 'we do not train on or retain your data' assurance; until then BLOCKED for the corpus. Sequencing: unblocks PLATFORM P5 design only - do not implement until P0-P4 (async jobs, storage/compute abstractions) ship, per owner directive.",

"confidence": "medium",

"caveats": [

"Pricing is volatile and sourced largely from aggregator blogs plus some primary pages; treat all dollar-per-hour figures as +/-15-25% and re-check live pricing pages before committing budget.",

"Biggest UNVERIFIED gap: VIDEO/multimodal fine-tuning support and cost. Most data found is LLM-text-centric. Qwen2.5-VL VIDEO fine-tuning (frames as image tokens) can balloon token counts and VRAM far beyond text-7B numbers - the \$10-\$50/one-A100 estimate is for a modest LoRA run and must be re-benchmarked on real clip-QA data with actual frame counts before trusting it for Option B.",

"Together AI, Lambda Labs, and Azure ML privacy terms could NOT be confirmed to include an explicit no-train-on-your-data clause from public pages; flagged as needing a signed DPA before the corpus touches them.",

"Vertex/GCP nuance: the published gen-AI residency region list (excludes asia-south1) governs Google's MANAGED gen-AI APIs, not custom training on your own weights via Compute Engine GPUs - asia-south1 GPUs exist and the no-train principle applies, but confirm A100/H100 quota/capacity in asia-south1 (capacity-constrained).",

"RTX 2070 Super: 4-bit 7B text inference fits ~7.1GB and QLoRA training of a 7B can fit ~7GB in ideal text-only conditions, but a 7B-VLM with video frames realistically needs 16-24GB; treat the 2070S as dev/inference-only. Turing also lacks bf16/modern kernels.",

"India-region: only Vertex/GCP, AWS (ap-south-1), Azure (Central India) offer in-India residency among providers studied; Modal/RunPod/Lambda/Vast/CoreWeave/Paperspace do not. For batch TRAINING this is usually irrelevant, but if a legal residency-at-rest requirement emerges only the three hyperscalers qualify.",

"Compliance certs (SOC2/GDPR/HIPAA) attest to security controls, NOT a 'we won't train on your data' promise. The two clear written no-train/no-access assurances found are Modal (no-access) and Google/Vertex (no-train-without-permission); everything else needs contract review."

]

},

{

"dimension": "Model architecture for rally temporal localization/segmentation (action = rally; labels = [start,end] intervals) over per-frame features",

"summary": "For OUR situation ? per-frame features already in hand (WASB trajectory ~3-4 dims + optional audio/pose), only ~3-10 labeled matches, RTX 2070S/8GB, commercial intent, on-device-capable ? the right model class is a LIGHT feature-input temporal model, NOT a heavy end-to-end video network. Two viable families, both consuming pre-extracted per-frame feature sequences (so no RGB pipeline required and they ride directly on WASB output):\n\n(1) TEMPORAL ACTION LOCALIZATION (TAL) ? ActionFormer / TriDet / TemporalMaxer. These output (start, end, class) instances directly via a multi-scale feature pyramid + classification/regression heads. This is the most NATURAL fit for rally [start,end] + ending_reason: each rally is one detected instance; ending_reason becomes the instance class (multi-class head) or a second head. ActionFormer and TriDet are MIT and accept ARBITRARY input feature dimension via a learned input projection (confirmed: the projection maps any D-dim feature to the model width), so the WASB (X,Y,Visibility) trajectory can be fed directly ? I3D is NOT required, just the default choice on benchmarks.\n\n(2) TEMPORAL ACTION SEGMENTATION (TAS) ? MS-TCN++ / ASFormer / DiffAct. These do per-frame dense labeling (frame -> {rally, non-rally}); rally segments are recovered by merging contiguous 'rally' frames. ASFormer is the standout for OUR data-hunger: it is explicitly designed to train FROM SCRATCH on tiny datasets (GTEA = 28 videos, 11 classes) with only ~1.1M params, by injecting local convolutional inductive bias to compensate for the Transformer's missing inductive bias on small data. This is the closest published analogue to our n=3-10 regime.\n\nKEY LICENSE FINDING (a real trap): the ORIGINAL ms-tcn repo (yabufarha/ms-tcn) is MIT + Commons Clause = NON-commercial; do NOT vendor it. The MS-TCN++ repo (sj-li/MS-TCN2), ASFormer (ChinaYi/ASFormer), DiffAct (Finspire13/DiffAct), ActionFormer (happyharrycn/actionformer_release), and TriDet (dingfengshi/TriDet) are all clean MIT and commercially usable. TemporalMaxer's LICENSE file could not be confirmed (404 on raw fetch) ? treat as 'verify before vendoring'. \n\nBADMINTON PRECEDENT: published systems for the adjacent task (per-frame hit detection from shuttle trajectory + pose) use exactly the light-sequence-model recipe ? GRU-based recurrent nets over court/pose/shuttle features predicting per-frame hit/no-hit (the MonoTrack lineage), and rule/threshold pipelines over TrackNet+YOLO. None use

heavy end-to-end video models for the localization step. This validates: a SMALL sequence model over per-frame features is the field-standard approach for racket-sports temporal structure at our data scale.\n\nNONE of these are the 'Option B' end-to-end multimodal VLM (Qwen-VL/Gemma) from the grounding brief ? they are the bridge between today's heuristic+LogisticRegression and that future VLM. They are the right FIRST owned model precisely because they learn from the labels and features we ALREADY have, with no paid-LLM data and no large pretraining corpus.",

```
"options": [
  {
    "name": "ASFormer (TAS) ? RECOMMENDED first owned model",
    "what": "Transformer for action segmentation that adds local convolutional inductive bias (dilated temporal convs) to a vanilla Transformer so it trains from scratch on small data. Per-frame dense classifier: every frame -> {rally, non-rally(, ending_reason)}; rally [start,end] recovered by merging contiguous rally frames + boundary cleanup. Input = per-frame feature sequence of arbitrary dim (we feed WASB X,Y,Visibility +/- audio-onset +/- pose). ~1.1M params.",
    "pros": [
      "Explicitly engineered for SMALL training sets ? trains from scratch on GTEA (28 videos); the only surveyed model whose paper makes the small-data claim its central thesis, matching our n=3-10 regime",
      "Tiny (~1.1M params) -> trivially fits RTX 2070S 8GB; trains in minutes; on-device/edge inference is easy",
      "Consumes per-frame features directly ? rides on existing WASB trajectory output; no RGB/I3D extraction pipeline needed; trivially fuses audio-onset + pose by concatenating feature channels (directly enables Option A modular fusion)",
      "Clean MIT (ChinaYi/ASFormer) ? commercially usable, fine-tunable, wrappable",
      "Per-frame formulation is forgiving of label noise and variable rally length; ending_reason maps to extra per-frame classes or a per-segment head",
      "Beats our current ceiling plausibly: it LEARNS temporal structure rather than threshold-tuning, and degrades gracefully where the LogisticRegression+windowing baseline plateaued (~F1 0.73)"
    ],
    "cons": [
      "Per-frame segmentation needs a post-merge step to produce clean [start,end] intervals (minor engineering; over/under-segmentation is the classic TAS failure mode)",
      "No native 'instance' concept ? two rallies separated by a 1-frame gap can merge; needs a small gap/duration prior (we already have the net-crossing gate to reuse)",
      "Standard recipe assumes I3D features; using raw low-dim WASB features is off-the-beaten-path and may need a feature-smoothing / windowing front-end to give the convs enough signal",
      "Boundary precision is frame-granular, not sub-frame regressed (TAL models regress boundaries more precisely)"
    ],
    "cost_note": "Free/OSS. Training cost ~minutes on the existing RTX 2070S; no cloud GPU required at this data scale. Inference is CPU-feasible.",
    "license": "MIT (ChinaYi/ASFormer) ? verified, commercially clean",
    "privacy_note": "Fully local; ~1.1M-param model runs on-device, user video/features never leave the box. No paid-LLM data anywhere in the loop.",
    "fit_for_us": "strong"
  },
  {
    "name": "ActionFormer (TAL) ? RECOMMENDED parallel/second model",
    "what": "Single-stage Transformer for temporal action localization. Multi-scale feature pyramid + shared classification/regression heads emit (start, end, class) action instances directly at every time step. Input = per-frame feature sequence; a learned input-projection layer accepts ARBITRARY feature dimension (confirmed), so WASB trajectory feeds in without I3D.",
    "pros": [
      "Most NATURAL label mapping: rally = one detected instance with regressed [start,end]; ending_reason = the instance class (multi-class head). Matches our CSV schema (rally_number,start,end,ending_reason) almost 1:1",
      "Sub-frame boundary regression -> more precise [start,end] than frame-merged TAS, directly attacking our F1 ceiling on boundary localization",
      "Input projection accepts any feature dim ? WASB (X,Y,Visibility) + audio + pose concatenated; no RGB pipeline",
      "Clean MIT (happyharrycn/actionformer_release); strong, well-maintained, widely forked
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codebase; TriDet/TemporalMaxer build on it so swapping later is cheap",
    "Standard, agent-runnable training recipe (config-driven) fits the 'one command +
report card' harness goal"
  ],
  "cons": [
    "More data-hungry than ASFormer ? designed/validated on THUMOS/ActivityNet (hundreds-
to-thousands of videos); benchmark papers do NOT claim few-shot, so at n=3-10 it risks
overfitting and may UNDERPERFORM ASFormer initially (same failure we saw when a learned model
lost to the baseline at n=2)",
    "Bigger than ASFormer (multi-scale pyramid + attention) though still small vs a VLM;
needs more careful regularization/augmentation on tiny data",
    "Default features are I3D; low-dim trajectory-only input is unproven and may need
feature engineering to give attention enough to attend to",
    "NMS / instance de-duplication tuning adds knobs"
  ],
  "cost_note": "Free/OSS. Trains on RTX 2070S; somewhat heavier than ASFormer but still no
cloud GPU needed at our scale.",
  "license": "MIT (happyharrycn/actionformer_release) ? verified, commercially clean",
  "privacy_note": "Fully local, small model, on-device inference; no paid-LLM data.",
  "fit_for_us": "strong"
},
{
  "name": "TriDet (TAL)",
  "what": "One-stage TAL built ON TOP of ActionFormer's codebase. Adds a Trident-head
modeling the action boundary as a relative probability distribution + Scalable-Granularity
Perception (SGP) layer replacing some self-attention. Same (start,end,class) instance output and
same arbitrary-feature input as ActionFormer.",
  "pros": [
    "Better boundary modeling than ActionFormer (the Trident-head is purpose-built for
precise [start,end] ? directly relevant to our boundary-localization ceiling)",
    "Drop-in once ActionFormer is wired (shares ActionFormer's data/feature plumbing), so
it's a cheap upgrade path, not a separate integration",
    "Clean MIT (dingfengshi/TriDet); SGP reduces reliance on full self-attention -> a bit
lighter/faster",
    "Releases VideoMAEv2 feature configs ? useful if we later add an RGB cue"
  ],
  "cons": [
    "Same small-data risk as ActionFormer (benchmark-scale validation; no few-shot claim)
? not the right thing to START with at n=3-10",
    "More moving parts than ActionFormer; marginal gains may be invisible until the corpus
grows",
    "Off-benchmark low-dim trajectory input equally unproven here"
  ],
  "cost_note": "Free/OSS; same hardware envelope as ActionFormer.",
  "license": "MIT (dingfengshi/TriDet) ? verified",
  "privacy_note": "Fully local; no paid-LLM data.",
  "fit_for_us": "medium"
},
{
  "name": "MS-TCN++ (TAS)",
  "what": "Multi-stage temporal convolutional network: stage 1 makes an initial per-frame
prediction, refinement stages iteratively smooth it. Pure dilated temporal convs (no attention).
Per-frame {rally/non-rally} -> merge to intervals. ~0.8M params.",
  "pros": [
    "Smallest/simplest (~0.8M params, conv-only) -> fastest to train, easiest on 8GB,
strongest edge/on-device story",
    "Conv inductive bias handles small data reasonably (precursor to ASFormer's design);
multi-stage refinement directly fights over-segmentation",
    "Clean MIT on the CORRECT repo (sj-li/MS-TCN2)",
    "Battle-tested, minimal dependencies, easy to read/modify -> good fit for the
'minimal/readable harness' goal"
  ],
  "cons": [
    "LICENSE TRAP: the original yabufarha/ms-tcn repo is MIT + Commons Clause (NON-
commercial). MUST vendor sj-li/MS-TCN2 (clean MIT), not the original ? easy to get wrong",

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        "Generally a notch below ASFormer in accuracy on the same benchmarks (no attention;
ASFormer was designed to beat it)",
        "Same per-frame->interval merge + over/under-segmentation concerns as all TAS",
        "Frame-granular boundaries, no sub-frame regression"
    ],
    "cost_note": "Free/OSS; cheapest to train/run of the whole set.",
    "license": "MIT ? but ONLY via sj-li/MS-TCN2. AVOID yabufarha/ms-tcn (Commons Clause,
non-commercial).",
    "privacy_note": "Fully local, tiny model, strong on-device fit; no paid-LLM data.",
    "fit_for_us": "medium"
},
{
    "name": "Plain 1D-CNN / GRU / small Transformer over features (DIY sequence model)",
    "what": "A from-scratch per-frame or windowed sequence classifier (1D-CNN, BiGRU, or a
small Transformer) over WASB+audio+pose features, emitting per-frame rally prob ->
threshold+merge -> [start,end]. This is the direct learned successor to today's
windowing+LogisticRegression and matches the published badminton hit-detection recipe (BiGRU
over court/pose/shuttle).",
    "pros": [
        "Maximum control + minimum dependencies; trivially small, trivially on-device, trains
in seconds",
        "Matches the field-standard racket-sports recipe (MonoTrack-style GRU hit detector
over trajectory/pose) -> lowest-risk, known-to-work pattern at our data scale",
        "No license questions at all (our own code, our own weights)",
        "Easiest to make deterministic/reproducible for the agent-runnable harness; cleanest
baseline to A/B against ASFormer"
    ],
    "cons": [
        "We essentially already have this (distilled LogisticRegression) and it UNDERPERFORMED
the heuristic at n=2 ? a bare GRU may not clear the ~0.73 bar without the multi-stage refinement
/ inductive-bias tricks ASFormer/MS-TCN++ bake in",
        "Reinventing temporal-modeling tricks (multi-scale, refinement, boundary regression)
that ASFormer/ActionFormer already implement and validate",
        "No community-tuned recipe -> more of our own tuning time; easy to under-build"
    ],
    "cost_note": "Free; negligible compute. Mostly costs our engineering time.",
    "license": "N/A (own code) ? no third-party weights/data",
    "privacy_note": "Fully local; no paid-LLM data.",
    "fit_for_us": "medium"
},
{
    "name": "TemporalMaxer (TAL)",
    "what": "Parameter-free max-pooling-only TAL on the ActionFormer codebase: drops self-
attention for local max-pooling. Outputs (start,end,class) instances. ~2.8x fewer GMACs and ~3x
faster inference than ActionFormer.",
    "pros": [
        "Cheapest/fastest TAL at inference (best edge/on-device latency of the TAL family);
fewer params than ActionFormer",
        "Shares ActionFormer's data/feature plumbing -> cheap to try alongside it",
        "Claims to beat ActionFormer on several TAL benchmarks despite simplicity"
    ],
    "cons": [
        "LICENSE UNCONFIRMED ? could not fetch a LICENSE file (raw 404); must verify it is
MIT/Apache before any commercial use",
        "Same benchmark-scale validation as ActionFormer; no few-shot claim -> not a starting
choice at n=3-10",
        "Max-pooling-only may lose the fine temporal context that matters for rally-boundary
precision on noisy amateur footage"
    ],
    "cost_note": "Free/OSS; lowest inference cost of the TAL set.",
    "license": "UNVERIFIED (likely permissive given ActionFormer base, but confirm the
LICENSE file before relying on it)",
    "privacy_note": "Fully local; no paid-LLM data.",
    "fit_for_us": "weak"
},

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{
  "name": "DiffAct (TAS, diffusion)",
  "what": "Diffusion-based action segmentation: iteratively denoises per-frame action
predictions conditioned on video features. ICCV 2023. Per-frame {rally/non-rally} -> merge.",
  "pros": [
    "SOTA-class accuracy on standard TAS benchmarks; principled boundary refinement via
denoising",
    "Clean MIT (Finspire13/DiffAct)",
    "Same per-frame feature input as MS-TCN++/ASFormer -> would consume WASB features"
  ],
  "cons": [
    "Most complex + most data-hungry of the TAS set; diffusion training is finicky and the
LAST thing you want at n=3-10",
    "Iterative denoising = slower inference -> weakest on-device story among TAS",
    "No few-shot / small-data claim; overkill until the golden corpus is large",
    "Highest harness/operability burden vs the 'minimal/readable, deterministic' goal"
  ],
  "cost_note": "Free/OSS but heaviest to train and tune; multi-step inference.",
  "license": "MIT (Finspire13/DiffAct) ? verified",
  "privacy_note": "Fully local; no paid-LLM data (but heavier model).",
  "fit_for_us": "weak"
}
],
"key_sources": [
  "https://arxiv.org/abs/2202.07925",
  "https://github.com/happyharrycn/actionformer_release",
  "https://github.com/dingfengshi/TriDet",
  "https://github.com/dingfengshi/TriDet/blob/master/LICENSE",
  "https://arxiv.org/abs/2303.09055",
  "https://github.com/TuanTNG/TemporalMaxer",
  "https://github.com/sj-li/MS-TCN2",
  "https://github.com/sj-li/MS-TCN2/blob/master/LICENSE",
  "https://github.com/yabufarha/ms-tcn/blob/master/LICENSE",
  "https://github.com/ChinaYi/ASFormer",
  "https://ar5iv.labs.arxiv.org/html/2110.08568",
  "https://arxiv.org/abs/2110.08568",
  "https://github.com/Finspire13/DiffAct",
  "https://github.com/Finspire13/DiffAct/blob/main/LICENSE",
  "https://arxiv.org/pdf/2006.09220",
  "https://pmc.ncbi.nlm.nih.gov/articles/PMC11244353/",
  "https://ar5iv.labs.arxiv.org/html/2204.01899"
],
"recommendation": "FIRST owned model = ASFormer (MIT, ChinaYi/ASFormer), as a TEMPORAL
ACTION SEGMENTATION model over our existing per-frame features. Rationale: it is the ONLY
surveyed architecture whose central published result is training FROM SCRATCH on a tiny dataset
(GTEA, 28 videos) with ~1.1M params, by adding convolutional inductive bias to compensate for
small-data overfitting ? exactly the failure mode we hit when a learned model lost to the
heuristic at n=2. It consumes per-frame feature sequences directly, so it rides on the WASB
trajectory we already produce (X,Y,Visibility) and trivially fuses audio-onset and pose by
concatenating channels ? which simultaneously delivers the Option A modular-fusion milestone. It
is tiny (fits the RTX 2070S, trains in minutes, runs on-device), clean MIT, fully local (no
paid-LLM data, preserving the privacy lever and all guardrails), and learns temporal structure
rather than threshold-tuning, giving it a credible path past the ~F1 0.73 ceiling.\n\nConcrete
plan:\n1. Map labels: per-frame target = rally(1)/non-rally(0) from the rally [start,end] CSVs;
recover predicted intervals by merging contiguous rally frames + a min-gap/min-duration prior
(reuse the existing net-crossing gate). Treat ending_reason as a SECOND, segment-level head (5
classes) once interval detection works ? do NOT try to learn it jointly first.\n2. Features:
feed WASB (X,Y,Visibility) per frame; add audio-onset and pose channels as they come online
(this IS Option A fusion). Smooth/window features so the dilated convs have signal on amateur
60fps footage.\n3. Split DISCIPLINE: split train/eval by MATCH, never by clip (prevents the
leakage the brief flags). Gemini silver labels stay eval/teacher-only, never in training.\n4.
Gate: ship only if held-out per-match F1 beats the heuristic+LogisticRegression baseline (~0.73)
on the golden set ? the nanochat-style pass/fail verdict.\n\nPARALLEL second model =
ActionFormer (MIT, happyharrycn/actionformer_release) as the TAL formulation: rally = one
regressed (start,end,class=ending_reason) instance ? the cleanest 1:1 map to our CSV schema and

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the better long-term boundary-precision play. Run it as the A/B challenger; expect it to LOSE to ASFormer at n=3-10 (it has no few-shot claim and is benchmark-scale) but to OVERTAKE once the Roora-fed golden corpus grows into the hundreds of matches. TriDet is the cheap drop-in boundary-precision upgrade on top of ActionFormer for that later stage.\n\nEXPLICITLY DEFER: DiffAct and TemporalMaxer (too data-hungry / license-unconfirmed for a first model). And note these models are the BRIDGE to the brief's Option B end-to-end Qwen-VL/Gemma VLM ? not a replacement for it; they unblock P5 design using only data and features we already own.\n\nLICENSE GUARDRAIL TO RECORD IN COMMERCIALIZATION.md: if MS-TCN++ is ever chosen, vendor ONLY sj-li/MS-TCN2 (clean MIT) ? the original yabufarha/ms-tcn carries a Commons Clause (non-commercial) and MUST NOT be used.",

"confidence": "high",

"caveats": [

"TemporalMaxer's license is UNCONFIRMED ? the raw LICENSE fetch 404'd and search did not surface a license. It is built on ActionFormer (MIT) so it is likely permissive, but verify the actual LICENSE file before any commercial vendoring. I marked it 'weak' partly for this reason.",

"LICENSE TRAP confirmed by hand: yabufarha/ms-tcn = MIT + Commons Clause (NON-commercial); sj-li/MS-TCN2 (the MS-TCN++ repo) = clean MIT. These are DIFFERENT repos for what is colloquially called 'MS-TCN' ? vendor the right one.",

"The few-shot evidence for ASFormer is GTEA = 28 videos, which are MULTI-CLASS densely-labeled cooking videos, not 3-10 single-action sports matches. The small-data claim is well-supported but our regime is even smaller and differently structured; expect to need feature smoothing, augmentation (time-warp/jitter on trajectories), and possibly leave-one-match-out CV. A from-scratch learned model still may not beat the ~0.73 heuristic until the golden corpus grows ? the brief itself notes the bottleneck is golden DATA + WASB recall, not the method. This recommendation lowers the floor and unblocks design; it does not guarantee a win at n=3-10.",

"All TAL/TAS benchmark recipes default to I3D (2048-d) features. Feeding raw ~3-4-dim WASB trajectory is OFF the validated path. ActionFormer's input-projection and ASFormer's input layer both accept arbitrary feature dim (so it is mechanically supported), but accuracy on such low-dim input is unproven in the literature ? a learned feature front-end or hand-built feature stack (velocity, acceleration, visibility-run-length, net-crossing flags) will likely be needed.",

"ending_reason (winner/forced_error/unforced_error/service_fault/let) is a SEMANTIC per-rally label that trajectory features alone almost certainly cannot predict well; treat it as a separate, later head and do not let it gate the core [start,end] detection milestone.",

"I did not benchmark these models against each other on OUR data (cannot run GPU/torch here per the environment note). The ranking is from architecture properties, published small-data behavior, and license ? empirical A/B on the golden set is still required and is the actual ship/no-ship gate.",

"These models are NOT the brief's Option B end-to-end multimodal VLM (Qwen-VL/Gemma). They are the intermediate owned model that beats the heuristic baseline using current assets; the VLM remains the 12-24mo target after scaffolding + corpus health, per the roadmap."

]

},

{

"dimension": "OPTION B ? Owned end-to-end multimodal model: open, video-capable Apache-2.0/MIT bases fine-tunable for rally temporal grounding (frames+audio+pose ? rally [start,end]).",

"summary": "For an OWNED, on-device-capable rally-temporal-segmentation model we must split candidates into two classes. (1) BASES we can legally own and ship commercially ? the only viable path ? are generic video-VLMs with clean Apache-2.0/MIT licenses, NOT the temporal-grounding-specialized research models. (2) The temporal-grounding specialists named in the brief (TimeChat, VTimeLLM, Momentor) and the classic video encoders (VideoMAE) are LICENSE-POISONED for a product: they are research-only / CC-BY-NC / LLaMA-derived-with-no-LICENSE, so they are usable ONLY as METHOD references (how to tokenize timestamps, how to format moment-localization QA), never as the weights we fine-tune and serve. Top approved bases: Qwen2.5-VL-7B (Apache-2.0, native timestamp grounding via mRoPE+absolute-time-alignment ? best-documented temporal-grounding VLM in the legal set), Qwen3-VL-4B/8B (Apache-2.0, explicit Text?Timestamp Alignment, 4B fits the on-device/privacy lever), Gemma 4 4B/12B (Apache-2.0 ? VERIFIED, a genuine reversal from Gemma 1/2/3's restrictive terms, native frames+audio+video without separate encoders, ideal for the Option-B multimodal fusion), and InternVL3-8B (MIT, built on Qwen2.5-7B Apache base). CRITICAL GUARDRAIL CONFIRMATIONS: Gemma 3 stays REJECTED (Gemma Terms of Use, viral) but Gemma 4 is Apache-2.0 and clean; Qwen2.5-VL-3B (Qwen Research, non-commercial) and Qwen2.5-VL-72B (100M-MAU cap) are REJECTED per-size ? only the 7B and 32B Qwen2.5-VL sizes are Apache-2.0. COMPUTE REALITY for our situation: the lone 8GB RTX 2070 Super CANNOT train any 7B-class video

VLM (video sequences push even QLoRA to ~20-24GB); it is an inference/eval box only. Training MUST be a rented GPU (one 24-48GB A100/L40S-hour-class job). DATA REALITY: 128 human-labeled rallies (96+32) is FAR below the 5k-20k clip?QA band needed for end-to-end fine-tuning ? so Option B is correctly a 12-24mo horizon; the near-term move is LoRA on a 4B Apache base as a PoC once Roora/rally-annotator grow the corpus, with the modular Option-A fusion shipping first.",

```
"options": [
  {
    "name": "Qwen2.5-VL-7B-Instruct",
    "what": "7B generic video-VLM; native video temporal grounding (mRoPE + absolute-time alignment lets it pinpoint relevant video segments / emit timestamps). The single best-documented temporal-grounding-capable model inside the Apache-2.0 fence. Built on the Qwen2.5-7B LLM with an optimized ViT (SwiGLU+RMSNorm).",
    "pros": [
      "EXACT license = Apache-2.0 at the 7B size (verified on the HF model card) ? commercial-clean, derivative-wrappable, plugs into the provider registry with no MAU cap",
      "Native timestamp/moment-localization ability out of the box ? directly the rally [start,end] task class; minimizes the grounding behavior we must teach",
      "Massive ms-swift / LoRA / QLoRA tooling + community fine-tuning recipes (the brief's chosen engine targets exactly this family)",
      "Mature, stable checkpoint (not a research preview) ? safe to standardize the harness on"
    ],
    "cons": [
      "7B cannot train on our 8GB RTX 2070 Super; LoRA needs ~20GB and even QLoRA on VIDEO sequences lands ~24GB ? training is rented-GPU only",
      "7B is heavy for the on-device privacy lever; local INFERENCE at 1080p/60fps video is feasible but not snappy on 8GB (expect quantized + low-fps sampling)",
      "Newer Qwen3-VL supersedes it on temporal alignment; choosing 7B trades cutting-edge accuracy for a battle-tested baseline",
      "Still needs 5k-20k clip?QA pairs to beat the F1?0.73 heuristic ceiling ? base license being clean does not solve the data scarcity"
    ],
    "cost_note": "Inference: runs locally on the 2070 Super quantized (8GB) for eval; comfortable on a 16GB+ rented card. Training: one LoRA run ~4-12h on a 24GB rented GPU (RunPod/Lambda L40S/A100 class), low-tens-of-USD per run.",
    "license": "Apache-2.0 (verified on huggingface.co/Qwen/Qwen2.5-VL-7B-Instruct). NOTE: 3B=Qwen Research (non-commercial, REJECT), 72B=Qwen License (100M-MAU cap, REJECT). Only 7B and 32B are Apache-2.0.",
    "privacy_note": "Local inference satisfies the privacy lever (video never leaves device) but 8GB is the binding constraint; per-user LoRA adapters are tiny and deletable (drop the adapter = delete the user's training).",
    "fit_for_us": "strong"
  },
  {
    "name": "Qwen3-VL-4B-Instruct (and 8B)",
    "what": "Newer Qwen VLM (released 2025-10) with explicit 'Text?Timestamp Alignment' (evolves beyond T-RoPE) for precise timestamp-grounded event localization; second-level video indexing, 256K-1M context. 4B is the small, on-device-friendly variant.",
    "pros": [
      "EXACT license = Apache-2.0 at BOTH 4B and 8B (verified on HF) ? commercial-clean",
      "4B is the best fit for the on-device/privacy lever: small enough to LoRA on a modest rented 24GB card and to run quantized locally for inference",
      "Temporal grounding is a headline feature (explicit textual-timestamp alignment) ? purpose-built for rally [start,end] emission",
      "Newest of the approved family ? strongest temporal modeling per the technical report; future-proofs the base lock"
    ],
    "cons": [
      "Still cannot train on the 8GB local box (4B QLoRA on video is ~12-16GB+); training is rented-GPU only, though cheaper than 7B",
      "Younger checkpoint ? fewer hardened community fine-tuning recipes than Qwen2.5-VL-7B (more harness debugging on us)",
      "'Thinking' variants add latency; the Instruct variant is the right pick for deterministic [start,end] output",
      "Same data-scarcity ceiling: 128 rallies is far below the training band"
    ]
  }
]
```

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    ],
    "cost_note": "Cheapest train of the strong options: 4B LoRA fits a single 24GB rented GPU comfortably, likely <$10/run. Local quantized inference plausible on the 2070 Super.",
    "license": "Apache-2.0 (verified: Qwen3-VL-4B-Instruct and 8B-Instruct on HF). Larger/MoE Qwen3-VL sizes must be re-checked per-size before any lock.",
    "privacy_note": "Best on-device candidate for the privacy lever given the 4B size; per-user adapters deletable.",
    "fit_for_us": "strong"
  },
  {
    "name": "Gemma 4 (4B / 12B, multimodal)",
    "what": "Google's 2026 open multimodal family (2B/4B/12B/26B/31B). Processes text+image+audio+VIDEO natively without separate encoder networks; 12B runs on a 16GB laptop. The native frames+audio path maps directly onto our Option-B fusion goal (frames+audio+pose ? rally).",
    "pros": [
      "EXACT license = Apache-2.0 ? VERIFIED, and a genuine reversal: Gemma 1/2/3 were under the restrictive 'Gemma Terms of Use', Gemma 4 (from the 2026-04 launch) is true Apache-2.0 with full commercial rights. This directly clears the guardrail that REJECTS Gemma 3",
      "Native audio+video in ONE model is uniquely aligned with Option B (collapse WASB-trajectory + audio-hit + pose into a single owned model) ? fewer modality bridges to build",
      "4B variant is genuinely on-device (16GB laptop / quantized smaller) ? strong privacy lever",
      "Backed by Google's tooling + ecosystem; broad fine-tuning support expected"
    ],
    "cons": [
      "Temporal-grounding/timestamp-emission is NOT as explicitly documented as Qwen's ? we'd likely have to teach moment-localization output format more from scratch",
      "Very new (mid-2026) ? fine-tuning recipes for the temporal-grounding task are immature; more harness risk",
      "Must KEEP verifying per-checkpoint: the brief's rejection was specifically Gemma 3; do not let 'Gemma' shorthand leak a Gemma-3 weight into the pool",
      "Audio-native is a future-facing bet; our near-term audio cue (E1) is AMBER, so the multimodal advantage is partly latent"
    ],
    "cost_note": "4B trains cheaply on a rented 24GB card; 12B runs locally for inference on a 16GB machine (our 8GB box would need the 2B/4B quantized).",
    "license": "Apache-2.0 (VERIFIED across multiple 2026 sources; replaces Gemma Terms of Use used by Gemma 1/2/3). Per-checkpoint re-verify before lock ? only Gemma 4 is clean; Gemma 3 remains REJECTED.",
    "privacy_note": "4B on-device satisfies the privacy lever; native local audio+video keeps the whole multimodal pipeline on-device ? strongest privacy story of all options.",
    "fit_for_us": "strong"
  },
  {
    "name": "InternVL3-8B",
    "what": "OpenGVLab VLM, MIT-licensed wrapper built on a Qwen2.5-7B Apache-2.0 LLM + InternViT vision encoder; supports interleaved image/video/text and extended video training samples.",
    "pros": [
      "EXACT license = MIT on the InternVL3 wrapper, and the underlying Qwen2.5-7B base is Apache-2.0 ? dual clean, commercial-safe",
      "Strong general video understanding; competitive VLM benchmarks",
      "MIT is the most permissive option ? zero ambiguity for derivative-wrapping",
      "Same Qwen2.5 lineage means ms-swift recipes largely transfer"
    ],
    "cons": [
      "Temporal-grounding/timestamp localization is LESS first-class than Qwen2.5-VL/Qwen3-VL (general video QA, not purpose-built moment localization) ? more to teach for rally [start,end]",
      "Per-SIZE base differs: 9B uses InternLM3 and 38B/78B use Qwen2.5-32B/72B ? the 72B-based 78B inherits the 100M-MAU cap; ONLY the 1B/2B/8B/14B/38B (Qwen2.5 ?32B) sizes are clean. Verify per size before lock",
      "8B has the same 8GB-box training impossibility as Qwen2.5-VL-7B",
      "Adds a vision-encoder dependency (InternViT) to the stack"
    ]
  }

```

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    ],
    "cost_note": "Comparable to Qwen2.5-VL-7B: rented-GPU LoRA, local quantized inference
only.",
    "license": "MIT (InternVL3 wrapper) + Apache-2.0 (Qwen2.5 base) for the 7B/14B/38B
sizes. REJECT InternVL3-78B (Qwen2.5-72B base = 100M-MAU cap). Verify per size.",
    "privacy_note": "Local inference feasible quantized; same adapter-deletion model as
Qwen.",
    "fit_for_us": "medium"
  },
  {
    "name": "VideoLLaMA2 / VideoLLaMA3 (DAMO)",
    "what": "DAMO video-LLMs; v3 (Jan 2025) is built on Qwen2.5 (7B and 1.5B/2B), code
Apache-2.0, with demonstrated temporal-grounding capability. v2 added explicit audio
understanding (relevant to Option B fusion).",
    "pros": [
      "CODE is Apache-2.0; v3 base LLM (Qwen2.5) is Apache-2.0; demonstrated temporal-
grounding notebook ? closest 'specialist behavior on a clean base' in the set",
      "v2's audio-understanding branch is a useful reference for our frames+audio fusion
target",
      "2B (Qwen2.5-1.5B) variant is small enough for the on-device lever"
    ],
    "cons": [
      "BLOCKER: the released VideoLLaMA3 SERVICE/checkpoint is declared a 'research preview
intended for non-commercial use ONLY', subject to extra Qwen + data-source terms ? so we CANNOT
ship DAMO's fine-tuned weights. We could only re-train from the Apache Qwen2.5 base ourselves
using their (Apache) code, i.e. it collapses to 'use Qwen2.5-VL/Qwen3-VL + their training
recipe'",
      "Therefore not a true 'owned base' ? it's a method/recipe reference plus an Apache
code dependency",
      "Mixed signal: 'Apache code' vs 'non-commercial weights' is exactly the trap to avoid
asserting as clean"
    ],
    "cost_note": "Same rented-GPU training profile; no advantage over directly using
Qwen3-VL/Gemma 4 as the base.",
    "license": "Code Apache-2.0; RELEASED WEIGHTS = research/non-commercial preview (NOT
shippable). Use as recipe reference only.",
    "privacy_note": "Moot for product use given the non-commercial weights; if re-trained
from Qwen2.5, privacy story = same as Qwen options.",
    "fit_for_us": "weak"
  },
  {
    "name": "TimeChat / VTimeLLM / Momentor (temporal-grounding specialists)",
    "what": "Academic video-LLMs purpose-built for temporal grounding / moment localization.
TimeChat (CVPR'24, time-sensitive), VTimeLLM (CVPR'24, 'grasp video moments'), Momentor (fine-
grained temporal reasoning, trained on Moment-10M).",
    "pros": [
      "State-of-the-art METHODS for exactly our task: timestamp tokenization, temporal
position encoding (Momentor's TPM), moment-localization instruction formats ? high-value as
design references for our QA-pair schema and the grounding head",
      "Papers/datasets (e.g. Moment-10M format) inform how to convert rally CSVs ? clip?QA
pairs"
    ],
    "cons": [
      "LICENSE-POISONED as bases: TimeChat = research-preview non-commercial; VTimeLLM = CC-
BY-NC-ND 4.0 (NonCommercial + NoDerivs ? cannot even fine-tune/redistribute); Momentor = built
on LLaMA with NO LICENSE file in-repo (legally unusable + LLaMA's own restrictions). NONE are
Apache-2.0/MIT",
      "Cannot be the owned, commercial, fine-tunable base under our hard guardrail ? full
stop",
      "Older LLM backbones (Vicuna/LLaMA-class) lag the approved Qwen3-VL/Gemma 4 bases on
raw capability anyway"
    ],
    "cost_note": "N/A as a base (cannot ship). Zero-cost as reading/method references.",
    "license": "TimeChat = research-only/non-commercial; VTimeLLM = CC-BY-NC-ND 4.0;
Momentor = no LICENSE + LLaMA-derived. ALL REJECTED as bases; method-reference only.",

```



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    "privacy_note": "N/A ? not deployable.",
    "fit_for_us": "weak"
  },
  {
    "name": "Video encoders: InternVideo2 / VideoMAE",
    "what": "Pure spatiotemporal video encoders (not generative VLMs) usable as a backbone
for a custom temporal-localization head (e.g. encoder ? CG-DETR-style grounding head, the
Option-A/modular or a from-scratch e2e route that avoids a full VLM).",
    "pros": [
      "InternVideo2-Stage2_6B = MIT (VERIFIED) and InternVideo2.5_Chat_8B = Apache-2.0 ?
clean encoder weights we CAN use commercially",
      "An encoder+grounding-head is far CHEAPER to train than a full VLM and could run on
the 8GB box at smaller sizes ? a realistic near-term owned-model PoC at our data scale",
      "InternVideo2 is a proven temporal-grounding backbone (evaluated on
QVHighlights/Charades-STA via CG-DETR)",
      "Sidesteps the 'generative VLM is huge' problem for the privacy/on-device lever"
    ],
    "cons": [
      "VideoMAE weights = CC-BY-NC 4.0 ? REJECTED for commercial use (research-only); only
its Apache-2.0 code components are reusable, not the pretrained checkpoints",
      "Encoder-only path is NOT the generative video-QA Option-B end-state ? it answers
'where is the rally' but not the rich rally semantics (ending_reason etc.); it's a
complementary/bridge architecture, not the VLM the brief ultimately wants",
      "Building+maintaining a custom grounding head is extra engineering vs. fine-tuning an
off-the-shelf VLM",
      "Need to confirm InternVideo2's largest checkpoints per-file (some sub-weights vary)"
    ],
    "cost_note": "Cheapest training of any option; a small InternVideo2 encoder + light head
is plausibly trainable on the 8GB box or a single small rented card ? the only option that fits
TODAY's 128-rally data scale.",
    "license": "InternVideo2-Stage2_6B = MIT; InternVideo2.5_Chat_8B = Apache-2.0 (both
VERIFIED, usable). VideoMAE checkpoints = CC-BY-NC 4.0 (REJECTED); VideoMAE code Apache-2.0
only.",
    "privacy_note": "Small encoders are the most on-device-friendly option ? strongest fit
for the local privacy lever at small data scale.",
    "fit_for_us": "medium"
  }
],
"key_sources": [
  "https://huggingface.co/Qwen/Qwen2.5-VL-7B-Instruct",
  "https://huggingface.co/Qwen/Qwen2.5-VL-72B-Instruct/blob/main/LICENSE",
  "https://huggingface.co/Qwen/Qwen2.5-VL-3B-Instruct/blob/main/LICENSE",
  "https://huggingface.co/Qwen/Qwen3-VL-8B-Instruct",
  "https://huggingface.co/Qwen/Qwen3-VL-4B-Instruct",
  "https://github.com/qwenlm/qwen3-vl",
  "https://blog.google/innovation-and-ai/technology/developers-tools/gemma-4/",
  "https://www.infoq.com/news/2026/04/google-gemm4/",
  "https://venturebeat.com/technology/googles-new-open-source-gemma-4-12b-analyzes-audio-
video-and-runs-entirely-locally-on-a-typical-16gb-enterprise-laptop",
  "https://huggingface.co/OpenGVLab/InternVL3-8B",
  "https://huggingface.co/OpenGVLab/InternVideo2-Stage2_6B",
  "https://huggingface.co/OpenGVLab/InternVideo2_5_Chat_8B",
  "https://github.com/DAMO-NLP-SG/VideoLLaMA3",
  "https://github.com/DAMO-NLP-SG/VideoLLaMA2/blob/main/LICENSE",
  "https://github.com/RenShuhuai-Andy/TimeChat",
  "https://github.com/huangb23/VTimeLLM",
  "https://github.com/DCDmllm/Momentor",
  "https://huggingface.co/MCG-NJU/videomae-base",
  "https://arxiv.org/html/2403.15377v1",
  "https://arxiv.org/html/2402.11435v1",
  "https://www.labellerr.com/blog/fine-tune-qwen-2-5-vl/",
  "https://kaitchup.substack.com/p/qwen3-vl-fine-tuning-on-your-computer"
],
"recommendation": "LOCK the base as a two-stage decision, not one model. (1) NEAR-TERM PoC
base = Qwen3-VL-4B-Instruct (Apache-2.0, explicit Text?Timestamp Alignment, smallest strong on-

```

device candidate) as the primary, with Qwen2.5-VL-7B-Instruct (Apache-2.0, best-documented temporal grounding, most mature ms-swift recipes) as the de-risking fallback. Both are clean for commercial ownership and the privacy lever. (2) WATCH/EVALUATE Gemma 4 4B (Apache-2.0, VERIFIED reversal from the rejected Gemma 3) as the Option-B end-state base ? its native frames+audio+video in one model is the best structural match for collapsing WASB-trajectory + audio + pose into a single owned model; lock it for Option B once a fine-tuning recipe for timestamp emission is proven. DO NOT use TimeChat/VTimeLLM/Momentor/VideoMAE as bases (all non-commercial or no-LICENSE) ? mine them only for the timestamp-tokenization and clip?QA-format methods. DO NOT touch Qwen2.5-VL-3B (Qwen Research), Qwen2.5-VL-72B / InternVL3-78B (100M-MAU cap), DAMO's released VideoLLaMA3 weights (research-preview), or any Gemma ?3. COMPUTE: treat the 8GB RTX 2070 Super as an inference/eval-only box; budget a single rented 24-48GB GPU (RunPod/Lambda L40S or A100 class) for every fine-tune ? even QLoRA on video exceeds 8GB. SEQUENCING (matches the brief's directive): ship modular Option-A fusion first; do NOT start any owned-model fine-tune until (a) the async job model / ComputeBackend P0 lands and (b) the golden corpus crosses ~thousands of clip?QA pairs (Roora + rally-annotator). At today's 128 rallies the only owned-model experiment that is data-realistic NOW is a small InternVideo2-encoder (MIT) + light grounding head as a bridge PoC ? the full VLM fine-tune is correctly a 12-24mo Option-B horizon.",

"confidence": "high",

"caveats": [

"Licenses change per-checkpoint and per-version. Re-verify the EXACT license file on the specific HF revision before any code lands ? especially: Qwen2.5-VL sizes (only 7B/32B Apache; 3B/72B rejected), InternVL3 sizes (78B inherits Qwen2.5-72B MAU cap), and ANY Gemma weight (only Gemma 4 is Apache; Gemma 3 stays REJECTED). The 'Gemma' shorthand is a live foot-gun.",

"Gemma 4's Apache-2.0 status is confirmed across multiple 2026 secondary sources and the official Google blog, but I did not open the raw LICENSE file on each Gemma 4 HF checkpoint ? do that before locking it as the Option-B base.",

"Compute/VRAM numbers are from community fine-tuning write-ups (Labellerr, Kaitechup, Unsloth), not a controlled benchmark on OUR footage. Real 1080p/4K-60fps video sequences with long context can push VRAM higher than the text/OCR LoRA figures cited; treat 24GB as a floor for a 7B video LoRA and validate empirically.",

"The 5k-20k clip?QA training band is the brief's own estimate; the true minimum to beat the F1?0.73 heuristic with an end-to-end VLM is unproven for this niche task and could be higher (or, with strong base priors + good QA formatting, somewhat lower). The honest n=2 result (learned model underperforms baseline) is the binding evidence that data ? not base choice ? is the current bottleneck.",

"Momentor having no LICENSE file is from a single fetch of the repo root; a LICENSE may exist in a subdir or have been added since ? but absence-of-license + LLaMA derivation is enough to disqualify it as a base regardless.",

"I treated 'on-device' against the stated 8GB RTX 2070 Super. If the owner provisions a 16-24GB local card later, the local-training calculus changes materially (4B QLoRA becomes locally trainable)."

]

},

{

"dimension": "LABEL & DATA REUSE / DATA ENGINEERING ? maximizing the value of our human rally CSVs + ending_reason + WASB trajectories across model generations (heuristics ? TAL ? owned video-VLM \"Option B\")",

"summary": "Our labels are far more valuable than their raw count (96 + 32 rallies ? 2 matches) suggests, IF we stop treating them as one-format eval files and start treating them as a model-agnostic \"labeled-corpus\" that we transcode into every consumer. (a) FORMATS: the rally [start,end]+ending_reason+sport rows already ARE the ActivityNet/ActionFormer schema modulo a transcode ? a `segment:[start,end]` (seconds) + `label` + per-video `duration/fps/subset` JSON; the SAME rows transcode into (i) candidate-window labels for our heuristic/LogReg distillation (already done via `training.label\_candidates`), (ii) ActivityNet-style JSON for any TAL net (ActionFormer/TriDet), and (iii) clip?QA text pairs (\"When does each rally happen?\" ? \"[start,end]?\") for the Option-B video-VLM. ending_reason becomes a second QA task for free. WASB trajectories are an auxiliary feature stream keyed by the same (video_id, frame). (b) AMPLIFICATION ? five legitimate levers that never touch paid-LLM outputs: (1) teacher-student / pseudo-labeling where the TEACHER is OUR OWN model or an open MIT model (WASB+heuristics, or a TAL net trained on the few labels), self-training on our ~26 GB unlabeled corpus ? SS-TAL is an active field and PLR/ALQA-style confidence-gated pseudo-labels are the standard recipe; (2) temporal data augmentation (speed/time-warp $\pm 20\%$ to 30% , temporal jitter, reverse, segment cut-mix/mixup) that multiplies effective segment count for any TAL/VLM trainer while preserving boundaries; (3) active learning ? spend the Roora budget where the model is

least certain (boundary entropy / disagreement vs WASB), measurably beating random labeling under a fixed budget; (4) timestamp/point supervision for cheaper future labels (~? the time of full segments, SkipTag ~?, reaching near-full-supervision accuracy) ? relevant for ending_reason and for scaling new sports; (5) for Option B specifically, RL/RFT (GRPO) is dramatically more LABEL-efficient than SFT for temporal grounding (VideoChat-R1: +31.8 temporal-grounding with \"limited samples\", and quality-filtered 13K beat unfiltered 56K in VTG-R1) ? meaning our small hand-verified set is the right asset class. (c) HOW MANY MATCHES: heuristic threshold tuning needs ~3?5 matches for stable cross-val (we're under that at n?2, which is exactly why the learned model underperforms baseline); a from-scratch TAL net (ActionFormer-class) wants ~50?200+ labeled videos to beat heuristics; an Option-B VLM fine-tune is the sweet spot ? single-digit-thousands of clip?QA pairs (literature uses 13K?18K), which our ~128 rallies x augmentation x QA-task-multiplexing can approach without thousands of new matches, and RFT can cold-start from hundreds. (d) The reusable format: one canonical JSONL \"rally record\" per (video_id, rally_ordinal) carrying [start,end], ending_reason, sport, source/provenance, split tag, + a pointer to the WASB trajectory slice; deterministic transcoders emit the heuristic-label view, the ActivityNet-TAL view, and the VLM clip?QA view, so EVERY label labeled once feeds all three model classes and all future generations.

```
"options": [
  {
    "name": "(a) Format transcoders ? one corpus, three consumer formats",
    "what": "Treat the existing rally CSV rows
(rally_number,start_time,end_time,ending_reason,sport) as the source of truth and write
deterministic transcoders to each model class's expected format, instead of re-labeling per
consumer. View 1 (heuristic/LogReg distillation): we ALREADY do this ?
`backend/eval/training.py::label_candidates` runs the same IoU `match_intervals` to turn rally
intervals into positive/negative labels on WASB candidate windows; keep it. View 2 (TAL nets):
emit ActivityNet/ActionFormer JSON ? top-level `database`, per-video `{subset, duration, fps,
annotations:[{segment:[start,end] in SECONDS, label:'rally', label_id:0}]}` (verified against
ActionFormer's thumos14 loader). Our seconds-based [start,end] maps 1:1; we only add per-video
duration/fps (already in ffprobe_meta) and a train/val/test `subset`. View 3 (Option-B VLM):
emit clip?QA text pairs ? sample a window, ask 'List the [start,end] of every rally in this
clip' ? answer is the rally intervals normalized to clip-relative seconds; ending_reason yields
a SECOND free task ('Why did rally N end?' ? winner/forced_error/...). One row, three formats,
zero re-labeling.",
    "pros": [
      "Zero new labeling: every consumer reads the same ~128 rallies; labels labeled once
feed heuristics + TAL + VLM and every future model generation",
      "Our schema is already 90% of the ActivityNet TAL schema ? `segment:[start,end]` in
seconds is exactly what ActionFormer/TriDet ingest; transcode is a thin map, not a re-
annotation",
      "ending_reason is a near-free second supervised task (rally-outcome QA) that
multiplies QA-pair yield from the SAME intervals",
      "Reuses the existing deterministic scorer (`metrics.match_intervals`) as both the
heuristic-labeler and the TAL/VLM eval grader ? one spine, no metric drift",
      "WASB trajectory CSV (Frame,Visibility,X,Y) attaches as an auxiliary per-frame feature
stream keyed by the same video_id+frame ? usable as TAL input features OR as a 'trajectory-
token' modality for Option B"
    ],
    "cons": [
      "VLM clip?QA generation needs a windowing/sampling policy (clip length, stride,
negative clips with no rally) ? a design choice that affects label balance",
      "ActivityNet-JSON path assumes you extract per-clip FEATURES (I3D/VideoMAE) for
classic TAL nets; that is extra preprocessing, though it's open-model and one-time",
      "Clip-relative vs absolute timestamps must be handled carefully in the VLM answer
format (Qwen2.5-VL aligns MROPE to absolute time ? pick one convention and lock it)"
    ],
    "cost_note": "Engineering-only; no labeling spend. A few hundred lines of transcoder +
tests; reuses existing metrics/annotations modules.",
    "license": "N/A (our own data + open formats). ActivityNet JSON schema and ActionFormer
loader are open/permissive references.",
    "privacy_note": "Transcoders run locally on licensed/owned corpus only; no user video,
no egress. Minors excluded at the corpus-membership layer before transcoding.",
    "fit_for_us": "strong"
  },
  {

```

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    "name": "(b1) Self-training / teacher-student / pseudo-labeling ? open-teacher ONLY",
    "what": "Amplify the few labels by pseudo-labeling our ~26 GB unlabeled corpus with a
teacher that is OUR OWN model or an open MIT/Apache model (NEVER Gemini). Concretely: train a
first-pass rally model (heuristics+WASB, or a small TAL net) on the ~128 labeled rallies, run it
over unlabeled matches, keep only high-confidence pseudo-rallies (confidence-gated, à la
ALQA/PLR semi-supervised TAL), retrain on labeled+pseudo, iterate (noisy-student style with
temporal augmentation as the 'noise'). This is the canonical SS-TAL recipe; it directly attacks
the n?2 data starvation that currently makes the learned model lose to the baseline.",
    "pros": [
        "Stays inside guardrails: teacher is WASB/our-model/open-MIT ? paid-LLM outputs are
never a training signal",
        "SS-TAL with confidence-gated pseudo-labels (ALQA, PLR) is an established, published
recipe for exactly our 'few labeled + many unlabeled videos' regime",
        "Noisy-student framing (student ? teacher + augmentation noise) is a proven data-
efficiency multiplier (Xie et al. CVPR'20 lineage)",
        "Uses the corpus we already own; no per-rally vendor spend to grow training signal",
        "Pseudo-labels can bootstrap Option-B VLM SFT data (clip?QA pairs from confident
pseudo-rallies) without human labeling"
    ],
    "cons": [
        "Pseudo-label noise: low-quality pseudo-rallies degrade boundaries ? MUST confidence-
gate and ideally human-spot-check a sample (the PLR/ALQA papers exist precisely because naive
pseudo-labels hurt)",
        "Teacher blind spots propagate (WASB recall ceiling becomes the pseudo-label recall
ceiling) ? confirmed bottleneck in our own results",
        "Boundary precision from pseudo-labels is weaker than human [start,end]; treat pseudo
as recall-boosters, keep human labels as the boundary authority",
        "Iteration adds GPU cost on the single RTX 2070 Super (8 GB) ? feasible but slow;
cloud burst may be needed"
    ],
    "cost_note": "Mostly compute on owned GPU + occasional cloud burst. No new human
labeling required to START; small human audit budget to validate pseudo-label quality.",
    "license": "Teachers restricted to Apache-2.0/MIT/BSD (WASB=MIT, our own LogReg, open
TAL nets). Explicitly excludes Gemini/OpenAI/Anthropic outputs as training data.",
    "privacy_note": "All pseudo-labeling on owned/licensed corpus, local. Pseudo-labeled
clips inherit the same minors-excluded, training-opt-in constraints.",
    "fit_for_us": "strong"
},
{
    "name": "(b2) Temporal data augmentation ? multiply effective segments",
    "what": "Apply temporal augmentations that preserve rally semantics while changing the
signal: speed/time-warp ( $\pm 20\%30\%$ , rescale [start,end] accordingly), temporal jitter of
boundaries, VideoReverse (rally played backward still a rally), and segment-level
CutMix/MixUp/FadeMixUp to synthesize new rally/non-rally compositions. For VLM clip?QA, also
vary clip windows/stride so the same rally appears in many contexts. Scale-aware sampling
(randomize action timescale) is reported superior to spatial-only aug for video.",
    "pros": [
        "Multiplies effective labeled-segment count with ZERO new annotation ? directly
addresses tiny-n",
        "Speed/time-warp + scale-aware sampling are validated to improve temporally-
localizable features (RandAugment-T, FadeMixUp lineage)",
        "Boundary-preserving by construction (you transform the timestamps with the signal),
so it doesn't corrupt our hard-won [start,end] authority",
        "Cheap and deterministic; runs in the data loader for both TAL nets and VLM SFT",
        "Window/stride variation for clip?QA is a free QA-pair multiplier for Option B"
    ],
    "cons": [
        "Augmentation realism risk: extreme speed changes can break shuttle-physics cues WASB
relies on ? keep ranges moderate",
        "CutMix/MixUp across rallies can create unnatural boundaries; FadeMixUp (soft) is
safer than hard cut-paste for localization",
        "Doesn't add NEW information (same matches, same players, same court) ? complements
but cannot replace corpus diversity from new matches/venues",
        "Gains are incremental, not a substitute for the ~3?5 real matches the heuristic tuner
needs for stable cross-val"
    ]
}

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    ],
    "cost_note": "Compute-only, in-loader; negligible marginal cost.",
    "license": "N/A ? transforms our own data.",
    "privacy_note": "Local, owned corpus; no egress.",
    "fit_for_us": "strong"
  },
  {
    "name": "(b3) Active learning ? label the most-informative match/segment next",
    "what": "Spend the Roora vendor budget where it buys the most: rank unlabeled
matches/clips by model uncertainty (boundary entropy, prediction disagreement between WASB-
heuristic and a TAL net, or high alignment-cost vs labeled prototypes) and send THOSE to Roora
first, rather than random/sequential matches. Active-learning-for-TAL papers (uncertainty/TPE
scoring, boundary-centric acquisition) show this beats random labeling under a fixed budget.",
    "pros": [
      "Maximizes IoU-lift-per-rupee on the constrained vendor budget ? exactly the solo-
founder cost constraint",
      "Published SS-TAL+active-learning results show uncertainty-based selection beats
random under fixed label budget",
      "Pairs naturally with (b1): pseudo-label everything, then human-label only the
segments the model is LEAST sure about (the highest-value oracle queries)",
      "Boundary-centric acquisition targets our actual weak point (dStart/dEnd boundary
error), not just presence/absence",
      "Disagreement signal is free ? we already run WASB-heuristic AND can run a TAL net;
route their disagreements to humans"
    ],
    "cons": [
      "Needs a deployable uncertainty signal first; our LogReg gives probabilities but a
richer TAL net gives better entropy estimates ? small upfront build",
      "Cold-start problem: with n?2 the model's uncertainty is itself unreliable; first 3?5
matches should still be diverse/random to bootstrap, THEN switch to active selection",
      "Risk of sampling bias (always querying hard cases) can skew the train distribution ?
mix in some random/diverse picks",
      "Operationally couples the eval/training loop to the vendor pipeline (must re-rank
between Roora batches)"
    ],
    "cost_note": "Saves vendor money net-net; small engineering cost for the scoring/ranking
step. Highest ROI once n?~5.",
    "license": "N/A.",
    "privacy_note": "Selection runs locally; only the SELECTED owned/licensed clips go to
the vendor through the existing single DPA-gated provider boundary.",
    "fit_for_us": "strong"
  },
  {
    "name": "(b4) Timestamp / point supervision ? cheaper labels for scale-out",
    "what": "For FUTURE labeling (new sports, ending_reason, scaling beyond Roora), prefer
point/timestamp supervision (one click per rally + outcome) over dense [start,end] where
boundary precision isn't the bottleneck. User study on Breakfast: timestamp supervision ? ? the
annotation time of full supervision, SkipTag ? ?, and timestamp-supervised models reach near-
full-supervision accuracy (e.g. 64.1% vs 68.0%). Point-level TAL is a mature sub-field.",
    "pros": [
      "Cuts annotation time substantially (timestamp ? ?, SkipTag ? ? of full) ? more
matches per rupee",
      "Near-full-supervision accuracy is achievable from point labels with the right losses
(HR-Pro, point-TAL methods)",
      "ending_reason is naturally point-like (one outcome per rally) ? fits timestamp
supervision perfectly",
      "Lowers the barrier to expand to tennis/pickleball/table-tennis (collector's 5-sport
shape) without dense re-labeling each sport",
      "Our existing dense [start,end] golden set becomes the high-quality 'boundary
authority' that calibrates the cheaper point-labeled bulk"
    ],
    "cons": [
      "Point labels alone give weaker boundaries ? keep a dense-labeled golden CORE for eval
and boundary calibration",
      "Requires point-supervision-aware training losses (not just feeding points to a full-

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supervision trainer)",
    "Mixing dense + point labels needs careful handling in the corpus format (label_type
field) and the trainer",
    "Our current dense CSVs are already collected ? this is a forward-looking lever for
the NEXT thousands of labels, not the existing 128"
  ],
  "cost_note": "Reduces future labeling cost ~33?66%. Existing dense labels unaffected.",
  "license": "N/A.",
  "privacy_note": "Same owned-corpus + vendor-DPA path; minors excluded.",
  "fit_for_us": "medium"
},
{
  "name": "(b5) Option-B specific: RL/RFT (GRPO) over SFT for label efficiency",
  "what": "When training the owned video-VLM (Qwen2.5-VL-7B / Qwen3-VL-8B / InternVL2.5,
all Apache-2.0/MIT), prefer reinforcement fine-tuning (GRPO/RFT) with a temporal-IoU reward over
pure SFT for the temporal-grounding objective. RFT is markedly more LABEL-efficient:
VideoChat-R1 reports +31.8 on temporal grounding with 'limited samples' and GRPO > SFT in their
ablation; VTG-R1 shows a quality-filtered 13K cold-start set beating an unfiltered 56K set. Our
reward = temporal IoU of predicted [start,end] vs our golden rally labels (the metric we already
compute).",
  "pros": [
    "RFT is data-efficient ? matches our exact situation: small, high-quality, hand-
verified label set is the RIGHT asset for RL, not a liability",
    "Reward = temporal IoU vs golden labels = our EXISTING deterministic scorer
(`metrics`) repurposed as the RL reward ? no new eval machinery",
    "Quality-over-quantity is empirically validated (filtered 13K > unfiltered 56K), so
our small clean set + augmentation can punch above its weight",
    "GRPO needs no separate reward model (uses verifiable IoU reward) ? operationally
simple, fits the nanochat-style one-command harness goal",
    "Bases are guardrail-clean (Qwen2.5-VL-7B/32B Apache-2.0; InternVL2.5 MIT; Gemma 4 if
Apache-2.0 confirmed)"
  ],
  "cons": [
    "RFT has higher COMPUTE cost than SFT (multiple rollouts per sample) ? heavy for one 8
GB RTX 2070 Super; likely needs cloud burst (RunPod/Lambda/Modal)",
    "Still wants a quality SFT cold-start first (cold-start data matters per VTG-R1) ? so
you need BOTH a clean SFT set AND the RL loop",
    "Reward hacking risk if IoU reward is naive (model emits over-long intervals) ? needs
precision/recall-balanced reward shaping",
    "VLM temporal grounding is still maturing; absolute accuracy on amateur single-camera
footage is unproven vs the broadcast-trained literature"
  ],
  "cost_note": "Compute-heavy (RL rollouts); plan cloud-GPU burst. But LABEL cost is low ?
the whole point. Benchmark self-hosted 7B vs Gemini per-call before committing (per
COMMERCIALIZATION C7).",
  "license": "Bases Apache-2.0/MIT only (Qwen2.5-VL-7B/32B, Qwen3-VL-4B/8B, InternVL2.5;
Gemma 4 pending license verify; Gemma 3 REJECTED). Reward = our own IoU, no paid-LLM signal.",
  "privacy_note": "Training data = owned/licensed corpus, training-opt-in only, minors
excluded. Final model runs locally (privacy lever).",
  "fit_for_us": "medium"
},
{
  "name": "(c) How many labeled matches per model class ? realistic bands",
  "what": "Heuristic threshold tuning (current WASB+windowing+gate): ~3?5 labeled matches
for STABLE leave-one-video-out cross-val; at n?2 the variance is too high, which is exactly why
our learned model loses to the baseline (this is a data-starvation artifact, not a method
failure). Small distilled classifier (our LogReg on trajectory features): also needs n?~5
diverse matches before it reliably beats threshold tuning. From-scratch TAL net
(ActionFormer/TriDet class): wants ~50?200+ labeled videos to be competitive (these nets are
tuned on THUMOS14/ActivityNet-scale sets) ? NOT realistic for us soon, so deprioritize training
one from scratch. Option-B VLM fine-tune: the sweet spot ? single-digit-thousands of clip?QA
pairs (literature: 13K?18K), reachable from ~128 rallies x temporal augmentation x QA-task
multiplexing (rally-localization + ending_reason) x pseudo-labeled clips; RFT can cold-start
from HUNDREDS of high-quality pairs.",
  "pros": [

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    "Sets honest, non-aspirational targets per model class ? prevents over-investing in a
from-scratch TAL net we can't feed",
    "Explains the current 'learned < baseline at n=2' result as expected data starvation,
not a dead end ? the fix is data, confirmed by our own docs",
    "Identifies the VLM path as the most label-EFFICIENT owned-model route given our asset
shape (few, clean labels + pretrained base)",
    "Gives a concrete corpus-growth milestone: get to ~5 matches to unblock honest
heuristic tuning NOW; build toward low-thousands QA pairs for Option B",
    "Aligns with the staged roadmap (heuristics ? Option A modular fusion ? Option B e2e)"
],
"cons": [
    "The ~50?200 for from-scratch TAL is a literature-scale heuristic
(THUMOS/ActivityNet), not measured on our footage ? treat as order-of-magnitude",
    "'Single-digit-thousands of QA pairs' for the VLM assumes augmentation + multiplexing
reach it; raw human rallies alone (128) are not enough without amplification",
    "Numbers are domain-transfer-sensitive: amateur single-camera footage may need MORE
labels than broadcast-trained literature implies",
    "Pose/audio-fused Option-A may shift these bands (more signal per label) ? revisit
after E1/pose probes"
],
"cost_note": "Planning guidance; the actionable spend is getting from n?2 to n?5 matches
(small Roora batch) to unblock the heuristic loop immediately.",
"license": "N/A.",
"privacy_note": "N/A.",
"fit_for_us": "strong"
},
{
    "name": "(d) Model-agnostic 'labeled-corpus' format ? the reusable spine",
    "what": "Define ONE canonical JSONL record per (video_id, rally_ordinal): {video_id
(file MD5), rally_ordinal, sport, start, end (seconds), ending_reason, source
(human|pseudo|vendor), annotator/provenance, label_type (segment|point), split (train|val|test,
assigned BY MATCH not by clip), trajectory_ref (pointer to the WASB CSV slice for this video)}.
Store it once (extends the existing collector export + candidate_segments lineage). Then
deterministic, tested transcoders emit: (1) the heuristic-label view (candidate-window pos/neg
via `match_intervals`), (2) the ActivityNet/ActionFormer TAL-JSON view (`database`?per-
video?`annotations:[{segment:[s,e],label,label_id}]`), (3) the VLM clip?QA view (rally-
localization + ending_reason tasks). Split is enforced by MATCH to prevent leakage
(CODE_MAP/PLATFORM 'split by match, not clip'). This is the asset the moat actually lives in.",
    "pros": [
        "Label once, consume everywhere and forever ? survives model-generation churn
(heuristics?TAL?VLM?whatever's next) because consumers read transcoder output, not raw labels",
        "Provenance + source + label_type fields make pseudo-labels, vendor labels, and human
labels first-class and auditable (critical for the no-paid-LLM + minors-excluded guardrails ?
each record carries who/what produced it and whether training-opt-in applies)",
        "split-by-match is baked into the format ? kills the data-leakage failure mode the
platform doc calls out (train|eval split by match not clip)",
        "Reuses existing infra: candidate_segments table is already the detector-agnostic
extension point; collector export already shares (video_id, ordinal, start, end) shape across 5
sports",
        "trajectory_ref keeps WASB as an attachable modality without bloating the rally record
? feeds TAL features AND Option-B multimodal tokens",
        "Same record powers eval (golden set) and training ? one corpus, graded by one scorer"
    ],
    "cons": [
        "Requires upfront schema discipline + migration of current CSVs into the JSONL (one-
time; the collector bridge/ADR-002 already points this way)",
        "Provenance/consent fields add bookkeeping the current flat CSV doesn't have ? but
they're non-negotiable given the guardrails",
        "Two timestamp conventions already exist in-repo (generic start,end vs golden
rally_number,start_time,end_time) ? the canonical record must absorb both and the transcoders
own the conversion",
        "trajectory_ref coupling means trajectory CSVs must be retained/versioned alongside
labels (storage, but cheap)"
    ],
    "cost_note": "One-time engineering: schema + migration + 3 transcoders + tests. No

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labeling spend. Pays back on every future model.",
  "license": "Our own schema; emits open formats (ActivityNet JSON). No third-party
license entanglement.",
  "privacy_note": "Provenance/consent/minors-excluded/training-opt-in are FIELDS in the
record, enforced at corpus build; the format is the enforcement point for the legal
guardrails.",
  "fit_for_us": "strong"
}
],
"key_sources": [
  "https://arxiv.org/html/2407.07673v1",
  "https://pmc.ncbi.nlm.nih.gov/articles/PMC11798530/",
  "https://openaccess.thecvf.com/content/ICCV2023/papers/Xia_Learning_from_Noisy_Pseudo_Labels_for_Semi-Supervised_Temporal_Action_Localization_ICCV_2023_paper.pdf",
  "https://github.com/happyharrycn/actionformer_release/blob/main/README.md",
  "https://arxiv.org/abs/2208.14856",
  "https://joonyoung-cv.github.io/assets/paper/18_eccv_what_do.pdf",
  "https://arxiv.org/html/2604.15173v1",
  "https://arxiv.org/pdf/2103.06669",
  "https://www.ecva.net/papers/eccv_2022/papers_ECCV/papers/136640276.pdf",
  "https://arxiv.org/abs/2504.06958",
  "https://arxiv.org/html/2507.18100v1",
  "https://arxiv.org/pdf/2502.13923",
  "https://openaccess.thecvf.com/content_CVPR_2020/papers/Xie_Self-Training_With_Noisy_Student_Improves_ImageNet_Classification_CVPR_2020_paper.pdf",
  "https://arxiv.org/abs/2211.04888",
  "https://arxiv.org/html/2008.05721",
  "https://arxiv.org/pdf/2308.12608"
],
"recommendation": "Stop treating the rally CSVs as eval-only files; promote them to a single
model-agnostic \"labeled-corpus\" JSONL (d) ? one record per (video_id, rally_ordinal) carrying
[start,end], ending_reason, sport, provenance/source, label_type, split-by-match, and a WASB-
trajectory pointer ? with deterministic transcoders to three consumer formats (heuristic-window
labels, ActivityNet/ActionFormer TAL JSON, VLM clip?QA pairs). This is the highest-leverage,
lowest-cost move and it IS the moat. Then, in priority order: (1) IMMEDIATE ? grow from n?2 to
n?5 matches via a small Roora batch; below ~5 matches the heuristic/LogReg tuner is data-starved
and that, not the method, is why the learned model currently loses to the baseline. (2) NEAR-
TERM, free ? temporal augmentation (moderate speed/time-warp, jitter, reverse, FadeMixUp) +
open-teacher pseudo-labeling (WASB/our-model only, confidence-gated à la PLR/ALQA) over the
owned ~26 GB corpus to multiply effective labels without violating the no-paid-LLM guardrail.
(3) ONGOING ? route the Roora budget by active learning (uncertainty/disagreement/boundary
entropy) once n?5, and adopt timestamp/point supervision for cheap scale-out to new sports and
ending_reason. (4) DEFER but design-for-now ? Option B: fine-tune a guardrail-clean base
(Qwen2.5-VL-7B Apache-2.0 / InternVL2.5 MIT) with a quality SFT cold-start THEN GRPO/RFT using
temporal-IoU-vs-golden as the verifiable reward ? the data-efficiency literature (VideoChat-R1
+31.8 with limited samples; VTG-R1 filtered-13K > unfiltered-56K) says our small, clean, hand-
verified set is exactly the right asset for RL, so we do NOT need thousands of new matches, we
need amplification + a few clean ones. Do NOT invest in training a from-scratch TAL net
(ActionFormer-class wants ~50?200+ labeled videos we won't have soon). Reuse
`metrics.match_intervals` as the single spine for heuristic-labeling, TAL eval, and the RL
reward to avoid metric drift and data leakage.",
  "confidence": "medium",
  "caveats": [
    "The 'how many matches' bands are calibrated from public TAL/VLM literature
(THUMOS14/ActivityNet scale, VTG datasets of 13K?18K) and our own n=2 results, NOT measured on
our amateur single-camera 60fps footage ? domain transfer from broadcast-trained literature is
the dominant unknown and likely pushes our requirements UP (our own research doc flags this
exact broadcast?amateur gap).",
    "'Single-digit-thousands of clip?QA pairs' for Option B assumes augmentation + QA-task
multiplexing + confident pseudo-labels reach it; ~128 raw human rallies alone are NOT enough ?
amplification is load-bearing, not optional.",
    "Pseudo-labeling inherits WASB's recall ceiling (our confirmed bottleneck): pseudo-rallies
can only be as complete as the teacher, so they boost recall on what WASB sees and must be
confidence-gated + human-spot-checked; treat human [start,end] as the boundary authority.",
    "RFT/GRPO is LABEL-efficient but COMPUTE-heavy (multiple rollouts/sample) ? infeasible to

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run fully on one 8 GB RTX 2070 Super; budget cloud-GPU burst (RunPod/Lambda/Modal) and benchmark self-hosted-7B vs Gemini per-call before committing (COMMERCIALIZATION C7).",

"Timestamp-supervision savings (~? full time; SkipTag ~?; 64.1% vs 68.0% acc) come from the Breakfast action-SEGMENTATION user study, not racket-sports rally detection ? directionally relevant but not a measured number for our task.",

"Per the owner sequencing directive, this is a DESIGN blueprint: no Option-B implementation starts until P0?P4 scaffolding is solid and the golden-set corpus is healthy; the corpus-format (d) and n?5 (c) moves are the only items safe to act on now, and even those should get owner sign-off before code lands (CLAUDE.md: do not start implementation without explicit approval).",

"Base-model license must be re-verified per-checkpoint at build time (Qwen2.5-VL Apache-2.0 only at 7B/32B; 3B/72B custom; Gemma 4 Apache-2.0 status pending confirmation; Gemma 3 REJECTED) ? licenses change."

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]
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},
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{
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"dimension": "COST + LICENSING + PRIVACY/COMPLIANCE for an owned, on-device-capable rally temporal-segmentation model (and its Option-B end-to-end multimodal successor) for a Bangalore-based solo-founder racket-sports highlight indexer.",

"summary": "Three findings dominate. (1) COST: the two training paths are an order of magnitude apart. A small TAL/segmentation head (ActionFormer-class) trains on ~4.5 GB VRAM, so it fits the existing RTX 2070 Super (8 GB) locally at ~\$0 marginal cost, or a few dollars of rented GPU per full re-train; this matches today's bottleneck (golden data, not method). LoRA fine-tuning a 2-7B video VLM needs ~16-24 GB VRAM ? it does NOT fit the 2070, so it is cloud-only: realistic ranges are roughly \$5-40 for a single LoRA run on rented GPU (RunPod A100 80GB at ~\$1.19-1.39/hr, ~4-30 GPU-hrs depending on clip count/epochs) or \$2-30 on managed token-priced backends (Together/Fireworks LoRA SFT ?16B at ~\$0.48-0.50/M tokens). Full multimodal fine-tuning is far heavier (~15 GPU-hrs/epoch on 4xA100 reported for a 7B VLM ? roughly \$70-300/epoch rented). (2) LICENSING: Apache-2.0-clean bases exist but the size matters enormously for Qwen ? verified per-checkpoint: Qwen2.5-VL-7B = Apache-2.0 (clean), but 3B = Qwen Research License (NON-commercial) and 72B = custom Qwen License (commercial-OK but 100M-MAU gate + 'Built with Qwen' attribution + competing-clause concerns). Qwen3-VL 4B/8B/30B/235B are Apache-2.0. InternVL2.5 is MIT but some variants embed a Qwen-3B research-licensed component ? verify the specific checkpoint. Gemma 3 is correctly REJECTED (custom Gemma Terms + Prohibited Use Policy + viral distribution-flowdown). Gemma 4 reportedly ships Apache-2.0 but I could only confirm this via secondary/editorial sources (not a primary Google page) and it is image-multimodal, with no confirmed VIDEO capability ? treat as AMBER. Training on your OWN footage + human rally labels is clean (you own the footage; labels are facts/timestamps, not copyrightable expression; WASB code is MIT). (3) PRIVACY: Local inference is the strongest privacy lever (video never leaves device ? no cross-border issue at all). For training/fallback backends: Fireworks and Together both contractually state they do NOT train on customer data, both are SOC2 Type II (Fireworks also HIPAA, BAAs), both offer Zero Data Retention; Google Vertex AI states it 'won't use your data to train or fine-tune any models without your prior permission,' with 24h in-memory-only caching and a ZDR path. India's DPDP Act 2023 is permissive on residency (transfers allowed except to a govt 'negative list' ? currently empty as of Feb 2026), but minors are under-16 and require verifiable parental consent ? reinforcing the committed 'exclude minors from training pool' guardrail. The owned-model 'no paid-LLM outputs as training data' rule is sound: Gemini/OpenAI/Anthropic terms restrict using outputs to build competing models, so they remain eval/teacher-only.",

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"options": [
```

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{
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"name": "Path 1 ? Small TAL/segmentation head, trained LOCALLY (Option A, near-term)",

"what": "ActionFormer-class or your distilled classifier on WASB-trajectory + (later) audio/pose features ? rally [start,end]. ~4.5 GB training VRAM fits the RTX 2070 Super (8 GB) + WSL2. No VLM, no per-call fees.",

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"pros": [
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"Effectively \$0 marginal training cost on existing hardware; full re-train on a new golden batch costs a few \$ if you rent a GPU at all",

"Fastest iteration loop ? re-train per labeled batch, CI-runnable, matches nanochat-style operability goal",

"All deps permissive (ActionFormer MIT, WASB code MIT, torch BSD) ? zero licensing exposure",

"Directly attacks today's real bottleneck (golden-data volume + WASB recall), not the modeling method",

"Cheapest at inference ? best steady-state margin per COMMERCIALIZATION C7"

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    ],
    "cons": [
        "Inherits WASB candidate-window recall ceiling (the structural limit Option B is meant to remove)",
        "Not end-to-end multimodal ? caps eventual quality vs a frames+audio+pose model",
        "Distilled learned model currently UNDERPERFORMS the threshold baseline at n=2 videos ? needs more data before it pays off"
    ],
    "cost_note": "~$0 on the local 2070 (8 GB) for the head; if rented, a full re-train is single-digit dollars (e.g. <1-3 RunPod GPU-hrs at $0.34 RTX-4090 to $1.39 A100). Inference is near-free.",
    "license": "MIT/BSD/Apache-2.0 across the stack ? clean.",
    "privacy_note": "Trains and infers fully local; user video never leaves device. No cross-border/DPDP/GDPR transfer issue at all. Strongest privacy posture.",
    "fit_for_us": "strong"
  },
  {
    "name": "Path 2 ? LoRA fine-tune a 2-7B video VLM on RENTED cloud GPU (Option B bridge, self-managed)",
    "what": "ms-swift/Unsloth LoRA/QLoRA on Qwen2.5-VL-7B (Apache-2.0) or Qwen3-VL-8B (Apache-2.0) over 5k-20k clip?QA pairs, rented on RunPod/Lambda/Modal. ~16-24 GB VRAM ? A100 40/80GB class.",
    "pros": [
        "Removes WASB recall ceiling; moves toward the owned end-to-end multimodal target",
        "Per-run cost is modest and pay-as-you-go ? no standing GPU bill",
        "Apache-2.0 base (at 7B/8B) keeps fine-tuned weights private, gateable, monetizable",
        "RunPod is cheapest self-serve (A100 80GB ~$1.19-1.39/hr; H100 PCIe ~$1.99/hr; spot 40-50% off); you keep full control of weights + data"
    ],
    "cons": [
        "Does NOT fit the local 2070 (8 GB) ? must rent; adds a second stack to operate",
        "You own MLOps: data prep, leakage guards (split by match not clip), eval gate, reproducibility",
        "Self-managed rental providers offer thin/no DPA vs managed APIs ? keep training data to owned/consented footage only (which is already the plan)",
        "Needs 5k-20k curated QA pairs to be worth it; current 96+32 labeled rallies are far below that band"
    ],
    "cost_note": "Single LoRA run ? $5-40 (?4-30 A100-hrs at ~$1.2-1.4/hr; 7B LoRA over ~1k-10k examples ? 0.5-5 hrs/epoch on one A100). Full multimodal FT is ~10-15x heavier (~$70-300/epoch on 4xA100). Spot pricing cuts ~40-50%.",
    "license": "Qwen2.5-VL-7B = Apache-2.0 (VERIFIED clean). Qwen3-VL-8B = Apache-2.0. AVOID Qwen2.5-VL-3B (Qwen Research License, non-commercial) and 72B (custom license, 100M-MAU gate + attribution).",
    "privacy_note": "Training data = your owned/consented footage only; no user uploads, no paid-LLM outputs. Self-managed rental ? verify the provider's own DPA/retention; prefer the providers' ZDR/non-training defaults where offered. Inference can still be run locally after training.",
    "fit_for_us": "medium"
  },
  {
    "name": "Path 3 ? Managed fine-tuning backend (Together AI or Fireworks AI)",
    "what": "Token-priced LoRA SFT on a supported open base, run as a managed job. You upload the QA dataset; they return tuned weights/an endpoint.",
    "pros": [
        "Strong, explicit privacy contracts: Fireworks states it does NOT train on customer content and is ZDR-by-default + SOC2 Type II + HIPAA/BAA; Together states no training without opt-in, offers ZDR, SOC2 Type II, BAAs",
        "Cheap for the small band: LoRA SFT ?16B ~$0.48-0.50/M tokens ? a 5k-20k QA-pair run is typically single-digit to low-tens of dollars",
        "Zero MLOps for the training run itself; fits the 'agent runs a fine-tune unattended' operability goal",
        "Together fine-tuning catalog includes Qwen variants ? keeps you on an Apache-2.0 base"
    ],
  },

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    "cons": [
        "Less control over exact base checkpoint/version and weight portability than self-hosting",
        "Per-run economics only beat rented GPU at low/medium volume; high sustained volume favors owning the GPU",
        "Need to confirm the specific base offered is the Apache-2.0 size (e.g. Qwen 7B/8B, not 3B/72B) before committing",
        "Data residency: confirm region ? these are US-centric; DPDP permits the transfer today but Significant-Data-Fiduciary localization powers exist"
    ],
    "cost_note": "Together LoRA SFT: ?16B $0.48/M, 17-69B $1.50/M, 70-100B $2.90/M tokens (full FT ~2.5x). Fireworks fine-tuning from ~$0.50/M (?16B). A realistic 5k-20k clip?QA-pair run lands in the ~$2-30 range.",
    "license": "Base must be an Apache-2.0/MIT open model (Qwen 7B/8B, InternVL2.5 verified-clean variant). Provider tooling is fine; the weights' license is what governs monetization.",
    "privacy_note": "Best contractual privacy of the cloud options: Fireworks 'does not use customer Content to train/improve models' + ZDR default; Together 'no training without opt-in' + ZDR setting. Still: route only owned/consented footage, never user uploads or paid-LLM outputs, and prefer in-region/ZDR config.",
    "fit_for_us": "medium"
  },
  {
    "name": "Path 4 ? Vertex AI managed tuning (the option the owner named)",
    "what": "Google Vertex AI tuning/Model Garden for an open base or Gemini tuning. Managed, integrated with GCP storage/compute.",
    "pros": [
        "Clear data-governance statement: Google 'won't use your data to train or fine-tune any AI/ML models without your prior permission or instruction'; 24h in-memory-only caching; ZDR path available",
        "Mumbai (asia-south1) region available ? can keep data in-India, easing DPDP/Significant-Data-Fiduciary concerns",
        "Mature MLOps, integrates with the existing Gemini fallback already in the provider registry"
    ],
    "cons": [
        "GUARDRAIL TRAP: tuning Gemini itself does NOT yield an OWNED, on-device base ? it stays Google-served; only tuning an OPEN base in Model Garden gives portable weights",
        "Tends to be pricier and more lock-in-prone than RunPod/Together for small runs",
        "Open-base catalog/licensing on Vertex must be checked per-model; don't assume Apache-2.0",
        "Heavier setup than a single rented GPU for a solo founder's first runs"
    ],
    "cost_note": "GPU/tuning priced above the cheapest self-serve rentals; exact tuning cost depends on the base + tokens. Use only if in-India residency or GCP integration outweighs the premium.",
    "license": "Acceptable ONLY if you tune an Apache-2.0/MIT open base with portable weights. Gemini-tuned models are not owned/on-device ? fails the owned-model goal (fine as a fallback, not as 'the owned model').",
    "privacy_note": "Vertex paid tier: no training on customer data without permission; asia-south1 (Mumbai) keeps data in-India. Good privacy + residency story; weaker on the 'owned/on-device' requirement.",
    "fit_for_us": "weak"
  }
],
"key_sources": [
    "https://www.runpod.io/pricing",
    "https://lambda.ai/pricing",
    "https://www.spheron.network/blog/gpu-cloud-pricing-comparison-2026/",
    "https://www.together.ai/pricing",
    "https://fireworks.ai/pricing",
    "https://huggingface.co/Qwen/Qwen2.5-VL-7B-Instruct",
    "https://huggingface.co/Qwen/Qwen2.5-VL-3B-Instruct/blob/main/LICENSE",
    "https://huggingface.co/Qwen/Qwen2.5-VL-72B-Instruct/blob/main/LICENSE",
    "https://github.com/QwenLM/Qwen3-VL/blob/main/LICENSE",
    "https://huggingface.co/OpenGVLab/InternVL2_5-8B-MPO",

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"https://github.com/OpenGVLab/InternVL/issues/900",
"https://ai.google.dev/gemma/terms",
"https://ai.google.dev/gemma/prohibited_use_policy",
"https://www.mindstudio.ai/blog/what-is-gemma-4-apache-2-license-commercial-ai-deployment",
"https://docs.cloud.google.com/vertex-ai/generative-ai/docs/data-governance",
"https://docs.fireworks.ai/guides/security_compliance/data_security",
"https://www.together.ai/blog/soc-2-compliance",
"https://www.meity.gov.in/static/uploads/2024/06/2bf1f0e9f04e6fb4f8fef35e82c42aa5.pdf",
"https://www.cyrilshroff.com/wp-content/uploads/2025/12/FAQs-DPDPA.pdf"
],
"recommendation": "Stage it. (1) NOW: keep building the small local TAL/segmentation head (Path 1) ? it costs ~$0 on the existing RTX 2070 Super, attacks the real bottleneck (golden-data volume + WASB recall), and carries zero licensing/privacy exposure. Spend the budget on golden labels (Roora/rally-annotator), not GPUs. (2) WHEN the corpus reaches the 5k-20k clip?QA-pair band: prototype the owned VLM with a LoRA fine-tune of Qwen2.5-VL-7B or Qwen3-VL-8B (both VERIFIED Apache-2.0) ? first on RunPod rented A100 80GB (Path 2, ~$5-40/run, cheapest + full weight control) for the proof-of-concept; graduate to a managed backend (Path 3) if you want the stronger DPA + zero-MLOps and the per-run economics hold. Treat Vertex (Path 4) as in-India-residency or fallback infra, NOT as the owned model. (3) HARD licensing locks to encode in the C14 gate: APPROVE Qwen2.5-VL-7B/32B, Qwen3-VL-4B/8B/30B, InternVL2.5 (verified-clean variant); REJECT Qwen2.5-VL-3B (research/non-commercial), Qwen 72B (custom license), Gemma 3 (custom viral terms). Hold Gemma 4 as AMBER until you confirm Apache-2.0 from a primary Google source AND confirm video (not just image) capability. (4) PRIVACY: lean on local inference as the headline privacy lever; for any cloud training, enable ZDR/non-training configs (Fireworks/Together both contractually don't train on customer data; Vertex won't without permission), route ONLY owned/consented footage, keep minors out of the training pool (DPDP under-16 + verifiable parental consent), and keep paid-LLM outputs strictly eval/teacher-only.",
"confidence": "medium",
"caveats": [
  "GPU prices are volatile and provider-/region-/spot-dependent; the $ ranges (RunPod A100 80GB ~$1.19-1.39/hr, H100 PCIe ~$1.99/hr; Lambda H100 ~$3.29/hr) are 2026 snapshots ? re-quote before committing budget. RunPod adds ~$0.10-0.40/instance fees on top of GPU rates.",
  "Training-cost ranges depend heavily on dataset size, epochs, sequence/frame length, and LoRA vs full FT. The VLM LoRA figures assume the realistic 5k-20k clip?QA-pair band; full multimodal fine-tuning is roughly 10-15x heavier per epoch. Benchmark on your real workload before pricing the product (echoes COMMERCIALIZATION C7).",
  "Gemma 4 Apache-2.0 is the one license claim I could only confirm via secondary/editorial sources (mindstudio.ai), not a primary Google license page; and those sources describe it as image-multimodal with NO confirmed video capability. Verify both the license AND video support from ai.google.dev before relying on it as a base. Gemma 3 rejection is well-grounded (custom Gemma Terms + Prohibited Use Policy + distribution flow-down).",
  "InternVL2.5 is MIT, but a GitHub issue confirms some variants embed a Qwen-3B-Instruct component whose license can be the non-commercial Qwen Research License ? verify the EXACT checkpoint's bundled components before commercial use.",
  "Backend 'we don't train on your data' claims are from each vendor's own docs/blog (Fireworks, Together, Google) ? strong but self-asserted; get the signed DPA/BAA before sending any footage, and confirm the contracted region for India residency.",
  "India DPDP residency is permissive TODAY (empty negative list as of Feb 2026), but the govt retains Significant-Data-Fiduciary localization powers and the DPDP Rules are still settling ? re-check if you scale or get designated an SDF.",
  "The 'own footage + human labels = clean to train on' conclusion is a strong reading (you own the video; timestamps/reasons are non-copyrightable facts; WASB code is MIT) but is not formal legal advice ? the per-clip consent ledger + minors exclusion + separate training opt-in already planned are the right operational guardrails; confirm with counsel before scale."
]
}
]

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VERIFICATION:

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[
{
  "focus": "platform GPU prices + data-use/retention terms + India data residency",
  "checked_claims": [
    {

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    "claim": "Modal A100 80GB ~$3.72/hr and H100 ~$4.29/hr (per-second billing).",
    "verdict": "outdated",
    "evidence": "Modal's live pricing page shows A100 80GB at $0.000694/sec (~$2.50/hr) and H100 at $0.001097/sec (~$3.95/hr). The findings' figures are ~30-50% too high for the A100 and ~8% high for the H100. Modal is therefore notably CHEAPER than stated ? this strengthens (not weakens) the 'decouple from raw GPU price' thesis and the Modal-as-primary recommendation, but the specific dollar figures are stale.",
    "source": "https://modal.com/pricing"
  },
  {
    "claim": "Modal 'will never access your source code, the inputs/outputs to your Functions, or any data you store'; inputs/outputs deleted within 7 days; SOC2 Type II; HIPAA via BAA.",
    "verdict": "supported",
    "evidence": "Modal's security page verbatim: 'Modal will never access or use: your source code; the inputs (function arguments) or outputs (function return values) to your Modal Functions; any data you store in Modal.' Function inputs/outputs retained up to 7 days then deleted. SOC 2 Type 2 audit completed; HIPAA via BAA on Enterprise (with feature carve-outs: Volumes v1 / Images / Memory Snapshots out of scope for PHI). The load-bearing privacy claim is accurate.",
    "source": "https://modal.com/docs/guide/security"
  },
  {
    "claim": "Modal regional multipliers 1.25x-2.5x for non-US.",
    "verdict": "outdated",
    "evidence": "Modal's current pricing page states region selection is '1.5 - 1.75x base prices', not 1.25x-2.5x. The directional point (non-US regions cost more, and Modal has no India region) holds, but the multiplier range is wrong.",
    "source": "https://modal.com/pricing"
  },
  {
    "claim": "RunPod Secure Cloud A100 80GB ~$1.19-$1.29/hr, H100 ~$2.65/hr on-demand (spot ~$1.19).",
    "verdict": "outdated",
    "evidence": "RunPod's pricing page now shows A100 PCIe $1.39/hr, A100 SXM $1.49/hr, H100 PCIe $2.89/hr, H100 SXM $3.29/hr (figures shown are Community-tier; Secure Cloud separate figures not exposed on the page). Current A100 is ~$1.39-1.49 (above the cited $1.19-1.29) and H100 ~$2.89-3.29 (above the cited $2.65). Still the cheapest vetted-DC option, so the fallback recommendation stands, but the numbers are low/stale and Secure-Cloud-specific rates are no longer published.",
    "source": "https://www.runpod.io/pricing"
  },
  {
    "claim": "RunPod Secure Cloud: SOC2 Type II, ISO 27001, GDPR, HIPAA; ToS bars hosts inspecting pod data; Secure vs Community tier distinction.",
    "verdict": "supported",
    "evidence": "RunPod security docs: Secure Cloud uses 'vetted infrastructure partners who meet enterprise security standards including SOC 2, ISO 27001, and PCI DSS certifications'; 'Runpod is fully compliant with the General Data Protection Regulation (GDPR)'; and verbatim 'Runpod's terms of service prohibit hosts from inspecting your Pod/worker data or analyzing your usage patterns.' HIPAA confirmed via a separate press release ('independently audited and verified as meeting the standards set forth by HIPAA... and GDPR'). Secure Cloud (T3/T4 vetted DCs) vs Community distinction confirmed. Caveat: SOC2/ISO are PARTNER/datacenter certs ('compliance certifications apply at the platform level; data center compliance may vary'), and the docs did not surface an explicit AES-256-at-rest figure or a named DPA ? so 'AES-256 at rest' and 'has a DPA' are uncertain from primary sources.",
    "source": "https://docs.runpod.io/references/security-and-compliance ; https://www.runpod.io/press/runpod-meets-hipaa-and-gdpr-standards"
  },
  {
    "claim": "Google/Vertex: 'Customer data is never used by Google. We do not use customer data to train our models... without permission' ? a strong written no-train commitment.",
    "verdict": "supported",
    "evidence": "Confirmed from two Google-aligned sources: the gen-AI data-residency blog states 'Customer data is never used by Google. We do not use customer data to train our models,

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or models used by others, without permission.' Independently, Vertex AI data-governance docs describe a Section 17 'Training Restriction' in the Service Specific Terms: 'Google won't use your data to train or fine-tune any AI/ML models without your prior permission or instruction,' applying to all managed models, plus sole-customer-access to fine-tuned/customer-trained models. The strongest contractual no-train assurance among the cloud options, as claimed.",

"source": "https://cloud.google.com/blog/products/ai-machine-learning/google-cloud-generative-ai-data-residency-guarantees-for-data-stored-at-rest ; Vertex AI data-governance docs (docs.cloud.google.com/.../data-governance)"

},
{

"claim": "Vertex/GCP is the ONLY strong-privacy option with an India region: asia-south1 (Mumbai) has Compute Engine GPUs for genuine India residency-at-rest, and the published gen-AI residency region list (excluding asia-south1) applies only to MANAGED gen-AI APIs, not custom training.",

"verdict": "supported",

"evidence": "Two-part claim, both parts hold with one refinement. (1) The gen-AI data-residency blog lists residency for managed gen-AI APIs across US/Canada/NL/FR/UK/DE/BE/Japan/Singapore/Korea and explicitly does NOT include any India region ? confirming the 'managed gen-AI APIs exclude asia-south1' nuance. (2) asia-south1 DOES have GPUs for custom Compute Engine training: the GPU regions doc lists A3 Ultra (H100/H200-class) in asia-south1-b and A3 Edge + G2/G4 across Mumbai zones. REFINEMENT THAT COULD CHANGE A RECOMMENDATION: asia-south1 does NOT list the older A100 ? only newer A3 (H100/H200) and L4/T4-class. The findings' cost table implies an 'A100 in Mumbai' job; in practice an India-residency training run would be on H100-class A3 (pricier) or T4/L4, not A100. Capacity for A3 in Mumbai is also account-team gated.",

"source": "https://docs.cloud.google.com/compute/docs/regions-zones/gpu-regions-zones ; https://cloud.google.com/blog/products/ai-machine-learning/google-cloud-generative-ai-data-residency-guarantees-for-data-stored-at-rest"

},
{

"claim": "Vertex AI A100 ~\$3.37/hr all-in (compute + management fee).",

"verdict": "supported",

"evidence": "Corroborated: NVIDIA A100 at \$2.93/hr compute + \$0.44/hr Vertex management fee = \$3.37/hr. NOTE this is the 40GB A100; the 80GB a2-ultragpu-1g runs ~\$5.07/hr on Compute Engine (plus any Vertex fee), so for an 80GB-class job the all-in figure understates cost. Directionally correct (Vertex is the most expensive + overhead-heavy option), with the 40-vs-80GB caveat.",

"source": "https://cloud.google.com/vertex-ai/pricing (via aggregated 2026 pricing summaries: nops.io, cloudprice.net a2-ultragpu-8g)"

},
{

"claim": "Lambda Labs A100 80GB ~\$1.99-\$2.49/hr, H100 ~\$3.29-\$4.29/hr; free egress; no spot tier; US-only.",

"verdict": "outdated",

"evidence": "Lambda's pricing page shows A100 80GB at \$2.79/GPU/hr and H100 at \$3.99/GPU/hr (8x configs). The A100 figure (\$2.79) is ABOVE the cited \$1.99-2.49 range; the H100 (\$3.99) is within the cited range. 'No spot tier' is confirmed (none shown). 'Free egress' and 'US-only/no India region' could NOT be confirmed from the current pricing page (not stated there) ? treat those two sub-claims as uncertain pending Lambda's docs.",

"source": "https://lambda.ai/pricing"

},
{

"claim": "Together AI could NOT confirm a verbatim no-train clause or public DPA -> BLOCKED for the consented corpus until a signed DPA is obtained. Managed LoRA ~\$0.48/M tokens (<=16B).",

"verdict": "supported",

"evidence": "Verified on both counts. (1) Together's fine-tuning page asserts data/model OWNERSHIP, SOC2 Type II, HIPAA-aligned, regional residency (incl. Asia/ME) ? but contains NO explicit 'we do not train on your data' statement and no public DPA, exactly as the findings flag. The 'BLOCKED until DPA' posture is well-grounded. (2) LoRA SFT <=16B = \$0.48/M tokens is confirmed across multiple Feb-2026 sources (consistent with Together's own docs). Note: inference hosting is a separate ongoing charge (~\$6.49/hr H100 dedicated endpoint).",

"source": "https://www.together.ai/fine-tuning ; https://docs.together.ai/docs/fine-tuning-pricing (corroborated by eesel/cloudzero 2026 snapshots)"

},

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{
  "claim": "AWS Mumbai ap-south-1 supports SageMaker training with P4d (A100) for India residency.",
  "verdict": "outdated",
  "evidence": "India residency claim HOLDS and is now STRONGER, but the instance specifics are stale. As of 2026, ap-south-1 (Mumbai) supports P5 (single-GPU H100) via SageMaker Studio notebooks, On-Demand/Spot, and Flexible Training Plans. P4d (A100) availability in Mumbai is NOT explicitly confirmed in current AWS announcements (the public materials now center on P5/H100). So 'India residency via SageMaker' is supported, but 'P4d (A100)' specifically is uncertain/superseded by P5.",
  "source": "https://aws.amazon.com/about-aws/whats-new/2026/01/p5.4xl-new-launch-sagemaker-studio-notebooks/"
},
{
  "claim": "Azure Central India offers A100/H100; NC A100 v4 ~$3.67/hr on-demand / ~$0.82/hr spot; ND H100 v5 ~$6.98-$12/hr.",
  "verdict": "supported",
  "evidence": "Pricing confirmed: NC24ads A100 v4 = $3.673/hr on-demand, $0.8213/hr spot (matches the findings near-exactly). NC40ads H100 v5 = $6.98/hr on-demand, $2.19/hr spot; large ND96isr H100 v5 = $98.32/hr (i.e. ~$12.29/hr per H100 across 8 GPUs) ? both consistent with the cited '$6.98-$12/hr/GPU' band. India residency (Central India) is plausible but was NOT confirmed region-specifically from a primary Azure page in this check ? treat the region-availability sub-claim as uncertain pending the Azure region table. The 'could not confirm an Azure ML no-train statement' caveat remains valid (not checked here).",
  "source": "https://instances.vantage.sh/azure/vm/nc24ads-v4 ; https://instances.vantage.sh/azure/vm/nc40adsh100-v5"
},
{
  "claim": "vast.ai cheapest: A100 80GB ~$0.67/hr, H100 ~$1.87/hr, spot to ~$0.34/hr; Community tier unvetted vs Secure Cloud (ISO-27001+ vetted) ? Community FAILS the privacy requirement.",
  "verdict": "uncertain",
  "evidence": "vast.ai's pricing page now renders only 'Live GPU Prices' dynamically and exposes On-Demand/Interruptible/Reserved instance types ? it did NOT surface the static A100 ~$0.67 / H100 ~$1.87 figures, so those specific numbers could not be confirmed from the primary page (marketplace prices float). The qualitative claim (vast.ai is the cheapest marketplace; Community/peer-to-peer tier is unvetted and unsuitable for consented video; restrict to Secure tier) is consistent with vast.ai's known model but the security-doc Secure-vs-Community detail was not re-verified here. The recommendation (vast.ai for non-sensitive only) is unaffected.",
  "source": "https://vast.ai/pricing"
},
{
  "claim": "Gemma 4 is Apache-2.0 (a genuine reversal from Gemma 1/2/3's restrictive Gemma Terms); Gemma 3 stays REJECTED.",
  "verdict": "supported",
  "evidence": "Confirmed from a PRIMARY Google source: the Google Open Source Blog (opensource.googleblog.com, 'Gemma 4: Expanding the Gemmaverse with Apache 2.0', Mar/Apr 2026) states Gemma 4 is released under Apache 2.0, sizes 'from edge devices to 31B parameters'. The reversal from Gemma 1/2/3's custom 'Gemma Terms of Use' is well-documented. This load-bearing guardrail claim (Gemma 4 OK, Gemma 3 REJECT) is solid.",
  "source": "https://opensource.googleblog.com/2026/03/gemma-4-expanding-the-gemmaverse-with-apache-20.html"
},
{
  "claim": "Gemma 4 processes frames + audio + VIDEO natively in one model (the structural match for Option-B multimodal fusion).",
  "verdict": "uncertain",
  "evidence": "INTERNALLY INCONSISTENT in the findings and unconfirmed by primary sources. The official Google Open Source Blog confirms Apache-2.0 and multimodality but does NOT mention VIDEO or native audio. The 'native frames+audio+video' assertion rests on secondary/editorial sources (VentureBeat, InfoQ). One findings section asserts it as VERIFIED ('native frames+audio+video without separate encoders'), while another correctly AMBER-flags it ('image-multimodal, with no confirmed VIDEO capability'). The conservative AMBER framing is right; the VERIFIED framing is not supported by a primary source. This matters because the Gemma-4-for-Option-B recommendation leans on the native-video story ? confirm video support from

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ai.google.dev before locking Gemma 4 as the multimodal base.",
    "source": "https://opensource.googleblog.com/2026/03/gemma-4-expanding-the-gemmaverse-
with-apache-20.html ; https://blog.google/innovation-and-ai/technology/developers-
tools/gemma-4/"
    },
    {
        "claim": "Qwen2.5-VL-7B = Apache-2.0 with native temporal grounding; Qwen2.5-VL-3B =
non-commercial Qwen Research License (REJECT); Qwen2.5-VL-72B = Qwen License with 100M-MAU cap +
'Built with Qwen' attribution (REJECT).",
        "verdict": "supported",
        "evidence": "All three verified from primary HuggingFace sources. 7B model card:
'License: apache-2.0', with temporal grounding via 'mRoPE in the time dimension with IDs and
absolute time alignment' + 'dynamic FPS sampling'. 3B LICENSE: 'Qwen RESEARCH LICENSE
AGREEMENT... FOR NON-COMMERCIAL PURPOSES ONLY... If you are commercially using the Materials,
you shall request a license from us.' 72B LICENSE: 'If you are commercially using the Materials,
and your product or service has more than 100 million monthly active users, you shall request a
license from us', plus mandatory 'Built with Qwen'/'Improved using Qwen' attribution. The per-
size APPROVE/REJECT guardrail is accurate.",
        "source": "https://huggingface.co/Qwen/Qwen2.5-VL-7B-Instruct ;
https://huggingface.co/Qwen/Qwen2.5-VL-3B-Instruct/blob/main/LICENSE ;
https://huggingface.co/Qwen/Qwen2.5-VL-72B-Instruct/blob/main/LICENSE"
    }
],
"corrections": [
    "Modal GPU prices are STALE and too HIGH: current pricing is A100 80GB ~$2.50/hr (not
$3.72) and H100 ~$3.95/hr (not $4.29). This actually STRENGTHENS the Modal-as-primary
recommendation (Modal is cheaper than the findings claimed), but every Modal dollar figure
should be re-quoted. The regional multiplier is also wrong: it is now 1.5-1.75x, not
1.25x-2.5x.",
    "RunPod prices are stale and LOW: current page shows A100 ~$1.39-1.49/hr (not $1.19-1.29)
and H100 PCIe/SXM ~$2.89-3.29/hr (not $2.65). Also, RunPod no longer publishes separate Secure-
Cloud dollar figures (the page shows Community-tier rates), so any Secure-Cloud cost estimate is
now uncertain. RunPod's 'AES-256 at rest' and 'has a DPA' sub-claims were NOT confirmable from
the primary security docs (only the host-no-inspect ToS clause, GDPR, and partner SOC2/ISO/PCI
were).",
    "Lambda A100 is now $2.79/hr (above the cited $1.99-2.49) and H100 $3.99/hr; Lambda's
'free egress' and 'US-only / no India region' sub-claims could not be confirmed from the current
pricing page and are uncertain.",
    "India residency refinement that could change a recommendation: GCP asia-south1 (Mumbai)
does NOT offer A100 ? only A3 (H100/H200-class, account-team gated) and L4/T4/G2/G4. The
findings' cost framing implies an A100-in-Mumbai job; a real India-residency training run on GCP
would be on pricier H100-class A3 hardware (or cheaper T4/L4), not A100. AWS ap-south-1 likewise
now centers on P5/H100 (P4d/A100 not confirmed in Mumbai). The India-residency-at-rest
conclusion (only the three hyperscalers qualify) still holds; the specific in-India GPU
SKU/price does not.",
    "Gemma 4 'native video' is the one materially shaky data-use claim and it is internally
inconsistent: one findings section asserts native frames+audio+video as VERIFIED, another AMBER-
flags it. The primary Google source confirms only Apache-2.0 + multimodality, NOT video or
native audio. Treat Gemma-4-for-Option-B's native-video premise as UNVERIFIED and confirm from
ai.google.dev before locking it as the multimodal fusion base.",
    "vast.ai's specific per-hour figures (A100 ~$0.67, H100 ~$1.87) could not be confirmed ?
the pricing page is now a live/dynamic marketplace feed. The qualitative guidance (cheapest
marketplace; Secure tier only; never Community for consented video) is unaffected.",
    "Net effect on recommendations: NONE of the verified discrepancies overturn the core plan
(Modal primary on privacy grounds, RunPod Secure fallback, Vertex/GCP as the India-residency
bench option, paid-LLM/Gemma-3/Qwen-3B/72B reject gates). The pricing errors are within the
findings' own stated +/-15-25% volatility band and the privacy/license/residency load-bearing
claims all held. The two items worth acting on: (a) re-quote all GPU prices live before
budgeting; (b) verify Gemma 4 video support and the in-India GPU SKU (H100/A3, not A100) before
relying on either for an architecture decision."
]
},
{
    "focus": "Exact licenses of every named base/architecture model (Apache/MIT vs Gemma-
Terms/community/NC) and their video-capability claims, plus the most decision-critical

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privacy/pricing claims. Verified against HuggingFace model cards, GitHub LICENSE files, and official provider pages.",

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"checked_claims": [
  {
    "claim": "Qwen2.5-VL-7B-Instruct is Apache-2.0 and natively supports video temporal grounding (mRoPE + absolute-time alignment, can pinpoint video segments / emit timestamps).",
    "verdict": "supported",
    "evidence": "HF model card lists License: Apache-2.0. Card states it can 'comprehend videos of over 1 hour', uses dynamic-FPS sampling, and does temporal grounding to 'pinpoint the relevant video segments' via updated positional embeddings that 'learn temporal sequence and speed'. Both the license and the video-grounding capability are confirmed verbatim.",
    "source": "https://huggingface.co/Qwen/Qwen2.5-VL-7B-Instruct"
  },
  {
    "claim": "Qwen2.5-VL-3B is Qwen Research License (non-commercial, REJECT).",
    "verdict": "supported",
    "evidence": "LICENSE file is titled 'Qwen RESEARCH LICENSE AGREEMENT' and grants rights 'FOR NON-COMMERCIAL PURPOSES ONLY' with 'If you are commercially using the Materials, you shall request a license from us.' Not Apache-2.0. Rejection is correct.",
    "source": "https://huggingface.co/Qwen/Qwen2.5-VL-3B-Instruct/blob/main/LICENSE"
  },
  {
    "claim": "Qwen2.5-VL-72B is a custom Qwen License with a 100M-MAU commercial gate + 'Built with Qwen' attribution (REJECT per-size).",
    "verdict": "supported",
    "evidence": "LICENSE file is the 'Qwen LICENSE AGREEMENT' containing 'if ... your product or service has more than 100 million monthly active users, you shall request a license from us' and a mandatory 'Built with Qwen'/'Improved using Qwen' attribution clause for derivative AI models. Confirmed exactly as the finding states.",
    "source": "https://huggingface.co/Qwen/Qwen2.5-VL-72B-Instruct/blob/main/LICENSE"
  },
  {
    "claim": "Qwen3-VL-4B-Instruct and 8B-Instruct are Apache-2.0, with explicit Text-Stamp Alignment for video temporal grounding.",
    "verdict": "supported",
    "evidence": "Both HF cards list License: apache-2.0. The 4B card documents 'Text-Stamp Alignment' enabling 'precise, timestamp-grounded event localization for stronger video temporal modeling' plus 256K-1M context for hours-long video. License and capability both confirmed.",
    "source": "https://huggingface.co/Qwen/Qwen3-VL-4B-Instruct + https://huggingface.co/Qwen/Qwen3-VL-8B-Instruct"
  },
  {
    "claim": "Gemma 4 is Apache-2.0 (a genuine reversal from Gemma 1/2/3's restrictive Gemma Terms), with native frames+audio+video. (Finding #3 marked this VERIFIED; finding #5 conservatively marked it AMBER, 'could only confirm via secondary sources, no confirmed VIDEO capability'.)",
    "verdict": "supported",
    "evidence": "Now confirmed from PRIMARY sources, resolving the finding #5 AMBER caveat. Official Google blog (blog.google) states twice 'Gemma 4 is released under a commercially permissive Apache 2.0 license' and 'All models natively process video and images'; E2B/E4B add native audio. The official google/gemma-4-E4B HF card lists License: apache-2.0 (no Gemma Terms). VIDEO capability IS confirmed (contra finding #5's 'no confirmed video'). NOTE a size discrepancy: official sizes are E2B/E4B/26B-A4B/31B; a 12B was released June 3 2026. The finding's repeated '4B/12B' shorthand is imprecise on the smaller-size names but the license+video substance holds.",
    "source": "https://blog.google/innovation-and-ai/technology/developers-tools/gemma-4/ + https://huggingface.co/google/gemma-4-E4B"
  },
  {
    "claim": "Gemma 3 remains correctly REJECTED (custom Gemma Terms of Use, restrictive, viral distribution flow-down).",
    "verdict": "supported",
    "evidence": "ai.google.dev/gemma/terms confirms a custom 'Gemma Terms of Use' (NOT Apache-2.0) applying to Gemma 1/1.1/2/3/3n, with mandatory flow-down ('include the use

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restrictions ... as an enforceable provision in any agreement' + pass a copy of the Agreement
downstream) and Prohibited Use Policy. The page explicitly states Gemma 4 is excluded and uses
the separate Apache-2.0 license. Rejection of Gemma 3 is well-grounded.",
    "source": "https://ai.google.dev/gemma/terms"
  },
  {
    "claim": "InternVL3-8B is MIT (wrapper) on an Apache-2.0 Qwen2.5-7B base; supports
video; only sizes built on Qwen2.5 <=32B are clean, and InternVL3-78B inherits the Qwen2.5-72B
100M-MAU cap (REJECT 78B).",
    "verdict": "supported",
    "evidence": "Project is MIT-licensed and uses pre-trained Qwen2.5 (Apache-2.0 at 8B/7B)
as a component; video multi-round conversation is documented. Per-size check confirms:
InternVL3-38B = MIT + Apache-2.0 (Qwen2.5-32B); InternVL3-78B = MIT wrapper BUT Qwen2.5-72B
component under the Qwen License (the 100M-MAU-gated one); InternVL3-9B uses InternLM3. The
finding's per-size approve/reject logic is accurate. Minor: the InternVL3-8B card text itself
foregrounds 'Apache-2.0' (from the Qwen component); the MIT designation is the project/repo
license ? both true, not contradictory.",
    "source": "https://huggingface.co/OpenGVLab/InternVL3-8B +
https://huggingface.co/OpenGVLab/InternVL3-78B"
  },
  {
    "claim": "VideoLLaMA3 code is Apache-2.0 but the RELEASED WEIGHTS are a research
preview, non-commercial (not shippable).",
    "verdict": "supported",
    "evidence": "Repo confirms code under Apache 2.0, but states 'The service is a research
preview intended for non-commercial use ONLY', with the weights restricted. The finding's
framing (Apache code / NC weights, collapses to 'use Qwen2.5 base + their recipe') is
accurate.",
    "source": "https://github.com/DAMO-NLP-SG/VideoLLaMA3"
  },
  {
    "claim": "Temporal-grounding specialists are license-poisoned as bases: TimeChat =
research-only/non-commercial; VTimeLLM = CC-BY-NC-ND 4.0; Momentor = no LICENSE + LLaMA-
derived.",
    "verdict": "supported",
    "evidence": "VTimeLLM repo: 'licensed under a Creative Commons Attribution-
NonCommercial-NoDerivs 4.0 International License' (NC + ND, cannot even fine-tune/redistribute)
- confirmed. Momentor: no LICENSE file visible, acknowledges building on LLaMA - confirmed.
TimeChat: states 'just a research preview intended for non-commercial use only' - confirmed NON-
COMMERCIAL. ONE NUANCE the finding omits: TimeChat's underlying CODE is BSD-3-Clause (inherited
from LAVIS/MiniGPT4); the finding labels it simply 'research-only/non-commercial' which is
correct as to usability but doesn't mention the BSD code layer. The load-bearing conclusion (not
usable as a shippable base) is unaffected.",
    "source": "https://github.com/huangb23/VTimeLLM + https://github.com/DCDmllm/Momentor +
https://github.com/RenShuhuai-Andy/TimeChat"
  },
  {
    "claim": "VideoMAE pretrained weights are CC-BY-NC 4.0 (REJECTED for commercial use);
InternVideo2-Stage2_6B = MIT and InternVideo2.5_Chat_8B = Apache-2.0 (both usable).",
    "verdict": "supported",
    "evidence": "MCG-NJU/videomae-base HF card lists License: cc-by-nc-4.0 (non-commercial)
- rejection confirmed. InternVideo2-Stage2_6B card lists License: mit; InternVideo2_5_Chat_8B
card lists License: apache-2.0 - both clean/usable as claimed.",
    "source": "https://huggingface.co/MCG-NJU/videomae-base +
https://huggingface.co/OpenGVLab/InternVideo2-Stage2_6B +
https://huggingface.co/OpenGVLab/InternVideo2_5_Chat_8B"
  },
  {
    "claim": "LICENSE TRAP: yabufarha/ms-tcn = MIT + Commons Clause (non-commercial, DO NOT
vendor); sj-li/MS-TCN2 = clean MIT (vendor this one).",
    "verdict": "supported",
    "evidence": "yabufarha/ms-tcn LICENSE = MIT + Commons Clause: 'the License does not
grant to you, the right to Sell the Software' (non-commercial) - trap confirmed. sj-li/MS-TCN2
LICENSE = clean MIT, no Commons Clause, full 'use/copy/modify/...sell' grant - confirmed. The
finding's central license-trap warning is accurate and is the single most exploitable

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distinction in the report.",
    "source": "https://github.com/yabufarha/ms-tcn/blob/master/LICENSE +
https://github.com/sj-li/MS-TCN2/blob/master/LICENSE"
  },
  {
    "claim": "ActionFormer, TriDet, ASFormer, and DiffAct are all clean MIT (commercially
usable).",
    "verdict": "supported",
    "evidence": "ActionFormer (happyharrycn/actionformer_release) = MIT; TriDet
(dingfengshi/TriDet) LICENSE = MIT (c) 2023 Dingfeng Shi; ASFormer (ChinaYi/ASFormer) = MIT;
DiffAct (Finspire13/DiffAct) LICENSE = MIT (c) 2023 Daochang Liu. All four confirmed clean
MIT.",
    "source": "https://github.com/happyharrycn/actionformer_release +
https://github.com/dingfengshi/TriDet/blob/master/LICENSE + https://github.com/ChinaYi/ASFormer
+ https://github.com/Finspire13/DiffAct/blob/main/LICENSE"
  },
  {
    "claim": "ActionFormer/ASFormer accept arbitrary input feature dimension via a learned
input projection (so WASB ~3-4-dim trajectory can be fed directly without I3D).",
    "verdict": "uncertain",
    "evidence": "Could not confirm from primary source within the scope of this pass. The
ActionFormer README documents only specific pre-extracted features (I3D 2048-d, TSP, SlowFast)
and does NOT describe the input-projection architecture on the page; verifying the arbitrary-dim
claim requires reading the model code (libs/modeling). This is a code-architecture claim, not a
license claim, and the finding itself flags 'off-the-validated-path / unproven on low-dim input'
as a caveat - so the mechanism plausibly exists but is NOT primary-source-confirmed here. Does
not change a license recommendation.",
    "source": "https://github.com/happyharrycn/actionformer_release"
  },
  {
    "claim": "Modal has the strongest written no-access contract: 'will never access your
source code, the inputs/outputs to your Functions, or any data you store'; inputs/outputs
deleted within 7 days; SOC2 Type II.",
    "verdict": "supported",
    "evidence": "Modal security page confirms verbatim: 'Modal will never access or use:
your source code. the inputs ... or outputs ... to your Modal Functions. any data you store in
Modal...'; inputs/outputs 'deleted within a maximum TTL of 7 days'; 'successfully completed a
SOC 2 Type 2 audit.' All three sub-claims confirmed exactly.",
    "source": "https://modal.com/docs/guide/security"
  },
  {
    "claim": "Google/Vertex states customer data is never used to train Google's models
without permission (the no-train assurance).",
    "verdict": "supported",
    "evidence": "Google blog states verbatim: 'Customer data is never used by Google. We do
not use customer data to train our models, or models used by others, without permission.'
Confirmed.",
    "source": "https://cloud.google.com/blog/products/ai-machine-learning/google-cloud-
generative-ai-data-residency-guarantees-for-data-stored-at-rest"
  },
  {
    "claim": "Vertex/GCP is the ONLY strong-privacy option with a genuine India (asia-south1
Mumbai) region for residency-at-rest, unique among the strong options.",
    "verdict": "uncertain",
    "evidence": "Partially supported but softer than stated. The cited gen-AI data-residency
blog lists Asia residency only as Japan/Singapore/Korea and does NOT include asia-south1/Mumbai
for managed gen-AI services - consistent with the finding's own nuance that the gen-AI residency
list excludes asia-south1. Compute Engine GPUs in asia-south1 DO exist, BUT GCP docs indicate
asia-south1 high-end GPU (A100/H100) capacity is 'by invitation only' / 'contact your account
team' - so India-residency custom training is NOT a turnkey, guaranteed-available capability.
The 'unique India region among strong options' point is directionally right (AWS ap-south-1 and
Azure Central India are the only other in-India options and the finding lists them too), but
treat 'genuine India residency-at-rest' as capacity-gated, not confirmed-available. Re-quote
capacity before relying on it.",
    "source": "https://docs.cloud.google.com/compute/docs/regions-zones/gpu-regions-zones"
  }

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    },
    {
        "claim": "Together AI could NOT be confirmed to have a verbatim no-train clause or public DPA -> BLOCKED for the consented corpus until a signed DPA + no-train assurance is obtained.",
        "verdict": "supported",
        "evidence": "Together's fine-tuning page states 'Your data and models remain fully under your ownership' and 'AICPA SOC 2 Type II', BUT contains NO explicit 'we do not train on your data' statement and no public DPA or zero-data-retention language on that page. The finding's cautious 'BLOCKED until signed DPA + explicit no-train' is correct and appropriately conservative.",
        "source": "https://www.together.ai/fine-tuning"
    },
    {
        "claim": "RunPod Secure Cloud A100 80GB ~$1.19-$1.29/hr, H100 ~$2.65/hr on-demand.",
        "verdict": "outdated",
        "evidence": "Live pricing page now shows A100 80GB PCIe at $1.39/hr (SXM $1.49/hr) and H100 PCIe $2.89/hr (SXM $3.29/hr) - higher than the finding's cited $1.19-$1.29 (A100) and $2.65 (H100). The page does not cleanly separate Secure vs Community figures in the fetched extract. Direction/magnitude of the finding's recommendation (RunPod = cheapest self-serve) is unaffected, but the specific dollar figures are stale on the low side. The finding's own caveat ('treat all $/hr as +/-15-25% and re-check live') covers this; re-quote before budgeting.",
        "source": "https://www.runpod.io/pricing"
    }
],
"corrections": [
    "Gemma 4 should be UPGRADED from finding #5's AMBER ('confirmed only via secondary sources; no confirmed VIDEO capability') to GREEN/verified. Primary sources now confirm BOTH the Apache-2.0 license (official Google blog states it twice; official google/gemma-4-E4B HF card lists license: apache-2.0) AND native video ('All models natively process video and images'). Finding #5's caveat that video was unconfirmed is now resolved in favor of the claim. Finding #3's 'VERIFIED' status was correct.",
    "Gemma 4 size shorthand is imprecise across the findings. Official model sizes are E2B / E4B / 26B-A4B (MoE) / 31B (dense), with a separate 12B released June 3 2026. The findings repeatedly say 'Gemma 4 4B/12B' - the '4B' maps to E4B and the '12B' is a later release, but the '4B/12B' phrasing does not match the primary E2B/E4B/26B/31B family naming. Use the official size names before locking a checkpoint.",
    "TimeChat's license is incompletely characterized. The findings label it only 'research-preview non-commercial'. The underlying CODE is BSD-3-Clause (inherited from LAVIS/MiniGPT-4); it is the TimeChat model/service that adds the 'non-commercial research preview' restriction. The conclusion (not a shippable commercial base) is unchanged, but the BSD code layer should be noted for accuracy.",
    "The Vertex 'genuine India residency-at-rest, unique among strong options' claim is overstated as turnkey. The cited gen-AI residency list excludes asia-south1; and asia-south1 high-end GPU (A100/H100) capacity appears to be invitation-only / account-team-gated per GCP docs. India-residency custom training is capacity-gated, not guaranteed-available. This does not flip the recommendation (Vertex stays 'keep on bench') but the residency advantage should be stated as conditional on confirmed asia-south1 GPU quota.",
    "RunPod $/hr figures are stale on the low side: live page shows A100 80GB PCIe $1.39/hr and H100 PCIe $2.89/hr vs the finding's $1.19-$1.29 / $2.65. Together AI's no-train clause remains genuinely unconfirmed on the public page (no explicit 'we do not train on your data', no public DPA) - the 'BLOCKED until DPA' gate is correct. Neither changes a model-license recommendation; both are flagged by the findings' own volatility caveats.",
    "InternVL3-8B presentation nuance: the finding calls it 'MIT wrapper'; the HF card foregrounds Apache-2.0 (from the Qwen2.5 component). Both are true (repo=MIT, component=Apache-2.0) and not contradictory - no correction to the conclusion, just noting the card emphasizes the Apache component license.",
    "One claim could NOT be verified from a primary source: that ActionFormer/ASFormer accept arbitrary low-dim input features via a learned input projection (the README only documents I3D-2048/TSP/SlowFast features). This is a code-architecture claim the findings themselves flag as 'off the validated path / unproven on low-dim WASB input'. It is not a license issue and does not change a base-model recommendation, but the 'feed WASB trajectory directly, no I3D required' claim should be validated in code before relying on it architecturally."
]
},

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{
  "focus": "data-efficiency / few-match feasibility claims for the recommended architectures
(ASFormer/ActionFormer/MS-TCN++ TAL-TAS path; Qwen2.5-VL-7B / Qwen3-VL-4B / Gemma 4 /
InternVideo2 owned-base path; RFT/GRPO label-efficiency; cheap-label and compute-fit claims).
Each checked against the primary source (paper, HF card/LICENSE, vendor pricing page) rather
than the aggregator citations.",
  "checked_claims": [
    {
      "claim": "ASFormer is the only surveyed model whose central thesis is training from
scratch on a tiny dataset (GTEA = 28 videos, 11 classes), with ~1.1M params, using convolutional
inductive bias to compensate for small-data overfitting ? making it the best fit for the n=3-10
regime.",
      "verdict": "supported",
      "evidence": "Primary source (ar5iv of arXiv:2110.08568) confirms each load-bearing
number: GTEA = '28 videos of 11 action classes'; parameter count stated as 'ASFormer: 1.134'
million; and the small-data thesis verbatim ? the local-connectivity inductive bias 'constrains
the hypothesis space within a reliable scope, and is beneficial for the action segmentation task
to learn a proper target function with small training sets.' This is a genuine, paper-central
claim, not an inflated reading. IMPORTANT NUANCE: 'from scratch' applies to the segmentation
HEAD trained on GTEA; the INPUT features are Kinetics-pretrained I3D (2048-d), so 'from scratch'
does not mean no pretrained representations. The findings' own caveats acknowledge this. Does
NOT change the recommendation.",
      "source": "https://ar5iv.labs.arxiv.org/html/2110.08568 (and arXiv:2110.08568)"
    },
    {
      "claim": "ActionFormer is more data-hungry than ASFormer: it is designed/validated on
benchmark-scale sets (THUMOS14 / ActivityNet / EPIC-Kitchens) and makes NO few-shot or small-
data claim, so at n=3-10 it risks overfitting and may underperform ASFormer initially.",
      "verdict": "supported",
      "evidence": "ActionFormer GitHub/README confirms it trains/evaluates on THUMOS14 (66.83%
mAP), ActivityNet 1.3 (36.56%), EPIC-Kitchens 100, and Ego4D ? all hundreds-to-thousands of
videos ? with no few-shot or data-efficiency claim anywhere in the documentation. This validates
the contrast the recommendation rests on (ASFormer first, ActionFormer as the later A/B
challenger once the corpus grows). The secondary 'arbitrary input feature dimension via learned
projection' sub-claim is plausible (multiple feature types ? I3D/TSP/SlowFast/Omnivore/EgoVLP ?
are supported) but the README does not verbatim confirm a learned projection layer; the findings
already flag low-dim trajectory input as 'mechanically supported but unproven,' which is
honest.",
      "source": "https://github.com/happyharrycn/actionformer_release"
    },
    {
      "claim": "LICENSE TRAP: the original yabufarha/ms-tcn repo is MIT + Commons Clause (non-
commercial), while sj-li/MS-TCN2 (the MS-TCN++ repo) is clean MIT ? so vendor sj-li/MS-TCN2,
never yabufarha/ms-tcn, for commercial use.",
      "verdict": "supported",
      "evidence": "Fetched both LICENSE files directly. yabufarha/ms-tcn/LICENSE contains the
Commons Clause: 'the License does not grant to you, the right to Sell the Software' with the
standard Commons Clause 'Sell' definition (includes hosting/consulting for a fee) ? this makes
it non-commercial. sj-li/MS-TCN2/LICENSE is plain MIT ('Permission is hereby granted, free of
charge ... without restriction'), copyright (c) 2019 yabufarha, with no Commons Clause. The
findings' directional guidance is correct and is a real, easy-to-miss commercial trap worth
recording in COMMERCIALIZATION.md.",
      "source": "https://github.com/yabufarha/ms-tcn/blob/master/LICENSE and
https://github.com/sj-li/MS-TCN2/blob/master/LICENSE"
    },
    {
      "claim": "RFT/GRPO is dramatically more label-efficient than SFT for temporal grounding
? VideoChat-R1 reports +31.8 on temporal grounding with 'limited samples' ? so a small, hand-
verified label set is the RIGHT asset for RL, not a liability.",
      "verdict": "supported",
      "evidence": "Primary paper arXiv:2504.06958 (VideoChat-R1) abstract states RFT is 'a
data-efficient method to enhance video reasoning on specific tasks' and reports, vs
Qwen2.5-VL-7B, a temporal-grounding boost of +31.8 (and +31.2 object tracking) via multi-task
RFT with 'limited samples.' The +31.8 figure and the 'limited samples' / data-efficiency framing
are verbatim accurate. Caveat: this is an absolute mAP/metric delta on benchmark VTG data, not a

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demonstration on amateur single-camera rally footage ? the findings already flag the broadcast?amateur domain-transfer gap.",

"source": "https://arxiv.org/abs/2504.06958"

},

{

"claim": "A quality-filtered 13K cold-start set beats an unfiltered 56K set for video temporal grounding (VTG-R1), evidencing quality-over-quantity for our small clean corpus.",

"verdict": "supported",

"evidence": "Primary paper (arXiv:2507.18100, 'Datasets and Recipes for Video Temporal Grounding via Reinforcement Learning') confirms VTG-R1 (initialized from VTG-Coldstart-13K) significantly outperforms VTG-R1-U (unfiltered 56K cold-start), and that without a high-quality cold start, models on unfiltered data struggle to learn. The 13K>56K quality-over-quantity claim is accurately represented.",

"source": "https://arxiv.org/abs/2507.18100 (arxiv.org/html/2507.18100v1)"

},

{

"claim": "Qwen license is clean per-size: Qwen2.5-VL-7B = Apache-2.0 (commercial-clean); Qwen2.5-VL-3B = Qwen Research License (non-commercial, REJECT); Qwen2.5-VL-72B = custom license with a 100M-MAU cap (REJECT). Qwen3-VL-4B = Apache-2.0 with explicit Text?Timestamp Alignment.",

"verdict": "supported",

"evidence": "Verified each per-checkpoint against HF: Qwen2.5-VL-7B-Instruct card = 'License: apache-2.0'. Qwen2.5-VL-3B LICENSE = 'Qwen RESEARCH LICENSE AGREEMENT ... FOR NON-COMMERCIAL PURPOSES ONLY' (commercial requires a separate request) ? non-commercial as claimed. Qwen2.5-VL-72B LICENSE = 'Qwen LICENSE AGREEMENT' with the clause 'if ... your product or service has more than 100 million monthly active users, you shall request a license from us' ? the 100M-MAU cap is real. Qwen3-VL-4B-Instruct = Apache-2.0 and documents 'Text?Timestamp Alignment ... precise, timestamp-grounded event localization.' All per-size license gates are accurate and load-bearing for the owned-base feasibility.",

"source": "https://huggingface.co/Qwen/Qwen2.5-VL-7B-Instruct ; /Qwen2.5-VL-3B-Instruct/blob/main/LICENSE ; /Qwen2.5-VL-72B-Instruct/blob/main/LICENSE ; /Qwen3-VL-4B-Instruct"

},

{

"claim": "InternVideo2 encoder weights are commercially usable (InternVideo2-Stage2_6B = MIT), and VideoMAE checkpoints are CC-BY-NC (REJECTED for commercial use) ? so a small InternVideo2 encoder + light grounding head is the only owned-model PoC that is data-realistic at today's ~128 rallies.",

"verdict": "supported",

"evidence": "HF cards confirm: OpenGVLab/InternVideo2-Stage2_6B = 'License: mit'; MCG-NJU/videoMAE-base = 'License: cc-by-nc-4.0' (non-commercial). Both license facts are exactly as stated, and they are the basis for routing VideoMAE to 'code-only / reject weights' and InternVideo2 to 'usable encoder.' The downstream 'data-realistic at 128 rallies' judgement is an engineering inference, not a sourced number, but the license premises are solid.",

"source": "https://huggingface.co/OpenGVLab/InternVideo2-Stage2_6B and https://huggingface.co/MCG-NJU/videoMAE-base"

},

{

"claim": "Gemma 4 is Apache-2.0 (a genuine reversal from Gemma 1/2/3's restrictive terms) AND processes video natively ? directly clearing the Gemma-3 rejection guardrail and being the best structural match for the Option-B frames+audio+video fusion.",

"verdict": "supported",

"evidence": "Primary Google source (ai.google.dev/gemma/docs/core/model_card_4) confirms BOTH: 'License: Apache 2.0' and native 'Video Understanding ? Analyze video by processing sequences of frames' (max 60s at 1fps), plus audio ASR on E2B/E4B/12B Unified. This DIRECTLY REFUTES the more cautious framing in the COST/LICENSING finding (#5), which marked Gemma 4 as AMBER, 'confirmed only via secondary/editorial sources,' and 'image-multimodal with NO confirmed VIDEO capability.' Both the Apache-2.0 license AND native video are now confirmable from the primary Google model card ? the AMBER caveat is outdated and should be upgraded.",

"source": "https://ai.google.dev/gemma/docs/core/model_card_4"

},

{

"claim": "Gemma 4 ships in the sizes 4B and 12B (used throughout the findings, e.g. 'Gemma 4 4B/12B', '4B variant is genuinely on-device').",

"verdict": "refuted",

"evidence": "Primary Google model card lists the actual Gemma 4 sizes as E2B, E4B, 12B

Unified, 26B A4B (MoE), and 31B Dense ? there is no plain '4B' dense model; the small variants are E2B/E4B (effective-parameter naming). Secondary coverage (ghacks) lists 2B/4B/26B/31B. The repeated '4B/12B' shorthand in the findings is imprecise: the on-device small model is E4B (not a '4B'), and audio is limited to E2B/E4B/12B-Unified (not the 26B/31B). This does NOT change the recommendation (a small on-device Gemma 4 variant with native video+audio exists), but the exact size labels and the per-size audio availability should be corrected before any base lock to avoid pulling the wrong checkpoint.",

"source": "https://ai.google.dev/gemma/docs/core/model_card_4"

},
{

"claim": "Together AI managed LoRA fine-tuning of models ?16B is ~\$0.48/M tokens, making a 5k-20k clip?QA-pair run land in single-digit to low-tens of dollars.",

"verdict": "supported",

"evidence": "Multiple 2026 pricing sources corroborate LoRA SFT ?16B = \$0.48/M tokens, with the tier ladder up to \$3.20/M (full FT 70-100B) and no job minimum; cost = (train tokens × epochs + eval tokens) × rate. The headline \$0.48/M figure is accurate. NOTE: I could not load Together's own docs page verbatim (it redirects to the live pricing page) ? the \$0.48/M is confirmed via several independent 2026 aggregators rather than a single fetched primary page, and the findings' own caveat that VIDEO/multimodal token pricing is unverified remains valid (the \$0.48/M is a text-LLM LoRA rate; video token counts can balloon cost).",

"source": "Together AI pricing (docs.together.ai/docs/fine-tuning-pricing ? together.ai/pricing); corroborated by 2026 aggregators"

},
{

"claim": "Timestamp/point supervision is a cheap-label data-efficiency lever ? timestamp-supervised models reach near-full-supervision accuracy (e.g. 64.1% vs 68.0% on Breakfast) at a fraction of the labeling effort.",

"verdict": "supported",

"evidence": "Independently corroborated: the timestamp-supervision literature reports 'timestamp supervision halves the gap to full supervision (Acc=64.1% vs 68.0% on Breakfast)' and that only ~0.032% of frames need labels. The exact 64.1%/68.0% pair the findings cite is accurate. The companion '~2/3 annotation time, SkipTag ~1/3' figure from the 2018 ECCV 'What do' paper could NOT be parsed (PDF binary did not extract), so that specific time-fraction is uncertain ? but the core near-full-accuracy-from-sparse-labels claim is well supported. Domain caveat (action-segmentation user studies, not rally detection) is already noted by the findings.",

"source": "Timestamp-supervised TAS literature (arXiv:2103.06669 and survey sources); ECCV'18 'What do' (arXiv could not be parsed)"

},
{

"claim": "Compute feasibility: an 8GB RTX 2070 Super CANNOT train a 7B-class video VLM (QLoRA on video lands ~20-24GB) ? it is inference/eval-only; training MUST be a rented ~24-48GB GPU. By contrast ASFormer/MS-TCN++ (~0.8-1.1M params) train locally in minutes.",

"verdict": "supported",

"evidence": "Community fine-tuning evidence (Unsloth, Labellerr, HF discussions) consistently shows 7B-VLM inference needs ~24GB and QLoRA fine-tuning on a 24GB card is 'feasible but tight' ? well above 8GB, confirming the 2070S is inference-only for the 7B class and rented 24GB+ is required for training. The TAS-head local-training claim follows directly from the verified ~1.1M param count (ASFormer 1.134M). Caveat the findings already make: VRAM figures are from community write-ups on text/OCR LoRA, not a controlled benchmark on real 1080p/60fps rally video ? true video-sequence VRAM can exceed these, so 24GB is a floor, not a guarantee.",

"source": "huggingface.co/Qwen/Qwen2.5-VL-7B-Instruct/discussions/18 ; Unsloth/Labellerr fine-tuning write-ups"

},
{

"claim": "VideoLLaMA3 is usable as an owned commercial base because its code is Apache-2.0.",

"verdict": "refuted",

"evidence": "Confirmed the findings' OWN weak-fit verdict (so the recommendation is correct), but the underlying reason is even stronger than stated and intersects a hard project guardrail: the released VideoLLaMA3 weights are 'a research preview intended for non-commercial use ONLY, subject to the model Licenses of Qwen, Terms of Use of the data generated by OpenAI and Gemini.' So (a) the weights are NOT shippable, and (b) they were trained on OpenAI/Gemini-generated data ? which directly violates the project's non-negotiable 'paid LLMs are never a

```

training-data source' guardrail. Correctly relegated to method-reference-only; flagging because
the paid-LLM-data entanglement is a sharper disqualifier than 'non-commercial weights' alone.",
    "source": "https://github.com/DAMO-NLP-SG/VideoLLaMA3"
  }
],
"corrections": [
  "Gemma 4 should be UPGRADED from AMBER to confirmed-clean for the owned-base decision. The
  COST/LICENSING finding (#5) marks Gemma 4 as AMBER ? 'Apache-2.0 confirmed only via
  secondary/editorial sources, no confirmed VIDEO capability.' Both are now confirmable from the
  PRIMARY Google model card (ai.google.dev/gemma/docs/core/model_card_4): 'License: Apache 2.0'
  AND native 'Video Understanding ? Analyze video by processing sequences of frames' (60s @ 1fps)
  plus audio on E2B/E4B/12B. The internal contradiction across findings (#3 'strong/native video'
  vs #5 'AMBER/no video') resolves in favor of #3. The remaining honest caveat is that Gemma 4's
  timestamp-EMISSION format for grounding is less documented than Qwen's, not that video is
  unsupported.",
  "Gemma 4 size labels are wrong throughout. The findings repeatedly say 'Gemma 4 4B/12B'
  and 'the 4B variant.' Primary card lists E2B, E4B, 12B Unified, 26B A4B (MoE), 31B Dense ? there
  is no plain dense '4B'; the on-device small model is E4B, and native audio is limited to
  E2B/E4B/12B-Unified (NOT 26B/31B). Correct the size/audio mapping before any base lock so the
  right checkpoint is pulled.",
  "ASFormer's 'from scratch on tiny data' is real but should be stated precisely: the
  ~1.1M-param SEGMENTATION HEAD trains from scratch on GTEA's 28 videos, but its INPUT is
  Kinetics-pretrained I3D 2048-d features. This matters because the recommendation feeds raw
  ~3-4-dim WASB trajectory instead of pretrained 2048-d features ? an off-the-validated-path
  substitution that the small-data result does NOT directly cover. The findings' caveat flags
  this; keep it front-and-center as the single biggest empirical unknown for the ASFormer-first
  plan.",
  "Together's $0.48/M LoRA rate is a TEXT-LLM rate. It is confirmed for ?16B text LoRA, but
  the findings' own open question stands: VIDEO/multimodal clip?QA token counts can balloon far
  beyond text, so the 'single-digit-to-low-tens of dollars' estimate for a 5k-20k clip?QA run is
  unverified for video and should be re-benchmarked on real frame counts before being trusted for
  budgeting.",
  "VideoLLaMA3's disqualifier is sharper than 'non-commercial weights': its released weights
  were trained on OpenAI/Gemini-generated data, which directly trips the project's hard 'no paid-
  LLM outputs as training data' guardrail. Even re-training from the Apache Qwen2.5 base, do NOT
  reuse VideoLLaMA3's instruction data. Method-reference-only is the correct call, but record the
  paid-LLM-data reason explicitly.",
  "None of the verified facts change the headline architecture recommendations (ASFormer-
  first TAS over WASB features; Qwen3-VL-4B / Qwen2.5-VL-7B / Gemma 4 as clean owned bases;
  RFT/GRPO as the label-efficient Option-B route). The few-match feasibility claims and per-size
  license gates all hold against primary sources. The only recommendation-affecting change is the
  Gemma 4 upgrade (AMBER?clean, native video confirmed), which makes it a stronger Option-B
  candidate than finding #5 concluded ? and the persistent caveat that low-dim trajectory input +
  amateur-footage domain transfer remains empirically unproven (the actual ship/no-ship gate is an
  A/B on the golden set, which cannot be run in this environment)."
```

2. assistant (2026-06-08T13:17:45.122Z)

I'll synthesize the research, verification corrections, and grounding into a single decision-useful study. Let me first check the existing strategy docs to ensure roadmap alignment, then produce the structured output.

3. user (2026-06-08T13:17:45.928Z)

```

docs\EVALUATION.md
docs\TRACKNET_WSL_SETUP.md
docs\GOLDEN_SET_IMPLEMENTATION.md
docs\GOLDEN_SET_PHASE1_VIDEOS.md
docs\HOW_RALLY_DETECTION_WORKS.md
docs\CODE_MAP.md
docs\DEFERRED_HARDENING_PLAN.md
docs\PMF_MARKET_RESEARCH.md
```


docs\ROADMAP.md
docs\COMMERCIALIZATION.md
docs\PLATFORM_ARCHITECTURE.md
docs\BACKLOG.md
docs\GOLDEN_SET_VENDORING.md
docs\README.md
docs\VIDEO_RECORDING_GUIDELINES.md

4. user (2026-06-08T13:17:46.055Z)

docs\EVALUATION.md
docs\TRACKNET_WSL_SETUP.md
docs\GOLDEN_SET_IMPLEMENTATION.md
docs\GOLDEN_SET_PHASE1_VIDEOS.md
docs\HOW_RALLY_DETECTION_WORKS.md
docs\archives\COMMERCIAL_READINESS_REVIEW.md
docs\archives\MONETIZATION_AUDIT.md
docs\archives\README.md
docs\CODE_MAP.md
docs\DEFERRED_HARDENING_PLAN.md
docs\PMF_MARKET_RESEARCH.md
docs\ROADMAP.md
docs\archives\checkpoints\2026-06-eval-baseline\README.md
docs\archives\checkpoints\2026-06-logging-reporting\README.md
docs\archives\checkpoints\2026-06-scaffolding-complete\README.md
docs\archives\decisions\DECISIONS.md
docs\archives\past_projects\low-regret-moves\LOW_REGRET_MOVES_PLAN.md
docs\archives\past_projects\README.md
docs\archives\research\AUDIO_E1_FINDINGS.md
docs\archives\research\AUDIO_RALLY_DETECTION_PLAN.md
docs\archives\research\RALLY_DETECTION_RESEARCH.md
docs\COMMERCIALIZATION.md
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docs\BACKLOG.md
docs\GOLDEN_SET_VENDORING.md
docs\README.md
docs\VIDEO_RECORDING_GUIDELINES.md
docs\archives\checkpoints\2026-06-strategy-foundation\README.md
docs\archives\quality-iterations\2026-06-human-gt-tuning\README.md
docs\archives\quality-iterations\2026-06-initial\README.md
docs\archives\quality-iterations\README.md

5. assistant (2026-06-08T13:17:48.646Z)

Let me read the key roadmap docs to ensure alignment with the existing strategy framing.

6. user (2026-06-08T13:17:49.052Z)

```
1      # Commercialization Tracker
2
3      **Status:** ? LIVING ? track this in parallel with product/quality work.
4      **Owner:** avidullu . **Last refreshed:** 2026-06-07
5
6      **Purpose.** Single source of truth for the **commercial** dimension of the project:
7      licensing/legal posture, codec & output economics, hosting unit-economics, the
8      golden-set vendoring spend, packaging/distribution, and the product-market-fit
9      risks that gate monetization. Updated as decisions land ? see the
10     [Changelog](#changelog) at the bottom.
11
12     **Merged from / supersedes:**
13     - `archives/MONETIZATION_AUDIT.md` (2026-05-30 deep-research licensing & economics
14       audit) ? kept as the **research-of-record** (per-claim verification transcript,
15       primary sources). This doc carries its *current* status forward.
16     - `archives/COMMERCIAL_READINESS_REVIEW.md` (code-health + PMF assessment) ?
```

```

17     archived; its commercial findings + the quality-phase priorities live here now.
18
19     **Premise (unchanged, verified):** the product is a **closed-source hosted API** ?
20     users upload videos to a cloud NVIDIA GPU VM and get highlight reels back. We never
21     distribute code/binaries, so GPL/LGPL *distribution* copyleft is not triggered ? but
22     **AGPL §13 (network clause)**, **metered-API terms**, and **video-codec patent
23     royalties** all still apply.
24
25     ---
26
27     ## 1. Commercialization dashboard
28
29     Legend: ? done · ? in progress / watch · ? blocker / needs action · ?? needs legal
30     counsel · ? future
31
32     Priority: **P0** (do first) · **P1** · **P2** (lowest). Rows are grouped **Pending ? In-
33     flight ? Completed**, each sorted by priority. (C# IDs are stable identifiers ? they don't imply
34     order.)
35
36     | # | Workstream | Pri | Status | Where it stands | Next action / owner decision |
37     |---|---|---|---|---|---|
38     | | **? PENDING (needs action / not started)** | | | | |
39     | C5 | **Output codec royalties** | **P0** | ? | Highlights are still encoded **H.264 +
40     AAC** (`libx264` in `backend/pipeline/stitcher/compiler.py`) ? encode-side AVC patent exposure
41     remains. | Switch output to **AV1 + Opus** (royalty-free) on an **Ada-class GPU (L4/L40)** for
42     HW encode. See §4. Decision: provision Ada-class GPU? |
43     | C9 | **Packaging / distribution** | **P0** | ? | No `pyproject.toml`; the diagnostic
44     `setup.py` shadows the build system. | Design in `DEFERRED_HARDENING_PLAN.md` (#7) ? **awaiting
45     your approval** before code. |
46     | C12 | **SaaS foundations (async/auth/tenancy)** | **P1** | ? | Sync `/api/process`
47     blocks on long videos; no auth/tenancy. | Multi-user/platform strategy in
48     `PLATFORM_ARCHITECTURE.md` (pluggable storage/compute, tenancy, orchestration); async slice =
49     `DEFERRED_HARDENING_PLAN.md` (#16). |
50     | | **? IN-FLIGHT (in progress / watch)** | | | | |
51     | C4 | **Runtime AI strategy (A/B/C + owned model)** | **P0** | ? | Live = **C) Gemini
52     2.5 Flash, paid tier**. Long-term = **own a self-hosted generative video-QA model** (fine-tuned
53     open VLM) as primary, **paid Gemini always-ready as fallback** via the provider registry. | Make
54     "provider registry + paid fallback" a core principle; build the owned model **after** the
55     scaffolding/flywheel are solid. See `PLATFORM_ARCHITECTURE.md` §5A. |
56     | C13 | **Beachhead validation** | **P0** | ? | Market research
57     (`PMF_MARKET_RESEARCH.md`) recommends **India badminton academies** as the #1 beachhead.
58     Unvalidated. | Cheapest tests: 10?15 academy/coach discovery interviews; rally-detection P/R
59     benchmark on 100+ real amateur clips; fake-door pricing (B2C ~$99/yr, B2B academy ~$600/yr). See
60     §6. |
61     | C8 | **Golden-set vendoring spend** | **P1** | ? | ADR-006: **Roora (Bengaluru)**
62     selected for an initial paid pilot; licensed/owned video only (no user uploads to vendors). |
63     Confirm quality-per-dollar from the pilot; then GS-3 `HumanVendorProvider`. See
64     `GOLDEN_SET_VENDORING.md`. |
65     | C3 | **Shuttle-tracking weights provenance** | **P1** | ? ?? | WASB-SBDT **code is MIT
66     (clear)**; distributed **checkpoints are academic-data-trained** ? not covered by the MIT code
67     grant. MonoTrack/YOLO-NAS banned (noncommercial). | Train-own weights (Phase-4 roadmap) **or**
68     confirm per-checkpoint terms ?? before commercial deploy. |
69     | C14 | **Owned-model base licensing + training-data policy** | **P1** | ? ?? | Owned
70     video-QA model (PLATFORM §5A) must build on **Apache-2.0-clean, video-capable** bases ?
71     **Qwen2.5-VL-7B/32B** (Apache; 3B/72B are *not*), **Qwen3-VL-4B/8B**, or **Gemma 4** ? to keep
72     weights private, gate, and monetize. **Reject Gemma 3** (viral derivative clause ? permanent
73     Google PUP), Llama Vision (naming + 700M-MAU), `gpt-oss` (text-only). **Hard rule: never train
74     on paid-LLM (Gemini/OpenAI/Anthropic) outputs** (competing-model clause). The moat is the
75     **consented dataset + eval harness**, not the model. | Lock to Apache-2.0; **open-model teachers
76     only**; separate explicit **training opt-in**; **exclude minors** from the training pool; per-
77     clip consent ledger. See PLATFORM §5A/§7. |
78     | C6 | **Inbound HEVC decode** | **P2** | ? ?? | Decoding users' GoPro H.264/**HEVC**
79     uploads is unavoidable; HEVC pools are fragmented + highest-uncertainty. | Low/uncertain
80     residual; confirm decode-side exposure with counsel before launch. |
81     | C7 | **Hosting unit-economics** | **P2** | ? | Per-video cost is *cheap* for Gemini
82     (~$0.05 input for a 10-min clip); risk is volume × latency, not unit price. Self-host/offline

```

costs unmeasured. | ****Benchmark \$/video & latency on the target VM**** for A/B/C before pricing. See §5. |

46 | C10 | ****Value layer (product surface)**** | ****P2**** | ? | Minimal exports shipped (EDL/JSON/CSV + `basic\_stats`, PR #51). | Deepen: per-rally tags/notes that feed golden, per-player/match stats, clip gallery. See §6. |

47 | C11 | ****Input-quality enforcement**** | ****P2**** | ? | Per-video observability reports surface a customer verdict + footgun flags incl. audio-present (PR #50). | Turn more `VIDEO\_RECORDING\_GUIDELINES` dimensions into validators/report rules. |

48 | C2 | ****`supervision` ByteTrack pin**** | ****P2**** | ? | Pinned `<0.30.0` (deprecation ahead). | Migrate before lifting the pin (BACKLOG). Not a legal risk, a maintenance one. |

49 | | ****? COMPLETED**** | | | |

50 | C1 | ****AGPL / copyleft removal**** | ? | ? | ADR-005 shipped: AGPL `ultralalytics` YOLOv8 replaced with ****torchvision Faster R-CNN (BSD) + `supervision` ByteTrack (MIT)****. No AGPL code/weights in the stack. | Done. Keep new deps permissive. |

51

52 ---

53

54 ## 2. Licensing & dependency posture (refreshed risk table)

55

56 Verdicts updated against the current stack (`requirements.txt` + code). Original

57 2026-05-30 verdicts and the 25-claim verification transcript are in

58 `archives/MONETIZATION\_AUDIT.md`.

59

60	Dependency	License	SaaS obligation?	Current verdict	Notes
61	---	---	---	---	---
62	~~`ultralalytics` + `yolov8n.pt`~~	AGPL-3.0	Yes (§13)	? **REMOVED**	Replaced per ADR-005 ? no longer in the stack.
63	**torchvision** detectors (Faster R-CNN/RetinaNet)	BSD-3	No	? LIVE	The chosen replacement; person = COCO class (see `yolo_hybrid.py` `_PERSON_CLASS_ID`).
64	**`supervision`** (ByteTrack)	MIT	No	? LIVE	Pinned `<0.30.0` ? migrate before bump (C2).
65	**WASB-SBDT** (shuttle)	MIT (code)	No	? code clear	Weights provenance open (C3).
66	**MonoTrack / YOLO-NAS**	noncommercial	? ?	BANNED	Do not use.
67	**Gemini API** (paid tier)	service terms	Paid: no training/review	? LIVE WATCH	Must bill the paid tier to keep user video private (C4).
68	H.264 / HEVC / AAC codecs	patent pools	royalties on encode/decode	? ??	Separate from ffmpeg's license. Encode-side open (C5); decode-side residual (C6).
69	ffmpeg/ffprobe, google-genai, torch, opencv, numpy, fastapi/uvicorn/pydantic/jinja2/dotenv	LGPL/Apache/BSD/MIT	No	? CLEAR	Invoked as subprocess / imported; all permissive.
70	**`obsreport`** (reports)	permissive (private)	No	? soft dep	Pin the `git+ssh@vX` tag once the repo is published (BACKLOG).

71

72 ****Bottom line:**** the two hard blockers from the audit (Ultralytics; banned weights)

73 are resolved/avoided. Remaining commercial-legal items are ****codec royalties (C5/C6)****

74 and ****weights provenance (C3)****, plus the metered-API privacy posture (C4).

75

76 ---

77

78 ## 3. Runtime AI strategy ? A/B/C

79

80 The semantic rally-boundary step is the one external/metered dependency. Three paths

81 (from the audit), with current standing:

82

- 83 - **** (A) Eliminate ? offline learned segmenter. **** ActionFormer (MIT) / our own learned classifier on `candidate\_segments.stats`. ****Best steady-state margin**** (no per-call cost, no external dep) and matches the Phase-4 roadmap (substrate = WASB shuttle-motion, decided in ADR-004). ****Most engineering.**** ? ****target****
- 84 - **** (B) Self-host VLM. **** Qwen2.5-VL-7B/32B (Apache-2.0) or InternVL2.5-8B (MIT). No per-call fees; pays in GPU VRAM (~16?24 GB for 7B). ?? Qwen 3B is noncommercial.
- 85 - **** (C) Keep Gemini 2.5 Flash (paid). **** Fastest, cheap per video, video-native; ongoing metered cost + external dependency. ****? current live path.****

86

87

88

89

90

91

92 Hosted alternatives for (C) comparison: Anthropic Claude (no training on

```

93 inputs/outputs; customer owns outputs; **no native video** ? weaker here);
94 OpenAI (no training by default; ZDR available); Gemini remains strongest video-native.
95
96 ---
97
98 ## 4. Codec / output economics (the live gap ? C5)
99
100 **The trap:** ffmpeg being free (LGPL) does **not** grant the *patent* rights to the
101 codecs it implements. Royalties attach to the act of encode/decode.
102
103 - **Encode side (we control it):** highlights currently emit **H.264/AAC**. AVC (Via LA)
104   is first 100k units/yr royalty-free with a permanent "free to end-users over the
105   internet" provision ? a small SaaS likely falls under thresholds, but ?? confirm.
106 - **Mitigation:** emit **AV1 (SVT-AV1, BSD + AOMedia royalty-free patent grant) + Opus
107   (royalty-free)**. NVENC AV1 HW-encode needs **Ada-class GPUs (L4/L40)**; Ampere/Turing
108   must use **SVT-AV1 (CPU)**.
109 - **Decode side (unavoidable, C6):** inbound GoPro **HEVC** decode ? fragmented pools,
110   highest uncertainty; low residual.
111
112 **Recommendation:** provision an **Ada-class GPU (L4/L40)** and standardize highlight
113 output on **AV1 + Opus** ? removes the encode-side royalty question almost entirely.
114 *Implementation lives in `backend/pipeline/stitcher/compiler.py` (codec args).*
115
116 ---
117
118 ## 5. Hosting unit-economics (C7)
119
120 Reference workload: ~30-min 1080p match on a cloud NVIDIA GPU VM.
121
122 - **(C) Gemini 2.5 Flash (paid):** input $0.30/1M tokens, output $2.50/1M; video ~263
123   tok/s ? 10-min clip ? 158k input tokens ? **~$0.05 input**. Cheap per unit; the
124   economic risk is **volume × latency**. Flash-Lite is cheaper; Pro ~10×.
125 - **(B) Self-host Qwen2.5-VL-7B:** $0 API; cost = GPU-hours / throughput; needs ?24 GB
126   VRAM (L4/A10). $/video vs Gemini is **open ? measure**.
127 - **(A) Offline ActionFormer/learned:** cheapest at inference; cost = one-time training
128   + light GPU. **Best steady-state margin.**
129
130 > The A/B/C $/video comparison depends on GPU instance + batch throughput ?
131 > **benchmark on the target VM before pricing the product.**
132
133 ---
134
135 ## 6. Product-market fit & the quality/data phase
136
137 ### Beachhead (from market research, 2026-06-07 ? see
138 [ `PMF_MARKET_RESEARCH.md` ](PMF_MARKET_RESEARCH.md))
139 The static-camera constraint is a **B2B filter**: the buyer must already record from a
140 fixed angle, which points away from the median casual player and toward coaches/clubs.
141 Ranked beachheads:
142 1. **Badminton coaching academies & serious players, India** *(strongest)* ? 20M+
143   players,
144   formalizing academy ecosystem (Khelo India), genuine fixed-angle recording, coaches
145   who already monetize, badminton whitespace vs incumbents. Sell to the **academy**,
146   not
147   the individual; needs INR-tuned pricing.
148 2. **US/global pickleball players & clubs** ? fastest-growing sport (19.8M US, +45.8%
149   YoY)
150   but **SwingVision already owns single-camera tennis/pickleball AI** (USD 179.99/yr) ?
151   head-to-head, not greenfield.
152 3. **Badminton academies/clubs in East/SE Asia + Denmark** ? highest density; China
153   needs
154   local hosting/partners; Denmark/EU small but high-WTP, easy to pilot.
155
156 Pricing anchors *(estimate)*: software-only BYO-camera **B2C USD 60?180/yr**, **B2B
157 academy/club USD 300?1,500/yr** per location (undercuts hardware-bundled Veo/Pixellot/

```

```

153 Spiideo; comparable to SwingVision software-only).
154
155 ### PMF risks (from the readiness review)
156 - **Capture quality is the gate.** Accuracy is capture-conditional; broadcast/handheld
157 footage is a failure mode. The wedge (static, tripod, full-court) is defensible but
158 narrows the addressable market ? enforce/guide it (C11, `VIDEO_RECORDING_GUIDELINES`).
159 - **Value beyond raw segments is table stakes.** Time-saved editing alone is weak;
160 per-rally tags, stats, and a correction loop are what make it sticky (C10).
161 - **Golden labels are the recurring blocker** across the offline segmenter, audio
162 fusion, and measurable quality ? the highest-leverage investment (C8).
163
164 ### Refined priority order (quality/data phase ? "focus on quality once new videos
land")
165 Carried from the readiness review; commercialization items above are tracked in
parallel.
166
167 1. **Golden-set expansion + data-flywheel UX** (highest with new labeled videos):
168 ingest the new batch, expand calibration/CV/regression baselines, make label
169 contribution + safe retraining easy and non-contaminating (eval/training separation).
170 2. **Hybrid unification + observability hooks** (deferred item 7, now higher): extract
171 shared logic from `tracknet_hybrid`/`wasb_hybrid` before heavy AI timing/metrics
land.
172 3. **Detector/windowing quality iteration** with new data: re-tune velocity/merge/
173 min-duration + `rally_gate`; validate WASB on real video; trial audio fusion
(ADR-007).
174 4. **Deeper value layer** (beyond minimal exports, C10): per-rally tagging/notes ?
175 `candidate_segments`, per-player/match stats, clip gallery, human-correction loop.
176 5. **Input-quality enforcement & guidance polish** (C11): more guideline dimensions ?
177 validators/report rules; "your video is good enough" UX.
178 6. **Multi-sport maturation + recording best practices.**
179 7. **SaaS / multi-user / async foundations** (C12) ? lower until monetization is firm.
180 8. **Polish & distribution** (C9): installer story, first-run path, codec/output,
economics.
181
182 **Recommendation:** with the mid-week labeled-video batch, spend ~80% on #1 + #3
183 (flywheel + quality iteration on real data), then unification (#2) and deeper value
(#4).
184
185 ---
186
187 ## 7. Open questions needing owner / counsel sign-off ??
188
189 1. **Ultralitics Enterprise price** ? only relevant if a permissive detector ever proves
190 insufficient (fallback). Quote-only.
191 2. **Per-checkpoint / dataset licenses** for WASB / TrackNetV3 / any VLM ? train-own or
192 confirm each (C3).
193 3. **Exact Via LA (AVC) / Access Advance (HEVC) rates & free thresholds** ? confirm live
194 figures with counsel before relying on the "small SaaS under threshold" reading
(C5/C6).
195 4. **Real per-video GPU cost & latency** of self-hosted VLM vs Gemini paid at expected
196 throughput (C7).
197 5. **Provision Ada-class GPU (L4/L40)** for HW AV1 encode? (C5 ? gates the codec
switch.)
198
199 ---
200
201 ## Changelog
202
203 - **2026-06-07** ? Added the market-research-of-record (`PMF_MARKET_RESEARCH.md`):
204 ranked beachhead (India badminton academies #1), pricing benchmarks, and a C13
205 "beachhead validation" workstream. Wired the summary into $6.
206 - **2026-06-07** ? Created this living tracker. Merged the 2026-05-30 monetization
207 audit + the commercial-readiness review. **Refreshed since the audit:** C1 AGPL
208 removal now ? shipped (was the #1 blocker); added C5 as a ? (output is still H.264,
209 the audit's AV1 recommendation is not yet implemented); C9 packaging tied to

```

```

210     `DEFERRED_HARDENING_PLAN.md`; reflected PR #50/#51 (input-quality reports, value-layer
211     exports). Archived `COMMERCIAL_READINESS_REVIEW.md`.
212
213     ---
214
215     ## See also
216     - `PMF_MARKET_RESEARCH.md` ? market-study-of-record (sizing, segments, competitors,
217     beachhead; cited).
218     - `archives/MONETIZATION_AUDIT.md` ? licensing/economics research-of-record (sources +
219     verification transcript).
220     - `archives/COMMERCIAL_READINESS_REVIEW.md` ? superseded assessment (history).
221     - `archives/decisions/DECISIONS.md` ? ADR-005 (permissive stack), ADR-006 (vendor
222     process).
223     - `ROADMAP.md` ? Phase-4 offline segmenter (the C4 "(A) eliminate" path).
224     - `DEFERRED_HARDENING_PLAN.md` ? packaging (C9), async (C12) designs (pending approval).
225     - `GOLDEN_SET_VENDORING.md` ? vendor process + Roora pilot (C8).
226     - `BACKLOG.md` ? `supervision` migration (C2), `obsreport` pin, codec follow-ups.
227
228

```

7. user (2026-06-08T13:17:49.955Z)

```

1     # Platform Architecture ? Multi-User "Assistant Coach"
2
3     **Status:** ? Strategy / research ? for review. No code yet.
4     **Date:** 2026-06-07 . **Owner:** avidullu
5
6     **Purpose.** Map the path from today's **single-user, local** tool to a **multi-tenant
7     "assistant coach" platform**: users upload videos, get highlights **stored and
8     browsable by history**, and choose **where their data lives** (Google Drive / cloud
9     bucket / local) and **where the compute runs** (our GPU / their VM). The organizing
10    principle is to **decouple the layers** ? identity, ingestion, storage, orchestration,
11    execution/compute, the detection+highlight pipeline, and data-serving ? behind small
12    **pluggable interfaces**, reusing the registry pattern the codebase already uses for
13    segmenters/providers/validators/sports/annotations.
14
15    **Product framing (the north star *and* the v1 line).** This is an **assistant coach**,
16    not just a clipper ? but the strategy is **phased and unambiguous about scope**:
17    - **v1 (concrete, near-term).** **rally detection ? highlight-reel compilation ? per-
18    user
19    history ? a coach/user correction loop that feeds the golden set** ("assistant-coach
20    *lite"). Essentially today's pipeline made multi-user ? this is the real v1
21    deliverable.
22    - **Mid-term:** **rich rally/shot semantic metadata** (shot type, rally outcome) and the
23    **owned video-QA model** ($5A; $6 P4?P5).
24    - **North-star vision (strictly future ? *not* v1):** **gait / biomechanics, per-athlete
25    identifying templates, and deep coaching feedback under a supervising coach** ($6 P6).
26    Inspiring, but explicitly deferred ? and the part that raises the **biometric**
27    privacy
28    bar ($7).
29
30
31    The buyer is the **academy/coach** (beachhead in
32    [`PMF_MARKET_RESEARCH.md`](PMF_MARKET_RESEARCH.md)); **power users self-serve**. Two
33    consequences ripple through the design: (1) the data model keeps an **optional coach ?
34    athlete overlay over a generic core** ($4.1) ? not baked in; (2) gait/biometrics is
35    **special-category data**, gated and deferred ($7).
36
37
38    > **Competitive note (validated by research):** none of Veo / Hudl / SwingVision /
39    > Spiideo offer true BYO-storage or BYO-compute ? they're vendor-cloud SaaS
40    > (SwingVision uniquely runs *edge* compute on the user's iPhone). So **"your video,
41    > your storage, your compute"** is a genuine differentiator for privacy-sensitive
42    > academies ? and SwingVision proves user-side compute is commercially viable. All four
43    > share the same **teams ? athletes ? clips** data model, which we adopt ($4).
44
45
46    > Scope: this is a **direction-setting doc**. It does not start implementation; the
47    > first infra slice (an async job model) is the on-hold #16 in

```

```

42 > [ `DEFERRED_HARDENING_PLAN.md` ](DEFERRED_HARDENING_PLAN.md). Commercial constraints
come
43 > from [ `COMMERCIALIZATION.md` ](COMMERCIALIZATION.md) (C4 paid-LLM no-train; C12 SaaS
44 > foundations) and `archives/MONETIZATION_AUDIT.md`.
45
46 ## 1. Current state & gaps (what we're evolving from)
47
48 | Concern | Today | Gap for multi-user |
49 |---|---|---|
50 | Identity / tenancy | none | no users, orgs, auth, or isolation |
51 | Storage | local `filepath` string; output to `output_dir` | no abstraction; can't BYO
Drive/bucket; raw video must be on the box |
52 | Data model | `videos` keyed by file **MD5**, `play_segments`, `events`,
`candidate_segments` (SQLite) | no owner/athlete link; **highlights aren't a first-class
persisted entity** (compile writes a file and returns a path); no per-user history |
53 | Execution | pipeline runs **in-process, synchronously** in `/api/process` | blocks;
can't dispatch to a user's VM/GPU; no job status |
54 | Pipeline | **already pluggable** (segmenter/provider/validator/sport/annotation
registries) | reuse as-is ? this layer is the model for the rest |
55 | API | `/api/process`, `/api/query`, `/api/compile`, `/api/config`; no auth | needs
auth, per-tenant scoping, async job + history endpoints |
56
57 Grounding: `backend/infrastructure/database.py` (schema + `PRAGMA user_version`
58 migration ladder, v1), `backend/main.py` (endpoints, sync flow), the registry pattern
59 in `backend/pipeline/segmenters/base.py::SegmenterRegistry` and
60 `backend/providers/base.py`.
61
62 ## 2. Target layered architecture
63
64 Decouple into layers with a clean **control-plane vs data-plane** split ? the
65 orchestrator coordinates *metadata and jobs*; the heavy *bytes and GPU work* can live
66 elsewhere (even in the user's own cloud).
67
68 ...
69
70 CONTROL      ?????????????????????????????????????????????????????????
71 PLANE        ? Presentation / API (auth, per-tenant) ?
72              ? Identity & Tenancy (org?coach?athlete?user) ?
73              ? Orchestration / Jobs (submit?status?result) ?
74              ?????????????????????????????????????????????????????????
75              ? dispatch ? metadata
76              ?????????????????????????????????????????????????????????
77 DATA        ? Execution / Compute ? ? Data / Serving ?
78 PLANE        ? (local | BYO-VM | ? ? (per-tenant DB: ?
79              ? hosted GPU) ? ? videos, segments, ?
80              ? runs the Pipeline: ? ? highlights, jobs, ?
81              ? detect ? highlight ? ? history) ?
82              ?????????????????????????????????????????????????????????
83              ? read input / write artifacts
84              ?????????????????????????????????????????????????????????
85              ? Storage (local upload | Google Drive | S3 | ?
86              ? GCS | Azure) ? raw video + highlight outputs ?
87              ?????????????????????????????????????????????????????????
88
89 ...
90
91 The two **new pluggable seams** are **Storage** and **Compute/Execution** ; everything
92 else either exists (pipeline) or is net-new infra (identity, orchestration, serving). A
93 **model/training layer** ? the owned generative video-QA model ($5A) ? sits atop these,
94 fed by Storage (training corpus), Data (labels), and Compute (training + inference), and
95 is delivered through the existing **AI-provider registry with a paid fallback**.
96
97 ## 3. Pluggable abstractions (extend the existing registry pattern)
98
99 The codebase already proves the model: an `ABC` + a `Registry` singleton with
100 `register/get/list_available` (e.g. `SegmenterRegistry`). Add two more registries with
101 the same shape so they're plug-and-play and testable in isolation.

```

```

100
101   ### 3.1 `StorageBackend` registry
102   Replace the bare `video_path: str` that flows through the pipeline with a
103   **`StorageRef`** (a backend id + a key/URI), resolved by a backend.
104
105   ```python
106   class StorageBackend(ABC):
107       name: str
108       def open_read(self, ref: StorageRef) -> BinaryIO: ...      # stream large video
109       def download_to_local(self, ref: StorageRef, dst: str): ...  # for ffmpeg/cv2/WSL
110       def upload(self, local_path: str, ref: StorageRef): ...     # persist
highlight/artifacts
111       def signed_url(self, ref: StorageRef, ttl: int) -> str: ... # hand to LLM / browser
112       def stat(self, ref: StorageRef) -> StorageStat: ...
113       def list(self, prefix: StorageRef) -> list[StorageRef]: ...
114   ```
115
116   - **Recommended implementation:** build the backend over **`fsspec`** (one matrix ?
117     local / S3 (`s3fs`) / GCS (`gcsfs`) / Azure (`adlfs`) + Google Drive via `gdrivefs`,
118     Dropbox via `dropboxdrivefs`), reaching for **`smart_open`** as the robust large-MP4
119     byte-streaming primitive. Storage becomes one more registry. (Treat Drive/Dropbox as
120     add-ons ? less battle-tested than `s3fs`/`gcsfs`; cover with integration tests.)
121   - **Backends:** `local` (upload dir), `gdrive`, `s3`, `gcs`, `azure`. **BYO-storage**:
122     the tenant registers their own bucket/Drive; raw video **never leaves their account**.
123   - **Auth ? never persist users' long-lived keys.** For buckets, have the user create an
124     IAM role you **`AssumeRole`** into, scoped to one bucket/prefix, and mint **presigned
125     URLs** so raw video moves **browser ? the user's bucket directly, never through our
126     servers**. For Drive, OAuth with the narrow **`drive.file`** scope; for academics, a
127     **service account with domain-wide delegation**. Store any per-tenant creds encrypted.
128   - **Why it matters for the pipeline:** ffmpeg/OpenCV/WSL need a **local file**, while
the
129     LLM provider can take a **signed URL**. The backend exposes both `download_to_local`
130     (for compute) and `signed_url` (for the provider/browser), so the pipeline stops
caring
131     *where* the bytes live.
132   - **Migration path:** introduce `StorageRef` alongside the current local path; a `local`
133     backend wraps today's behavior so nothing breaks; adapters land incrementally.
134
135   ### 3.2 `ComputeBackend` / `ExecutionBackend` registry
136   Where the pipeline actually runs for a given job.
137
138   - **`local`** ? in-process / local subprocess (today's behavior; dev + self-host).
139   - **`hosted`** ? our GPU VM pool (the SaaS default).
140   - **`byo_worker`** ? the tenant runs a **pull-based worker** (a small agent) on their
own
141     VM/GPU. **Critical security property:** the worker *pulls* jobs scoped to its tenant
142     from the control plane, pulls input from the tenant's **own** storage, runs the
143     pipeline, and writes artifacts + results back ? so **the orchestrator never holds the
144     user's cloud credentials** and needs **no inbound access** to their network. This
145     mirrors the **self-hosted CI runner** pattern (GitHub Actions runners use a 50s
146     outbound-only HTTPS long-poll). This pull-based runner is the **cheapest, most
147     credential-safe v1** for BYO-compute.
148   - For users who want the platform to **provision GPUs in their *own* cloud account on
149     demand** (an academy with an AWS account but no standing GPU box), add **SkyPilot**
150     (BYOC: everything launches in the user's accounts/VPCs; creds stay with them).
151     **Modal** is the managed fallback for users wanting zero infra (accepting that raw
152     video then flows through Modal).
153   - **?? Operational / support burden (a real adoption risk).** For the primary beachhead
?
154     Indian badminton **academies/coaches** ? asking them to *run and maintain a worker on
155     their own VM/GPU* is a meaningful ongoing load (updates, monitoring, firewall/NAT, the
156     "it stopped after our IT change" ticket). So: **default academy users to `hosted`**,
and
157     treat `byo_worker`/SkyPilot as a **power-user / enterprise / privacy-sensitive tier**,
158     not the academy default. If/when BYO ships, plan for **auto-updating, self-healing

```



```

159     agents + health heartbeats + clear worker status in the UI**, and budget the support
cost.
160 - **Middle tier to consider ? "zero-infra BYO":** a managed-but-isolated path where the
161   **storage** stays in the user's bucket (signed URLs) while compute runs on **our**
GPUs
162   in a **strictly isolated, per-tenant, ephemeral sandbox** (nothing persists our side).
163   This gives the BYO *privacy* benefit (raw video never persists on our infra)
**without**
164   the user running a worker ? a pragmatic default between full-hosted and full-BYO.
165
166   The pipeline code is unchanged; only *where it's invoked* and *how it gets bytes* vary.
167
168   ### 3.3 Reuse, don't reinvent
169   The existing `segmenter` / `provider` / `validator` / `sport` / `annotation` registries
170   are the proven precedent and stay as the **pipeline** layer. The `annotation` registry +
171   `candidate_segments` table are also the natural seam for the **coach-correction / golden
172   -set** loop in the assistant-coach phase (§6, P4).
173
174   ## 4. Multi-tenant data model & serving layer
175
176   ### 4.1 Entities ? generic core, with the coach model as an optional overlay
177   The foundational schema is deliberately **domain-neutral** so we don't overfit to the
178   coach mindset ? the coach/athlete framing is a **market-fit hypothesis that will
iterate**
179   (see C13 / PMF), not a load-bearing assumption. **Generic core:**
180   ```
181   User ??< VideoCollection ??< Video ??< Highlight ??< HighlightMetadata
182               ???< PlaySegment ??< Event
183   Video / Highlight ??< Job (processing record: status, compute backend, timings)
184   ```
185   - **Generic & re-targetable:** a `VideoCollection` groups a user's videos (a season, an
186     event, a "folder") with no assumption of a coach or academy ? a power user, a coach,
or a
187     parent are all just `User`s with collections.
188   - **Highlights are first-class** (today compile only writes a file): a `highlights`
table
189     (segment ids + output `StorageRef` + params + created_at) powers "see past videos and
190     highlights by history," and **`HighlightMetadata`** carries the **rich semantic tags**
191     (shot type, rally outcome, etc. ? see §6 metadata stage) plus titles / ratings /
notes.
192   - **`video_id` stays the content MD5** (dedup within a tenant), scoped by `tenant_id`.
193   - **Coach overlay (optional, additive):** for the academy/coach GTM, layer
194     `Org/Academy ? Coach ? Athlete` *on top* (e.g. a `Membership`/role table + an
`athlete_id`
195     tag on collections/videos) **without** baking it into the core tables ? so the schema
196     stays usable for solo power users and the coach model can evolve, or be dropped, as
PMF
197     dictates.
198   - **History/serving API:** list collections/videos/highlights per user (and, with the
199     overlay, by athlete/coach/date); **query highlights by `HighlightMetadata`** (e.g.
shot
200     type / outcome); re-stitch from stored segments without reprocessing.
201
202   ### 4.2 Tenancy isolation ? options & recommendation
203   | Model | Isolation | Ops cost | Best for |
204   |---|---|---|---|
205   | **Shared DB + `tenant_id`** (row-level, ideally Postgres **RLS**) | logical | low |
**Hosted SaaS, many small tenants** ? *recommended default* |
206   | Schema-per-tenant | medium | medium | tens?hundreds of mid tenants |
207   | **DB-per-tenant** (incl. **SQLite-per-tenant**) | strong/physical | higher at scale |
**Self-hosted single-academy**, or high-isolation enterprise |
208
209   **Recommendation:** for the hosted product, **migrate SQLite ? Postgres with shared-DB +
210   `tenant_id` + Row-Level Security**; keep **SQLite-per-tenant** as the **self-hosted /
211   single-academy** deployment (it reuses today's code almost verbatim). Either way,

```

212 ****abstract DB access behind a repository layer now**** so the engine is swappable, and
213 graduate the `PRAGMA user\_version` migration discipline to ****Alembic**** under Postgres.

214

215 **## 5. Orchestration / async jobs**

216

217 This is the serving layer that makes multi-user real. It is the ****multi-user superset of**
218 `[`DEFERRED_HARDENING_PLAN.md`](DEFERRED_HARDENING_PLAN.md)` #16****** (async job model) ?

build

219 that first, single-tenant, then generalize.

220

221 - ****Job record**** (``jobs`` table): ``id, tenant_id, video_id, type (index|compile|analyze),`
222 `status (queued|running|done|failed), progress, compute_backend, result_ref, error,`
223 `timestamps``. ``/api/process`` returns **\*\*202 + job\_id\*\***; ``/api/jobs/{id}`` polls.

224 - ****Idempotency**** keyed on ``(tenant_id, video_id, pipeline_version)`` ? reuse the
225 content-hash discipline already in ``compute_video_id``.

226 - ****Dispatch:**** the orchestrator enqueues; a worker (hosted or ``byo_worker``) claims jobs
227 for its tenant, resolves input via the tenant's ``StorageBackend``, runs the pipeline,
228 writes artifacts + DB rows. ****Observability reports**** (already per-video) become the
229 job artifact surfaced in history.

230 - ****Queue/worker backend ? a decision to prototype in #16, *not* a default here.**** The
231 ***requirements*** are firm ? ****idempotency****, ****resumability**** (a 40-min GPU job mustn't
232 restart on a worker blip), ****progress states**** ? which rule out fire-and-forget

Celery/RQ

233 for the long jobs. The ***choice*** should be made by prototyping in the
234 `[`DEFERRED_HARDENING_PLAN.md`](DEFERRED_HARDENING_PLAN.md)` #16 slice, weighing:

235 - ****Start simplest:**** ****Arq**** (async-native, natural FastAPI fit) or even a ****plain**
236 **Postgres-backed queue + worker**** ? likely sufficient for the ***first*** single-tenant
237 async slice, with the lowest operational complexity and the easiest debugging.

238 - ****Durable-execution upgrade if/when needed:**** ****Hatchet**** (Postgres-backed, per-step
239 retries/timeouts/concurrency; pairs neatly with the pull-based worker) ? attractive
240 because it adds ****no new datastore**** once we're on Postgres, but it ***is*** a new
241 dependency + operational/debugging surface; adopt only when long-job resumability/
242 observability demands it.

243 - ****Heaviest:**** ****Temporal**** ? most mature durable workflows, operationally heavy;
244 reserve for genuinely complex multi-stage pipelines at scale.

245

246 Record the decision criteria in #16: operational complexity, maturity, ****debugging**
247 **experience for long GPU jobs****, and whether durability is worth a new dependency.

248 ****Bias:** start with the simplest thing that gives retries + status; graduate only on
249 evidence.**

250

251 **## 5A. Owned generative model (the video-QA assistant) & the data flywheel**

252

253 The north-star capability: an ****owned, self-hostable generative video model**** ?
254 fine-tuned on sports video (including ****consented, user-contributed**** footage) ? that
255 ****answers questions about rallies, shots, and technique****. This is the conversational
256 brain of the "assistant coach." Crucially, it does ****not**** replace the known stack; it
257 is introduced ****behind the same abstraction, with the paid API as an always-ready**
258 **fallback****.

259

260 > ****The moat is *not* the model.**** Anyone can download Qwen-VL/Gemma. The defensible
261 > assets are **** (a) the consented, badminton-specific video?QA dataset**** and **** (b) the**
262 > **eval harness****. Build and message ***those*** ? not "we own an LLM."

263

264 > ****? Committed guardrail (legal, non-negotiable).**** We respect the licensing findings
in

265 > full. ****Paid frontier models (Gemini / OpenAI / Anthropic) are used for *inference /**
266 > **serving / fallback* only ? never as a *training* data source.**** Harvesting their
outputs

267 > to train our owned model would breach their "no competing model" terms (a
terms/copyright

268 > violation), so it is prohibited in both engineering and legal process. ****Training**
teachers

269 > are open, Apache-2.0 models only****** (Qwen-VL / InternVL / Gemma 4). Detail in the
270 > distillation constraint below.

```

271
272     ### Core principle: AI is a pluggable provider with a paid fallback
273     The codebase already abstracts the cloud AI behind `provider_registry` /
274     `AIVideoProvider` (`backend/providers/`). Make that a first-class platform
principle**:
275     - Every generative capability routes through the provider registry.
276     - The owned model plugs in as a new provider (`owned_vlm`); Gemini (paid, via
277       Vertex AI) stays registered as the fallback (failover on error/low-confidence,
278       or as the quality ceiling/teacher reference).
279     - This lets us ship today on the known paid stack and swap in the owned model when
280       its quality clears a measured bar ? with zero architectural churn and the paid path
281       always one config flip away. (Mirrors the COMMERCIALIZATION C4 A/B/C strategy: the
282       owned model is the long-run "(A) eliminate the per-call dependency"; paid is "(C)".)
283
284     ### The model / training layer (new sub-layer, fed by the scaffolding)
285     ```
286     Data curation      Training      Eval      Serving
287     (consented user vid (fine-tune an open, (extend the existing (self-host inference
288     + golden set +      commercially-usable eval harness to QA as a provider /
289     rally/segment/event base VLM via      quality + the      ComputeBackend;
290     annotations ?      LoRA/QLoRA)      regression gate)      paid fallback)
291     video-QA pairs)
292     ```
293     This layer reuses every other layer: Storage holds the training corpus;
294     Data supplies labels (the `annotation` registry + `candidate_segments` +
295     golden set are already the labeling seam); Compute/Orchestration runs training and
296     inference jobs (even BYO-compute can serve inference on the tenant's GPU); the
297     eval harness already in `backend/eval/` extends to grade QA quality. In other
298     words, the rock-solid scaffolding is exactly what makes the model flywheel possible ?
299     which is why scaffolding comes first (see §6).
300
301     ### Base model & training (Apache-2.0 only ? licensing is decisive)
302     Recommended near-term base: `Qwen2.5-VL-7B-Instruct` ? Apache-2.0, native long-video
303     (dynamic-FPS sampling, temporal mRoPE), mature LoRA tooling. Upgrade path
304     Qwen3-VL-4B/8B (Apache-2.0); Gemma 4 multimodal (Apache-2.0, 2026) is a clean
305     Google-stack alternative to evaluate.
306     - ?? Lock to Apache-2.0-clean bases so you can fine-tune, keep weights private,
307       gate behind subscriptions, and never owe open-sourcing or branding. (Note Qwen2.5-VL
308       is Apache only at 7B/32B ? 3B/72B use the custom Qwen license.)
309     - ? Reject for the flagship: Gemma 3 ? custom terms with a viral derivative
310       clause (anything fine-tuned on it, or even trained on its synthetic outputs, stays
311       permanently bound to Google's *updatable* Prohibited-Use Policy); Gemma 4's switch
to
312     Apache-2.0 removes this. Llama 3.2 Vision ? custom license forces a "Llama-?"
313     model name + "Built with Llama" badge and a 700M-MAU ceiling. gpt-oss ?
Apache-2.0
314     but text-only, so usable only as an optional text-reasoning layer, never the video
315     encoder.
316     - Training method: QLoRA (? r=64, ?=128) on the 7B ? fits one prosumer GPU and
317     avoids catastrophic forgetting of general video skills. Build the dataset by
converting
318     the existing rally/shot/segment annotations + golden set into templated clip?QA
319     pairs, expand synthetically with open-model teachers only (see constraint
below),
320     and hold out a golden QA eval that gates every fine-tune. A few hundred gold pairs
?
321     5k?20k curated is a realistic band for this narrow domain.
322     - Serving economics: a 7B VLM ? 14 GB VRAM (FP16) / ~3.5 GB (INT4); a ?16 GB GPU is
a
323     comfortable serve target (vLLM for throughput). Self-hosting wins only once you
have
324     sustained video-QA volume **** a defensible consented dataset **** real per-call API
325     cost pain ? below that, the paid API is cheaper than a standing GPU and the second
stack
326     is overhead. Hence: paid-primary now ? owned-primary / paid-fallback later.

```

```

327
328     Base-model licensing + training-data policy are tracked as **COMMERCIALIZATION C14**.
329
330     ### Training backend ? managed service vs self-managed (the fine-tune "forge")
331     *Where* you run the LoRA/RL fine-tune. Weigh on **video support, weight
export/ownership,
332     smallest servable model, RL support, no-train DPA, and infra burden**.
333
334     **Decisive finding (researched 2026-06):** managed fine-tuning services are
335     **image-based** (JSONL + base64 images) ? for video you pre-sample frames and feed them
336     as image sequences. **Only the self-managed `ms-swift` explicitly advertises *native*
337     video (+audio) training.** So for true video-QA, a self-managed framework is the
338     strongest technical fit ? and it also lets you fine-tune the **small 7B** (cheap to
339     serve) and keep full ownership. Managed services trade some of that for zero infra.
340
341     | Option | Type | Video / VLM support | Smallest VL | RL | Export weights | Best for |
342     |---|---|---|---|---|---|---|
343     | **ms-swift** (ModelScope) | Self-managed (rented/own GPU) | **Native video+audio**;
Qwen3-VL/Omni, 300+ MLLMs | any incl **7B** | ? DPO/GRPO | ? full ownership | *** video-QA;
slots into BYO-compute ($3.2)** |
344     | **LLaMA-Factory** | Self-managed | Qwen3-VL + 100+ VLMs; video datasets | any incl 7B
| ? | ? | popular, well-documented |
345     | **Unsloth** | Self-managed (1-GPU optimized) | Qwen3-VL LoRA (vision+lang; frames) |
any incl 7B | ? GRPO | ? | cheapest first runs |
346     | **Tinker** (Thinking Machines) | Managed API (low-level) | Qwen3-VL **30B-A3B+ only**;
image input (video **unverified**) | 30B-A3B MoE | ? PPO/GRPO, custom loss | ? adapters + ckpt;
no-train ToS | zero-infra RL/research at MoE scale |
347     | **Together AI** | Managed (tune + serve) | Qwen3-VL / Gemma-3 / Llama-4; image data |
verify | partial | ? adapters downloadable | managed Qwen3-VL + serving |
348     | **Fireworks AI** | Managed (tune + serve) | Full Qwen2.5-VL (3B?72B SFT; LoRA@32B +
LoRA-upload) | **7B** | partial | ?? verify export (lock-in risk) | managed *small*-VL + serving
|
349     | **Predibase** (Rubrik) | Managed *** VPC self-host** | VLM instruct-tune; **LoRAX**
multi-adapter | verify | ? (VPC) | ? **RFT/GRPO** flagship | RL "creativity" + multi-adapter +
privacy |
350     | ~~OpenAI GPT-4o / Gemini tuning~~ | Closed | ? | ? | ? | ? no export | **non-fit ?
violates "own the model"*** |
351
352     *(Enterprise clouds ? Databricks Mosaic / SageMaker / Vertex on open models ? give full
353     control but are only worth it if you're already on that stack.)*
354
355     **Recommendation:**
356     - **Early iteration / true video-QA ?** self-managed **`ms-swift`** (or LLaMA-Factory /
357     Unsloth) on a **rented GPU** (RunPod / Lambda / Modal / SkyPilot-to-your-cloud):
native
358     video, any size incl. **7B** (cheap to serve), full ownership, and it drops into the
359     **ComputeBackend / BYO-compute** seam ($3.2). **Likely a better fit than Tinker** for
360     the video requirement *and* serving economics.
361     - **Managed, no GPU ops ?** **Together AI** (Qwen3-VL + downloadable adapters + serving)
362     or **Fireworks** (full Qwen2.5-VL incl. 7B + serving) ? confirm frame/video support
and
363     adapter export.
364     - **If RL "coaching creativity" + per-sport multi-adapter is central ?** **Predibase**
365     (RFT + LoRAX, VPC self-host for the biometric/privacy posture) or **Tinker** (lowest-
366     level RL primitives).
367     - **Tinker** remains a strong zero-infra RL-research option with clean weight export ?
just
368     not the cheapest path to a *small video* model (its VL floor is 30B-A3B).
369     - **Portability is the hedge:** you own the **data + recipe + exported weights**, so the
370     backend is swappable ? same ethos as the provider/registry pattern. Don't get locked
in.
371
372     **Verify before committing (any backend):** (1) **native video / frame-sequence**
373     ingestion + max frames per example; (2) **weight export** + a **commercial DPA**
covering

```

```

374     consented, minors-excluded video (biometric, $7); (3) the **inference plan** for the
size
375     you train (a self-managed **7B** is far cheaper to serve than a 30B-A3B MoE).
376
377     ### Training harness ? make it agent-runnable (nanochat-style operability)
378     The goal: **an agent (or CI) can run a fine-tune on a fresh batch of golden-labeled
video
379     and get a trustworthy ship / no-ship verdict, unattended.** Karpthy's **nanochat** is
the
380     reference for that *operability* ? one MIT repo, one `speedrun.sh` that runs
381     tokenizer?train?eval?serve, a single complexity dial, and an automatic **report-card**
382     score. It is **not** a usable engine here (text-only, from-scratch GPT-2-class), but its
383     five properties are exactly what make training agent-runnable, and we copy them:
384     ** (1) one command, end-to-end; (2) one complexity dial (sane auto-derived
defaults);
385     ** (3) an automatic **report card** ? a deterministic pass/fail vs a target (the crux);
386     ** (4) reproducible + deterministic + cheap; (5) minimal/readable.
387
388     **Engine (don't reinvent):** **`ms-swift`** already delivers this for VLMs ? CLI-driven,
389     **native video**, Qwen3-VL support, an eval backend (EvalScope), and reproducibility via
390     `args.json` + `full_determinism`. (LLaMA-Factory is the close second; HF **`nanoVLM`** ?
391     ~750 lines ? is the minimal reference to *learn* VLM training, not the production path.)
392
393     **The harness = thin glue over components we already have** ? one declarative,
394     agent-invokable job (call it `train_speedrun`):
395     1. **Build dataset** from golden labels ? clip?QA pairs (reuse the `annotation` registry
+
396     `candidate_segments` + golden set ? the labeling seam in `backend/`).
397     2. **Fine-tune** via ms-swift on a **ComputeBackend** GPU ($3.2), run as an
orchestration
398     job ($5 / DEFERRED #16): one config, idempotent, logged.
399     3. **Auto-eval + report card** ? extend the existing `backend/eval/` regression harness
to
400     **QA quality**; emit a `report.md` pass/fail vs the **held-out golden QA eval**.
401     4. **Register** the result as the `owned_vlm` provider (Gemini paid fallback stays)
**only
402     if it beats the bar**.
403
404     So: **golden labels ? ms-swift train (on a compute backend) ? eval gate ? register
405     provider**, behind one command an agent or CI runs. nanochat is the inspiration for the
406     wrapper; ms-swift is the engine; the **golden QA eval is the report card**.
407
408     > **Prerequisite (the crux):** the harness is only as trustworthy as the **golden QA
eval
409     > set**. nanochat's speedrun is meaningful *because* its eval (CORE score) is automatic
and
410     > trustworthy; ours is the golden QA eval ? **build that first**, or an agent-run loop
411     > optimizes blind. (Same "moat = dataset + eval", scaffolding-first principle.)
412
413     ### ?? Teacher / distillation constraint (the single biggest legal landmine)
414     You **cannot** train the owned model on outputs from the paid frontier APIs ? a general
415     video-QA assistant is squarely a **competing model**: **Gemini** ("may not use the
416     Services to develop models that compete"), **OpenAI** ("may not use Output to develop
417     models that compete"), **Anthropic** (bars using Outputs to train competing models). Two
418     safe lanes: ** (1) generate synthetic QA with **open-model teachers only** (Qwen-VL /
419     InternVL / Gemma 4 ? Apache-clean); (2) keep the paid API strictly as a **runtime
420     fallback** ? serving answers is fine, *harvesting its outputs to train the model that
421     replaces it is not*. Write a hard **"no training on competitor-API outputs"** rule into
422     eng + legal process. (This makes the ROADMAP's "Gemini as offline teacher only" precise:
423     the teacher is *open models*, not the paid APIs.)
424
425     ### Data flywheel, personalization & consent tiers
426     Two complementary loops, each with its **own** opt-in (training consent is separate from
427     processing consent ? $7):
428

```

```

429 - **Global model (shared ? raises the floor).** Train the base owned model on an
430 **explicitly-contributed, licensed** corpus ? *not* on whatever sits in a tenant's
bucket
431 (preserves "your video, your storage"). Flywheel: *more consented, labeled matches ? a
432 better shared model ? better highlights/answers ? more reason to contribute.* Higher
433 consent bar (pooled data, biometric/minor care ? $7); **minors excluded by default**.
434
435 - **Per-user adapters (personalized ? raises the ceiling) ? owner hunch, under
exploration.**
436 A small **per-user (or per-tenant) LoRA adapter, fine-tuned *only on that user's own
437 consented videos** on top of the shared base, served via **multi-LoRA** (LoRAX /
438 Together / Predibase). Why it's compelling:
439 - **Precision:** adapts to *that user's* court, camera angle, lighting, and players'
440 styles ? materially better detection/highlights for them.
441 - **Cleanest consent story:** "improve **your** detection using **your** data" is
far
442 lower-friction and lower-risk than contributing to a global model ? the data
improves
443 **only the user's own adapter**, with **no cross-user mixing**, sidestepping much of
the
444 pooled-corpus biometric/minor entanglement. Reinforces the differentiator: *your
445 video, your storage, your compute ? and now **your model**. With BYO-compute the adapter
can be
446 trained/served on the user's own GPU and **never leave their boundary**; **deletion
=
447 drop the adapter** (clean erasure).
448 - **Open questions (still a hunch):** cold-start (need enough of the user's labeled
data
449 before a personal adapter beats the base ? fall back to the shared model meanwhile);
a
450 **per-user held-out eval** to prove the adapter helps (don't ship a regression);
base +
451 optional global-adapter + per-user-adapter stacking; per-user training cost at
scale.
452 Worth a spike once the harness + golden eval exist.
453
454 **Two consent tiers, then:** *(a)** "improve **my own** detection with **my** data"
455 (per-user, low friction) and *(b)** "contribute to the **shared** model" (global,
higher
456 bar). **Default for both is off.**
457
458 ## 6. Phased roadmap
459
460 | Phase | Deliverable | Build / reuse |
461 |---|---|---|
462 | **P0** | **Async job model, single-tenant** (DEFERRED #16) | `jobs` table via
`user_version` ladder; worker; 202+status endpoints |
463 | **P1** | **Identity + multi-tenant data model + upload + highlight persistence +
history** | auth (build vs Auth0/Clerk/Supabase ? decide); Postgres + `tenant_id`/RLS;
`highlights` table; repository layer; history API + UI |
464 | **P2** | **`StorageBackend` registry + BYO-storage** | new ABC+registry;
`local`?`gdrive`/`s3`/`gcs`; replace `video_path` with `StorageRef`; encrypted per-tenant creds
|
465 | **P3** | **`ComputeBackend` registry + BYO-compute** | `local`/`hosted`/`byo_worker`;
pull-based worker agent; control/data-plane split |
466 | **P4** | **Rich rally/shot semantic metadata & tagging** *(the next value stage after
solid rally detection + highlights)* | shot type (smash / backhand drive / net), rally
**outcome** (winner / forced / **unforced error**), context (e.g. **open court but error**) ?
query-API filters + `HighlightMetadata` + a **human correction/review loop** feeding the golden
set via the `annotation` registry + `candidate_segments` |
467 | **P5** | **Owned generative video-QA model** ($5A) | global flywheel **+ per-user
adapters**; P4 metadata is training signal ? fine-tune an open VLM (Qwen-VL) via LoRA ? serve as
`owned_vlm` **provider with Gemini paid fallback**; extend the eval harness to QA quality |
468 | **P6** *(visionary)* | **Gait / biomechanics under a supervising coach** |

```

deliberately **back-burner**; technique feedback; highest Art. 9 bar ? gate behind explicit biometric + parental consent (§7) |

469

470 P0?P1 deliver the headline ask (per-user upload + stored highlights + history). P2?P3

471 deliver the "plug-and-play storage/compute" ask. **P4** (rich semantic metadata) is the

472 immediate next value stage once rally detection + highlights are solid ? e.g.

"rallies

473 that ended in a smash or backhand drive," **"rallies** where the court was open but the

474 player made an unforced error." **P5** owns the model; **P6** (gait) is the visionary

475 horizon, intentionally deferred."

476 **Sequencing principle (owner directive):** make the scaffolding (P0?P4) **rock-solid**

477 **first**, keep the **known paid stack** as an always-ready fallback, and only then shift

478 focus to **owned-model quality** (P5). The provider registry keeps the paid path one

479 config flip away.

480

481 **## 7. Privacy & legal** (gait raises the bar ? design for it now)

482

483 - **The load-bearing nuance:** biometric data is special-category **only** when used to

484 **uniquely identify** a person (GDPR Art. 9). **Highlight/shot detection** on players a

485 coach already knows is general analytics ? generally **not** Art. 9. But

486 **gait/biomechanics** that builds a per-athlete identifying template tips into Art. 9

487 and needs **explicit, granular consent** referencing biometric data. Engineer gait

488 with that boundary (consent gate, purpose limitation, no cross-account

identification);

489 decide whether it runs **locally/on-tenant only** (strong posture) vs our cloud. (EDPB

490 Guidelines 3/2019 on video devices apply.)

491 - **Minors** (e.g. academies). Parental/guardian consent and safeguarding are first-

order,

492 not an afterthought. Model consent explicitly per relationship (user?subject,

coach?athlete

493 where the overlay applies, parent?minor).

494 - **BYO-storage** is the central privacy/liability lever: if raw video stays in the

495 tenant's own bucket/Drive and we process via signed URL or their compute, we minimize

496 what we hold ? shrinking breach surface, residency, and erasure obligations.

497 - **Paid-LLM no-train (C4):** use **Gemini via Vertex AI** (contractual no-training),

498 **not the consumer/free tier**; set short/zero retention, document it in the DPA, and

499 surface "your footage is never used to train AI" as a trust feature for academies. For

500 the gait phase, prefer keeping biometric inference off third-party APIs entirely.

501 - **Training consent** is a distinct, higher bar (for the owned model, §5A). Training on

502 user video needs a **separate, explicit, opt-in training licence** ? not bundled with

503 processing consent ? and it **compounds** the biometric (Art. 9) and minor-consent

504 issues above. Train **only on explicitly contributed/licensed video**; never train on

505 what merely passes through a tenant's bucket. **Exclude minors' video** from the

training

506 pool by default, keep the training corpus a **separate, firewalled, opt-in sub-**

pool,

507 and log **per-clip consent/provenance** for revocation + audit. Incentives must not

make

508 consent non-free (the product stays fully usable without contributing). And the

509 **teacher-distillation rule** (§5A): never train on paid-LLM outputs. **Two consent**

tiers

510 (§5A): **(a)** "improve **my own** detection with **my** data" ? a per-user adapter

trained

511 only on the user's own video (lowest friction; no cross-user mixing; deletion = drop

the

512 adapter); **(b)** "contribute to the **shared** model" ? pooled corpus, higher bar.

Default

513 for both is **off**.

514 - **Right to erasure / retention:** per-tenant, per-athlete delete must cascade across

DB +

515 storage artifacts (the **ON DELETE CASCADE** discipline already in `database.py` extends

here).

516

517 **## 8. Key decisions for the owner** (the forks)

518

519 1. ****Hosted-first vs BYO-first.**** Recommended: ****hosted control-plane first**** (P0?P1)
with
520 ****BYO-storage soon after**** (P2); ****BYO-compute later**** (P3) as a power-
user/enterprise
521 option ? it's the highest-complexity, highest-security item.
522 2. ****Database now or later.**** Move to ****Postgres at P1**** for the hosted path, or stay
523 SQLite-per-tenant if self-hosted-single-academy is the first GTM. (Abstract DB access
524 regardless.)
525 3. ****Auth: build vs buy**** (Auth0 / Clerk / Supabase Auth / Cognito).
526 4. ****Orchestration tech**** (see research recommendation) ? bias to the simplest that
gives
527 retries + status.
528 5. ****Gait processing location**** ? local/on-tenant (privacy-max) vs cloud (capability-
max).
529 6. ****How far to push BYO-compute**** given its security/ops complexity.
530 7. ****Owned-model base + licensing (\$5A).**** Which commercial-OK video VLM to build on
531 (Qwen-VL Apache-2.0 vs Gemma 3 custom license vs ?) and confirm a gated commercial
532 derivative is permitted ? see COMMERCIALIZATION C14.
533 8. ****Training-data sourcing & consent.**** Opt-in contribution terms, incentives, and the
534 teacher-distillation constraint (whether paid-LLM outputs may seed the corpus).
535 9. ****When to start the owned model**** ? strictly after the scaffolding + flywheel are
536 solid (the explicit directive), with the paid provider as the standing fallback.
537
538 ****Risks:**** scope explosion (sequence strictly P0?P5); multi-tenant data-leak bugs (RLS +
539 tests); BYO-compute security; ****biometric/minor liability**** (get this right before any
540 gait feature ships); ****training on user video without proper opt-in/licence****, and
541 ****base-model / teacher-distillation licence violations****; cost of hosted GPU (training
542 *and* inference); ****premature model-building**** before data/eval are ready (mitigated by
543 keeping the paid fallback and gating P5 behind the flywheel).
544
545 **## 9. Sources**
546
547 External best-practices research (2026-06-07) backing the recommendations above. Each
548 underlying claim in the research was tagged fact-vs-synthesis; figures are as-of and
549 should be re-validated before committing budget.
550
551 ****Storage abstraction:**** [fsspec](https://fsspec.com/) ·
[cloudpathlib](https://github.com/drivendataorg/cloudpathlib) ·
[smart_open](https://github.com/piskvorky/smart_open) · [Apache
Libcloud](https://libcloud.apache.org/) · [gdrive-fsspec](https://github.com/fsspec/gdrive-
fsspec) · [dropboxdrivefs](https://github.com/fsspec/dropboxdrivefs) · [AWS S3 presigned
URLs](https://docs.aws.amazon.com/AmazonS3/latest/userguide/using-presigned-url.html) · [token-
based S3 with STS](https://oneuptime.com/blog/post/2026-02-12-token-based-access-s3-sts/view) ·
[signed URLs across clouds](https://medium.com/@shaikhsharwan49/signed-urls-in-aws-gcp-and-azure-
a-beginner-friendly-guide-e2e549425865)
552
553 ****Multi-tenancy:**** [Bytebase: multi-tenant DB
patterns](https://www.bytebase.com/blog/multi-tenant-database-architecture-patterns-explained/) ·
[Hunchbite: RLS vs schema-per-tenant](https://hunchbite.com/guides/multi-tenant-saas-
architecture) · [PlanetScale: tenancy in Postgres](https://planetscale.com/blog/approaches-to-
tenancy-in-postgres) · [DEV: DB-per-tenant vs shared schema](https://dev.to/young_gao/multi-
tenant-architecture-database-per-tenant-vs-shared-schema-1n2e)
554
555 ****BYO-compute / orchestration:**** [Pinecone BYOC](https://www.pinecone.io/blog/byoc/) ·
[Nuon: control/data-plane](https://nuon.co/blog/byoc-control-plane-data-plane-architectures) ·
[GitHub self-hosted runners (ARC)](https://docs.github.com/en/actions/hosting-your-own-
runners/managing-self-hosted-runners-with-actions-runner-controller/about-actions-runner-
controller) · [SkyPilot](https://github.com/skypilot-org/skypilot) · [SkyPilot AI control
plane](https://blog.skypilot.co/ai-job-orchestration-pt2-ai-control-plane/) · [Modal (AWS
Marketplace)](https://aws.amazon.com/marketplace/pp/prodview-j727623xqhh2k) · [Northflank:
Anyscale alternatives](https://northflank.com/blog/anyscale-alternatives-for-ai-ml-model-
deployment) · [DevPro: Python task queues
2025](https://devproportal.com/languages/python/python-background-tasks-celery-rq-dramatiq-
comparison-2025/) · [Judoscale: choosing a task queue](https://judoscale.com/blog/choose-python-
task-queue) · [Hatchet vs Temporal](https://hatchet.run/versus/hatchet-vs-temporal) ·

[Hatchet](https://github.com/hatchet-dev/hatchet)

556

557 **Privacy / legal:** [GDPR Art. 9](https://gdpr-info.eu/art-9-gdpr/) · [GDPR Local: biometric compliance](https://gdprlocal.com/biometric-data-gdpr-compliance-made-simple/) · [Legiscope: Art. 9](https://www.legiscope.com/blog/gdpr-article-9-special-categories.html) · [EDPB Guidelines 3/2019 (video)](https://www.edpb.europa.eu/sites/default/files/files/file1/edpb_guidelines_201903_video_devices_en_0.pdf) · [Fox Rothschild: EDPB video guidance](https://www.foxrothschild.com/publications/video-surveillance-under-gdpr-edpb-issues-final-guidance) · [ICO: children & UK GDPR](https://ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/childrens-information/children-and-the-uk-gdpr-old/what-are-the-rules-about-an-iss-and-consent/) · [EDPB Guidelines 05/2020 (consent)](https://www.edpb.europa.eu/sites/default/files/files/file1/edpb_guidelines_202005_consent_en.pdf) · [AI data-retention compared](https://ax-sentinel.com/blog/ai-data-retention-policies-compared) · [Xenoss: enterprise LLM platforms](https://xenoss.io/blog/openai-vs-anthropic-vs-google-gemini-enterprise-llm-platform-guide)

558

559 **Comparable products:** [Veo](https://www.veo.com/) · [Veo Help Center](https://support.veo.com/hc/en-us/articles/4459159526929-Overview-of-cam-veo-co-Record-manage-and-upload-footage-with-your-Veo-Cam-1) · [Lastly: Hudl competitors](https://www.lastly.org/hudl-competitors/) · [Apple: Behind the Design ? SwingVision](https://developer.apple.com/news/?id=0pg4dthn) · [SwingVision FAQ](https://swing.vision/faq) · [Sportsvisio: coach video-analysis comparison](https://www.sportsvisio.com/stories/video-analysis-software-for-coaches-complete-comparison)

560

561 **Training backends & harness (\$A):** [karpathy/nanochat (harness reference)](https://github.com/karpathy/nanochat) · [HF nanoVLM (minimal VLM)](https://github.com/huggingface/nanoVLM) · [nanoVLM blog](https://huggingface.co/blog/nanovlm) · [Tinker (GA + vision)](https://thinkingmachines.ai/blog/tinker-general-availability/) · [Tinker model lineup](https://tinker-docs.thinkingmachines.ai/model-lineup) · [Tinker ToS (weights/no-train)](https://thinkingmachines.ai/legal/tos/) · [Fireworks VLM fine-tuning](https://fireworks.ai/blog/vlm-tuning) · [Together AI fine-tuning](https://www.together.ai/fine-tuning) · [Predibase: fine-tune & serve VLMs](https://predibase.com/blog/how-to-fine-tune-and-serve-vlms-in-predibase) · [Predibase RFT/GRPO](https://docs.predibase.com/user-guide/fine-tuning/grpo) · [ms-swift (native video/audio)](https://github.com/modelscope/ms-swift) · [LLaMA-Factory (VLMs)](https://github.com/hiyouga/LlamaFactory) · [Unsloth (Qwen3-VL)](https://qwen.readthedocs.io/en/latest/training/unsloth.html) · [Fine-tuning tools 2026 (LoRA/QLoRA)](https://www.index.dev/blog/top-ai-fine-tuning-tools-lora-vs-qlora-vs-full)

562

563 **Owned model & licensing (\$A):** [Qwen2.5-VL](https://qwenlm.github.io/blog/qwen2.5-vl/) · [Qwen2.5 license split](https://qwenlm.github.io/blog/qwen2.5/) · [Qwen3-VL](https://github.com/QwenLM/Qwen3-VL) · [Gemma 3 (multimodal)](https://huggingface.co/blog/gemma3) · [Gemma Terms of Use](https://ai.google.dev/gemma/terms) · [Gemma license-risk analysis](https://wcr.legal/google-gemma-license-risks/) · [Gemma 4 / Apache-2.0](https://www.mindstudio.ai/blog/what-is-gemma-4-apache-2-license-commercial-ai-deployment) · [Llama 3.2 license (700M MAU + naming)](https://www.llama.com/llama3_2/license/) · [gpt-oss (text-only, Apache-2.0)](https://openai.com/index/introducing-gpt-oss/) · [Gemini API terms (compete clause)](https://ai.google.dev/gemini-api/terms) · [OpenAI terms](https://openai.com/policies/row-terms-of-use/) · [Anthropic: outputs & training](https://support.claude.com/en/articles/12326764-can-i-use-my-outputs-to-train-an-ai-model) · [QLoRA practical](https://apxml.com/courses/fine-tuning-adapting-large-language-models/chapter-4-parameter-efficient-fine-tuning/practice-qlora-fine-tuning) · [VLM LoRA cookbook](https://huggingface.co/learn/cookbook/en/fine_tuning_vlm_trl) · [GDPR Art. 9](https://gdpr-info.eu/art-9-gdpr/) · [athletes & biometric data](https://www.architectureandgovernance.com/applications-technology/biometric-data-and-athletes-privacy-law-and-compliance-implications/) · [federated learning + DP](https://research.google/blog/federated-learning-with-formal-differential-privacy-guarantees/) · [self-hosted LLM VRAM/cost](https://www.gpu-mart.com/guides/self-hosted-llm) · [Gemini API pricing](https://ai.google.dev/gemini-api/docs/pricing)

564

565 ## See also

566 - COMMERCIALIZATION.md ? C12 (SaaS foundations), C4 (paid-

LLM), economics.

```
567 - [ `DEFERRED_HARDENING_PLAN.md` ](DEFERRED_HARDENING_PLAN.md) ? #16 async job model (the
P0 infra slice), #7 packaging.
568 - [ `PMF_MARKET_RESEARCH.md` ](PMF_MARKET_RESEARCH.md) ? academy/coach beachhead (who this
serves).
569 - [ `CODE_MAP.md` ](CODE_MAP.md) ? current architecture + the registry extension points.
570 - [ `ROADMAP.md` ](ROADMAP.md) ? offline self-improving rally detection (the pipeline's
own trajectory).
571
```

8. assistant (2026-06-08T13:17:53.333Z)

I have strong grounding in the existing strategy docs. The research, verification corrections, and roadmap all align well. Let me also quickly check the eval harness and training code references to ground the label-reuse strategy in actual code.

9. user (2026-06-08T13:17:54.007Z)

```
backend\eval\__init__.py
backend\eval\annotations.py
backend\eval\classifier.py
backend\eval\metrics.py
backend\eval\training.py
backend\eval\calibration.py
backend\eval\gemini_refine.py
backend\eval\calibrate_local.py
backend\eval\calibrate_wasb.py
backend\eval\distill_local.py
backend\eval\export_wasb_segments.py
backend\eval\harness.py
backend\eval\regression.py
backend\eval\batch.py
backend\eval\eval_partial_labels.py
```

10. user (2026-06-08T13:17:54.370Z)

```
backend\eval\training.py
```

11. user (2026-06-08T13:17:57.812Z)

```
backend\eval\metrics.py:48:def match_intervals(preds: List[Interval], gts: List[Interval],
backend\eval\training.py:50:def label_candidates(candidate_intervals: List[Interval],
gt_intervals: List[Interval],
```

12. user (2026-06-08T13:17:58.267Z)

```
1      """Build a training set for the offline, API-free learned rally segmenter.
2
3      Phase 4 of the collector->indexer bridge (see docs/ROADMAP.md, ADR-004). The idea:
4      a detector (YOLO player-motion today, optionally WASB shuttle-motion later)
5      produces candidate rally windows with features; the collector's ground-truth
6      rally intervals tell us which windows are real. A lightweight classifier learns
7      that mapping, replacing the paid Gemini refinement with a local model.
8
9      This module is the *data layer* ? pure and unit-tested:
10
11      - `label_candidates` ? generic: label windows positive/negative by whether they
12        match a ground-truth rally (same IoU matching the evaluator uses, so training
13        labels are consistent with how we score).
14      - `extract_yolo_features` ? YOLO-specific feature vector from a candidate row's
15        persisted `stats`. Pluggable: a WASB substrate supplies its own feature_fn.
16      - `build_training_set` ? glue: rows + GT -> (X, y, intervals).
17
18      No model training or I/O here; that lives in the trainer/segmenter once the
```

```

19     substrate is chosen from the Phase 3 candidate-recall result.
20     """
21
22     from typing import Callable, Dict, List, Tuple
23
24     from backend.eval.metrics import Interval, match_intervals
25
26     # YOLO Stage-1 persists these per candidate window in `candidate_segments.stats`
27     # (see HybridYoloSegmenter). Duration is derived from the window bounds.
28     YOLO_FEATURE_NAMES = [
29         "duration", "peak_velocity", "mean_velocity", "active_frame_count",
30         "track_id_count",
31     ]
32
33     def extract_yolo_features(row: Dict) -> List[float]:
34         """Feature vector for one YOLO candidate row (dict from query_candidate_segments).
35
36         `stats` is already JSON-deserialized by the DB layer. Missing fields default
37         to 0.0 so a sparse/old row still yields a well-formed vector.
38         """
39         stats = row.get("stats") or {}
40         duration = float(row["end_time"]) - float(row["start_time"])
41         return [
42             duration,
43             float(stats.get("peak_velocity", 0.0)),
44             float(stats.get("mean_velocity", 0.0)),
45             float(stats.get("active_frame_count", 0)),
46             float(stats.get("track_id_count", 0)),
47         ]
48
49
50     def label_candidates(candidate_intervals: List[Interval], gt_intervals: List[Interval],
51                          iou_threshold: float = 0.5) -> List[int]:
52         """Label each candidate window 1 (real rally) or 0 (spurious).
53
54         A window is positive iff it is matched one-to-one to a ground-truth rally at
55         `iou_threshold` ? the *same* matching the evaluator uses for true positives,
56         so the classifier is trained on exactly what we score. Substrate-agnostic:
57         works for YOLO or WASB-derived windows alike.
58         """
59         mr = match_intervals(candidate_intervals, gt_intervals, iou_threshold)
60         positive = {m.pred_index for m in mr.matches}

```

13. assistant (2026-06-08T13:18:04.350Z)

The code confirms the label-reuse spine: `backend/eval/metrics.py::match\_intervals` is the single IoU scorer already reused by `backend/eval/training.py::label\_candidates`. This is exactly the "one spine, no metric drift" anchor the research recommends. I have everything I need to synthesize the study.

Let me apply the verification corrections (notably: Gemma 4 upgraded from AMBER to confirmed-clean with native video + Apache-2.0 per primary Google model card; corrected Gemma 4 size names E2B/E4B/26B/31B not "4B/12B"; updated GPU prices; GCP asia-south1 has H100-class A3 not A100, capacity-gated; VideoLLaMA3 disqualified for paid-LLM training data) and produce the structured output.

14. user (2026-06-08T13:21:13.817Z)

Structured output provided successfully

15. assistant (2026-06-08T13:21:28.124Z)

I've synthesized the complete roadmap-aligned training study and returned it via

StructuredOutput. The study integrates the research, applies all verification corrections, and respects every hard guardrail.

Key decisions delivered:

****Platform ranking (privacy+cost lens):**** Modal primary (strongest written no-access contract, per-second scale-to-zero) ? RunPod Secure Cloud fallback ? Vertex/GCP on the bench (only strong-privacy India region, but capacity-gated). The fine-tune is a ~\$10-50 sub-24h job, so privacy + ergonomics dominate, not GPU \$/hr.

****Phased model path:**** Gen-0 ASFormer (MIT) TAS head over WASB features that fits the RTX 2070 Super and learns temporal structure rather than threshold-tuning ? Gen-1 Option-A modular fusion into the same head ? Gen-2 Option-B end-to-end VLM (Qwen3-VL-4B primary, Qwen2.5-VL-7B fallback, Gemma 4 as the native-frames+audio end-state) via SFT cold-start + GRPO/RFT with a temporal-IoU reward.

****Label-reuse strategy:**** one canonical JSONL rally record ? three deterministic transcoders (heuristic-window labels, ActivityNet TAL JSON, VLM clip?QA), with `metrics.match\_intervals` as the single scorer for labeling, eval, and the RFT reward. Split-by-match leakage guard and provenance/consent fields baked into the format.

Verification corrections applied:

- ****Gemma 4 upgraded AMBER ? confirmed-clean**** (primary Google model card confirms both Apache-2.0 AND native video+audio); corrected size names to E2B/E4B/26B/31B (no plain "4B").
- GPU prices corrected (Modal cheaper than cited; RunPod/Lambda higher).
- GCP asia-south1 = H100-class A3, NOT A100, capacity-gated.
- VideoLLaMA3 weights doubly disqualified (non-commercial AND trained on OpenAI/Gemini data ? trips the paid-LLM guardrail).
- ASFormer's small-data result was on I3D-2048 features, not low-dim trajectory ? flagged as the dominant empirical unknown requiring a golden-set A/B (not runnable in this dev container).

The study is design-only and explicitly does not start implementation, per the owner sequencing directive (P0-P4 scaffolding first).

The two load-bearing code anchors I verified ? `backend/eval/metrics.py::match\_intervals` and `backend/eval/training.py::label\_candidates` ? confirm the label-reuse spine already exists in the codebase, so the strategy builds on real shipped state, not aspiration.